AIAA 3111 · 数据挖掘导论

L6 高级分类

分类：高级方法

Classification: Advanced Methods

中英对照 · 双页阅读



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第 6 讲 分类：高级方法

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AIAA 3111 - Introduction to Data Mining

Lec-6 Classification: Advanced Methods

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分类：高级方法

一条主线：使用什么证据、如何学习边界，以及标签或数据结构变化时会发生什么。

课次	核心主题	引导问题
1	特征选择 • 贝叶斯网络 • SVM	*如何选择证据并学习决策边界？*
2	规则 • 弱监督 • 复杂数据	*监督方式和数据结构如何改变分类器？*

设计原则

表示 → 学习目标 → 假设 → 评估

Classification: Advanced Methods

A single narrative: what evidence is used, how a boundary is learned, and what changes when labels or data structure change.

Session	Core topics	Guiding question
1	**Feature selection · Bayesian networks · SVM**	*How do we select evidence and learn a decision boundary?*
2	**Rules · Weak supervision · Rich data**	*How do supervision and data structure change the classifier?*

Design principle

Representation → learning objective → assumptions → evaluation

第 1 次课：选择、概率与间隔

模块	重点
1	特征选择与稀疏学习
2	贝叶斯网络与因子分解
3	支持向量机与核方法
4	综合

学习目标

解释每个模型的假设，以及其训练目标如何塑造预测。

Session 1: Selection, Probability and Margin

Block	Focus
1	Feature selection and sparse learning
2	Bayesian networks and factorization
3	Support vector machines and kernels
4	Synthesis

Learning objective

Explain what each model assumes and how its training objective shapes predictions.

特征选择与特征工程

- **特征选择**
  - 给定 p 个初始特征，如何选出少数最有效的特征?
  - 为什么?
    - 无关特征（用学号预测 GPA）
    - 冗余特征（月收入 vs. 年收入）
- **特征工程**
  - 给定初始特征，如何构造更有效的特征?
    - # of daily positive cases, # of daily tests, # of daily hospitalization → weekly positive rate
  - （传统上）领域知识是关键
  - 深度学习提供了自动化的方式

Feature Selection & Feature Engineering

- **Feature Selection**
  - Given a set of *p* initial features, how to select a few most effective ones?
  - Why?
    - Irrelevant features (student ID for predicting GPA)
    - Redundant features (monthly income vs. yearly income)
- **Feature Engineering**
  - Given the initial features, how to construct more effective ones?
    - # of daily positive cases, # of daily tests, # of daily hospitalization → weekly positive rate
  - (traditionally) domain knowledge is the key
  - Deep learning provides an automatic way

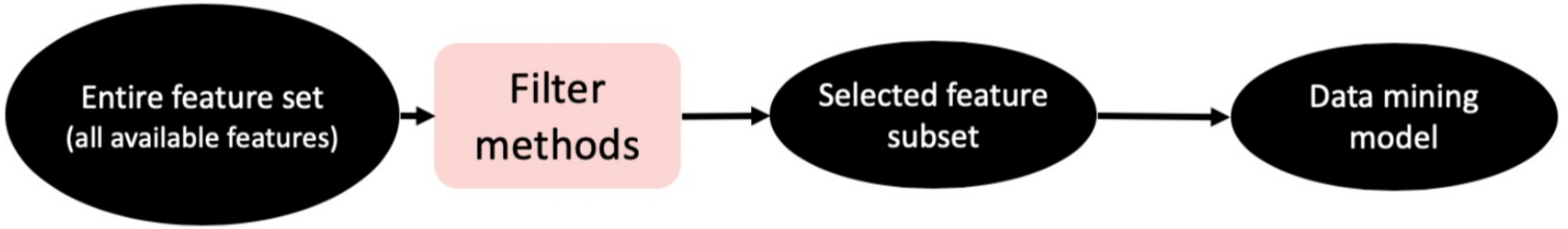
特征选择方法

- **过滤法**
  - 依据某种优良性度量选择特征
  - 与具体分类模型无关
- **包装法**
  - 把特征选择与分类模型构建两个步骤结合在一起，
  - 迭代地
    - 用当前选定的特征子集构建分类模型
    - 用当前分类模型更新所选特征子集。
- **嵌入法**
  - 同时构建分类模型并选择相关特征
  - 把特征选择步骤嵌入分类模型的构建过程中

Feature Selection Methods

- **Filter methods**
  - Select features based on some goodness measure
  - Independent of the specific classification model
- **Wrapper methods**
  - Combine the feature selection and classifier model construction steps together,
  - Iteratively
    - Use the currently selected feature subset to construct a classification model
    - Use the current classification model to update the selected feature subset.
- **Embedded methods**
  - Simultaneously constructs the classification model and selects the relevant features
  - Embed the feature selection step during the classification model construction step

过滤法

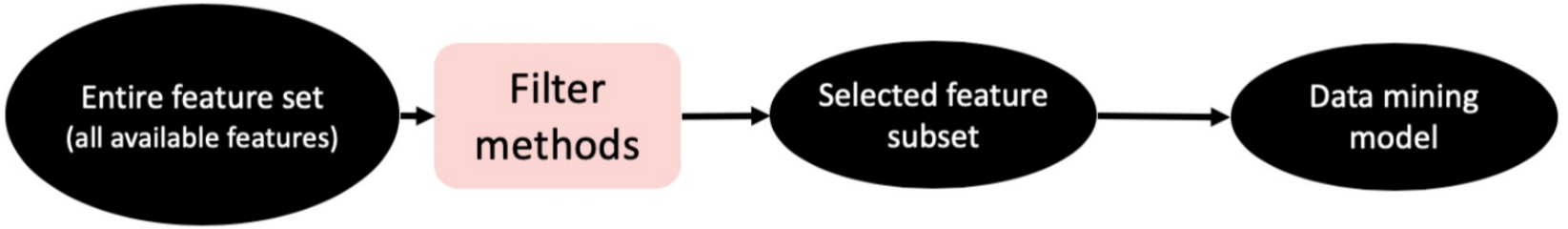


- **一般流程**
  - 依据某种优良性度量选择特征
  - 与具体分类模型无关
- **Fisher 分数**
  - **直觉**：若满足以下条件，特征 x （如收入）与类别标签 y （是否买电脑）强相关：
    - 买电脑的顾客的平均收入与不买电脑的顾客的平均收入显著不同，
    - 买电脑的顾客收入相近，并且
    - 不买电脑的顾客收入也相近。
  - **细节**

$$s = \frac{\sum_{j=1}^{c} n_j (\mu_j - \mu)^2}{\sum_{j=1}^{c} n_j \sigma_j^2}$$

- **其他优良性度量**
  - χ^2 检验（用于类别特征）；信息增益、互信息。

Filter Methods

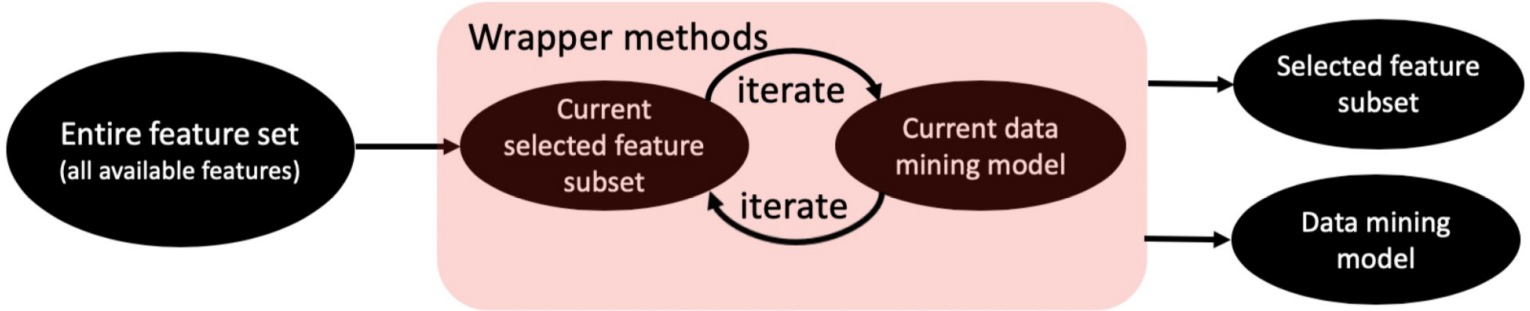


- **General Procedure**
  - Selects features based on some goodness measure
  - Independent of the specific classification model
- **Fisher Scores**
  - **Intuitions**: the feature x (e.g., income) is strongly correlated with the class label y (buy computer) if
    - the average income of all customers who buy a computer is significantly different from the average income of all customers who do not buy a computer,
    - all customers who buy a computer share similar income, and
    - all customers who do not buy a computer share similar income.
  - **Details**

$$s = \frac{\sum_{j=1}^{c} n_j (\mu_j - \mu)^2}{\sum_{j=1}^{c} n_j \sigma_j^2}$$

- **Other goodness measures**
  - χ^2 test (for categorical feature); information gain, mutual information.

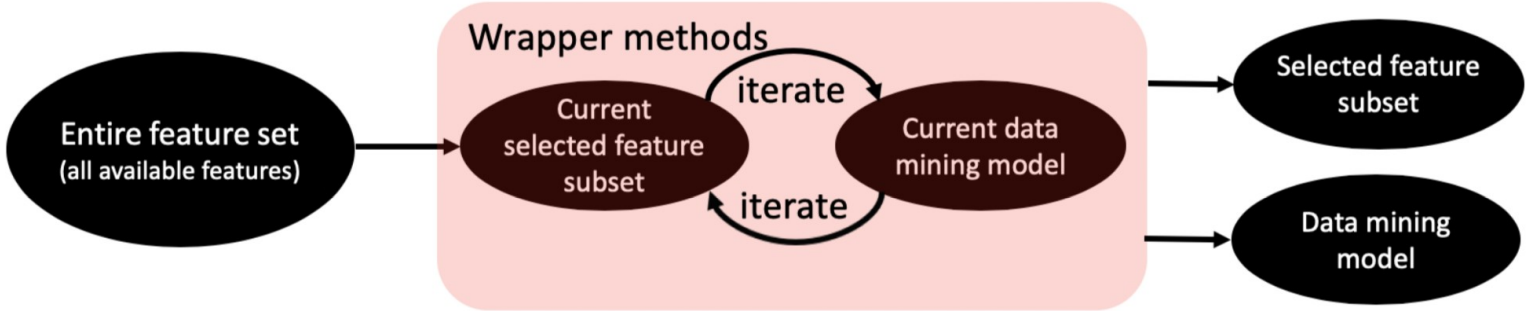
包装法



- **一般流程**
  - 把特征选择与分类模型构建两个步骤结合在一起，
  - 迭代地
    - 用当前选定的特征子集构建分类模型
    - 用当前分类模型更新所选特征子集。

- **关键：如何搜索最佳特征子集**
  - 穷举搜索： 2^p-1 （指数级）
  - 逐步前向选择：
    - 从空特征子集开始。
    - 每次迭代选择一个能最大程度提升性能的新特征
  - 逐步后向消除：从全集开始，每次删除一个特征
  - 混合方法

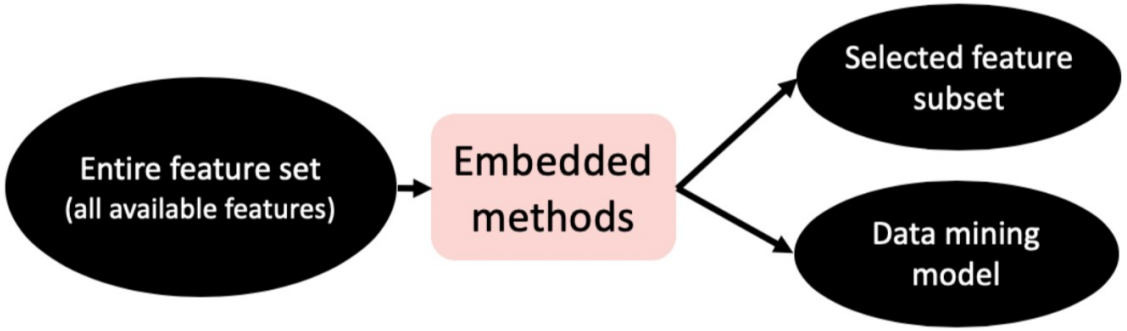
Wrapper Methods



- **General Procedure**
  - Combines the feature selection and classifier model construction steps together,
  - Iteratively
    - Use the currently selected feature subset to construct a classification model
    - Use the current classification model to update the selected feature subset.

- **Key: how to search for the best feature subset**
  - Exhaustive search: 2^p-1 (exponential)
  - Stepwise forward selection:
    - Start with an empty feature subset.
    - At each iteration, select an additional feature to improve performance most
  - Stepwise backward elimination: start with the full set, eliminate one feature at a time
  - Hybrid method

嵌入法



- **一般流程**
  - 同时构建分类模型并选择相关特征
  - 把特征选择步骤嵌入分类模型的构建过程中
- **LASSO**：最小绝对收缩与选择算子
  - 模型

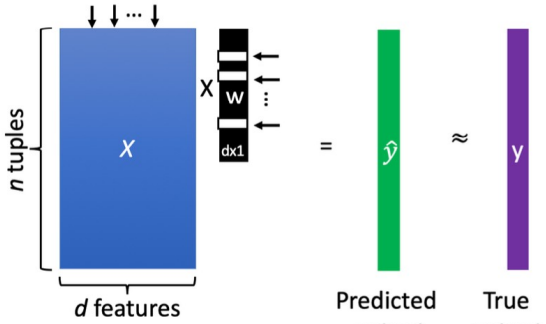
$|w|_0 = \cdot\mathrm{nnz}\{w_i, i = 1, ..., d\}$

$$\hat{L}(w) = \frac{1}{2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda\|w\|_1 = \frac{1}{2}\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \lambda\sum_{j=0}^{d}|w_j|$$

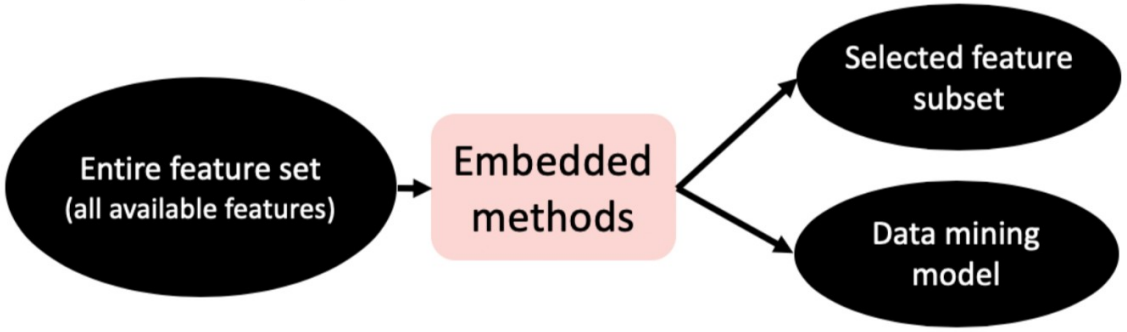
预测优良性

（凸）近似
所选特征数的

  - 训练：坐标下降



Embedded Methods



- **General Procedure**
  - Simultaneously constructs the classification model and selects the relevant features
  - Embed the feature selection step during the classification model construction step
- **LASSO**: Least Absolute Shrinkage and Selection Operator
  - Model

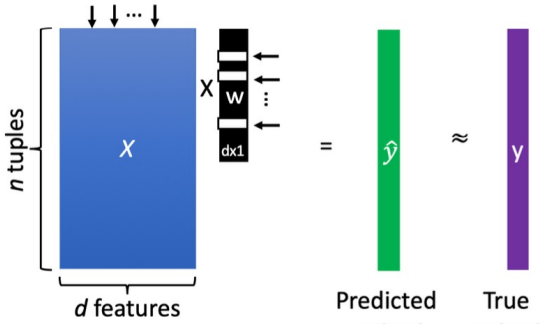
$$|w|_0 = \cdot\text{nnz}\{w_i, i = 1, ..., d\}$$

$$\hat{L}(w) = \frac{1}{2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda\|w\|_1 = \frac{1}{2}\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \lambda\sum_{j=0}^{d}|w_j|$$

Goodness of prediction

(convex) approximation of # of selected features

  - Training: Coordinate descent



Lasso 的坐标下降

- **LASSO**： 最小绝对收缩与选择算子

$$\hat{L}(w) = \frac{1}{2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda\|w\|_1 = \frac{1}{2}\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \lambda\sum_{j=0}^{d}|w_j|$$

  - 训练：每次迭代更新一个系数（w_t），固定其他系数
    - 求解一个最小二乘回归问题：一维输入（$x_{i,t}$），输出 r

$$\beta_t = \text{argmin}_{w_t}\frac{1}{2}\sum_{i=1}^{n}(r_i - w_t x_{i,t})^2$$

$$r_i = y_i - \sum_{j=0, j\neq t}^{d} w_j x_{i,j}$$

$$L(w_t) = \frac{1}{2}(w_t - \beta_t)^2 + \lambda|w_t|$$

    - 求解如下优化问题（二次、[illegible]）

假设每个特征已归一化： $\sum_{i=1}^{n} x_{i,t}^2 = x_t^T \cdot x_t = \|x_t\|^2 = 1$

Coordinate Descent for Lasso

- **LASSO**: Least Absolute Shrinkage and Selection Operator

$$\hat{L}(w) = \frac{1}{2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda\|w\|_1 = \frac{1}{2}\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \lambda\sum_{j=0}^{d}|w_j|$$

  - Training: at each iteration, update one coefficient (w_t) while fixing others
    - Solve a least square regression problem, single dimension input ($x_{i,t}$) and output r

$$\beta_t = \text{argmin}_{w_t}\frac{1}{2}\sum_{i=1}^{n}(r_i - w_t x_{i,t})^2$$

$$r_i = y_i - \sum_{j=0, j\neq t}^{d} w_j x_{i,j}$$

$$L(w_t) = \frac{1}{2}(w_t - \beta_t)^2 + \lambda|w_t|$$

    - Solve the following optimization p[illegible]dratic, single variable)

assume each feature is normalized: $\sum_{i=1}^{n} x_{i,t}^2 = x_t^T \cdot x_t = \|x_t\|^2 = 1$

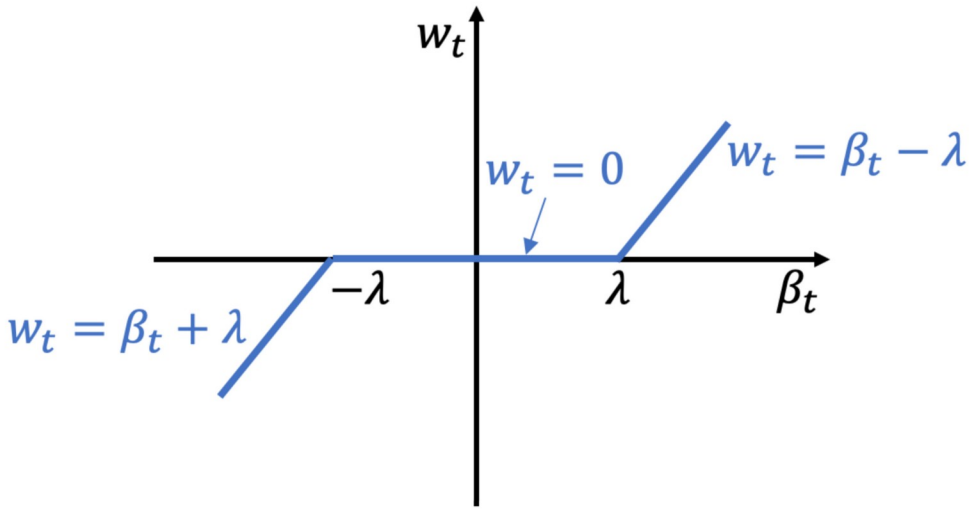
Lasso 的坐标下降

- 求解如下优化问题（二次、单变量）

$$L(w_t) = \frac{1}{2}(w_t - \beta_t)^2 + \lambda|w_t|$$

- 解：软阈值

$$w_t = \begin{cases} \beta_t - \lambda & \text{if } \beta_t \geq \lambda \\ \beta_t + \lambda & \text{if } \beta_t \leq -\lambda \\ 0 & \text{otherwise} \end{cases}$$



- 直觉：把 β_t 的大小推向 0

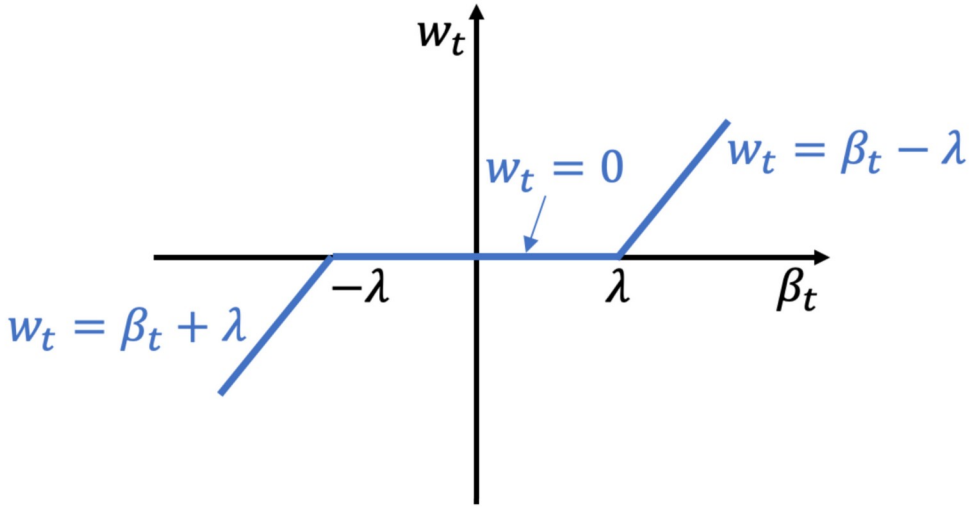
Coordinate Descent for Lasso

- Solve the following optimization problem (quadratic, single variable)

$$L(w_t) = \frac{1}{2}(w_t - \beta_t)^2 + \lambda|w_t|$$

- Solution: soft-thresholding

$$w_t = \begin{cases} \beta_t - \lambda & \text{if } \beta_t \geq \lambda \\ \beta_t + \lambda & \text{if } \beta_t \leq -\lambda \\ 0 & \text{otherwise} \end{cases}$$



- Intuition: push the magnitude of β_t toward 0

Lasso 之外：稀疏学习

- 弹性网：L1+L2 正则化（即 ridge + lasso）

$$\frac{1}{2}\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \lambda_1|w|_1 + \lambda_2\|w\|_2^2$$

- 组 Lasso $\frac{1}{2}\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \sum_{j=1}^{m}\lambda_j\|w_j\|_2$
  - 鼓励一组特征同时不被选中
  - w_j：一组特征（第 j 组）的系数向量
  - 组内用 L_2 范数，组间用 L_1 范数
- 融合 Lasso $\frac{1}{2}\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \lambda_1|w|_1 + \lambda_2\sum_{j=2}^{d}|w_j - w_{j-1}|$
  - 鼓励相邻特征同时被选中（或同时不被选中）
- 还有很多：如图引导 lasso、图 lasso、矩阵补全
  -

Beyond Lasso: Sparse Learning

- Elastic net: L1+L2 regularization (i.e., ridge + lasso)

$$\frac{1}{2}\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \lambda_1 |w|_1 + \lambda_2 \|w\|_2^2$$

- Group Lasso $\frac{1}{2}\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \sum_{j=1}^{m}\lambda_j \|w_j\|_2$
  - Encourage a group of features to be unselected simultaneously
  - w_j : coefficient vector for a group of features (group j)
  - L_2 norm within each group; L_1 norm across groups
- Fused Lasso $\frac{1}{2}\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \lambda_1 |w|_1 + \lambda_2 \sum_{j=2}^{d} |w_j - w_{j-1}|$
  - Encourage co-select (or co-un-select) adjacent features
- Many more: e.g., graph-guided lasso, graphical lasso, matrix completion

特征选择方法：区别在哪里？

三种方法都减少特征，但区别在于预测模型何时进入选择过程。

方法	选择信号	代价	典型风险
过滤法	模型拟合前的特征–目标得分	Low	可能漏掉特征交互
包装法	特征子集的验证性能	高	代价高；选择可能过拟合
嵌入法	选择发生在模型训练过程中	Medium	取决于模型与正则项

避免泄漏

特征选择*只在训练数据上*拟合；再把学到的变换应用于验证和测试数据。

Feature Selection Methods: What Changes?

All three approaches reduce features, but they differ in when the predictive model enters the selection process.

Method	Selection signal	Cost	Typical risk
Filter	Feature–target score before model fitting	Low	May miss feature interactions
Wrapper	Validation performance of feature subsets	High	Expensive; may overfit selection
Embedded	Selection occurs during model training	Medium	Depends on model and regularizer

Avoid leakage

Fit feature selection on the *training data only*; then apply the learned transformation to validation and test data.

快速检查：哪种特征选择方法？

问题：在训练任何分类器之前，按特征与类别标签的互信息对特征排序。这属于哪一类？

A	B	C	D
过滤法	包装法	嵌入法	特征工程

答案： A. 过滤法

该得分的计算与下游分类器无关。

Quick Check: Which Feature-Selection Method?

Question: **Features are ranked by mutual information with the class label before any classifier is trained. Which family is this?**

A	B	C	D
Filter	Wrapper	Embedded	Feature engineering

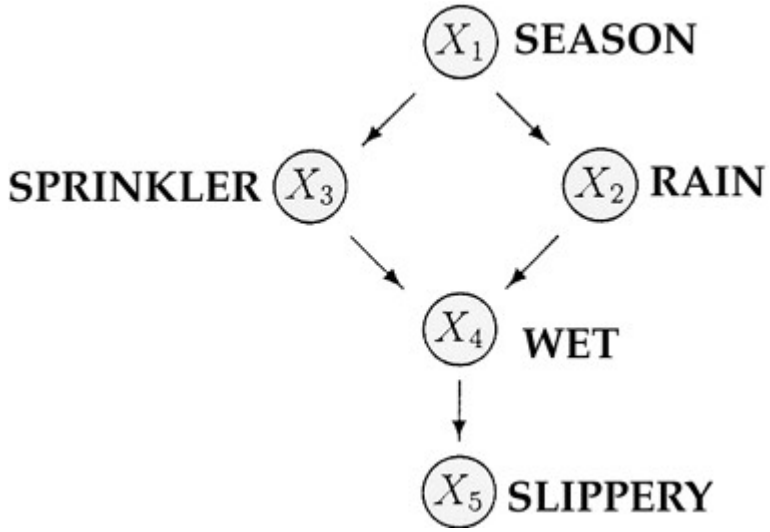
Answer: A. Filter

The score is computed independently of the downstream classifier.

从朴素贝叶斯到贝叶斯网络

- 朴素贝叶斯分类器**假设**在给定类别变量时，某个特征的取值与其他任何特征的取值**相互独立**
  - 这一假设往往过于简单，难以很好地刻画现实世界

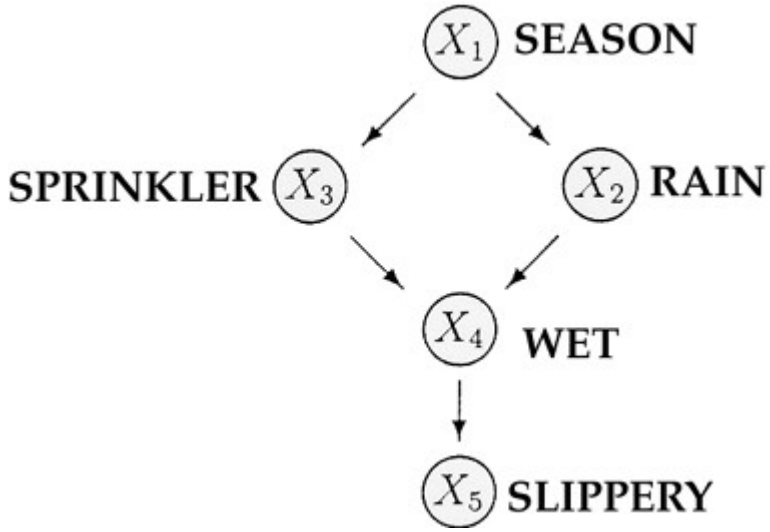
- **贝叶斯网络**（又称贝叶斯网、信念网络、贝叶斯模型或概率有向无环图模型）是一种**概率图模型**
  - 用*有向无环图* (DAG) 表示一组*随机变量*及*其条件依赖关系*
    - 例如：给定症状，可用该网络计算各种疾病存在的概率



From Naïve Bayes to Bayesian Networks

- Naïve Bayes classifiers **assume** that the value of a particular feature is **independent** of the value of any other feature, given the class variable
  - This assumption is often too simple to model the real world well

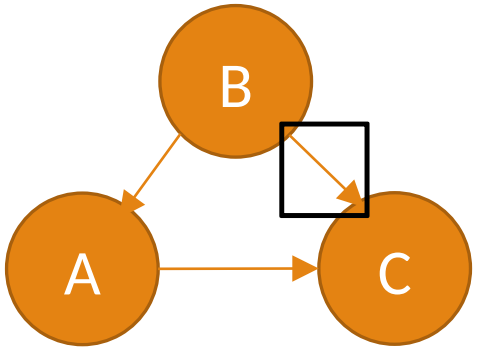
- **Bayesian network** (or Bayes network, belief network, Bayesian model or probabilistic directed acyclic graphical model) is a **probabilistic graphical model**
  - Represented by a set of *random variables* and *their conditional dependencies* via a *directed acyclic graph* (DAG)
    - E.g. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases



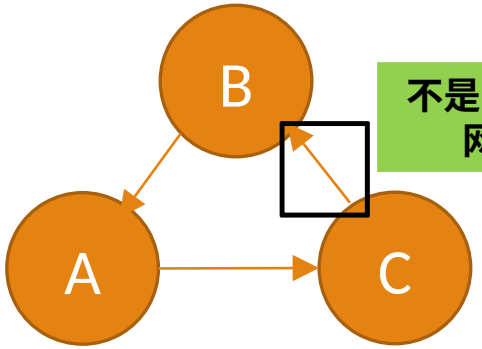
贝叶斯信念网络

- **贝叶斯信念网络**（或**贝叶斯网络、概率网络**）：
  - 允许变量的*子集*之间存在*类条件独立*
- 两个组成部分：
  - 一个*有向无环图*（称为结构）
  - 一组*条件概率表* (CPT)

结点：随机变量　　连边：依赖关系



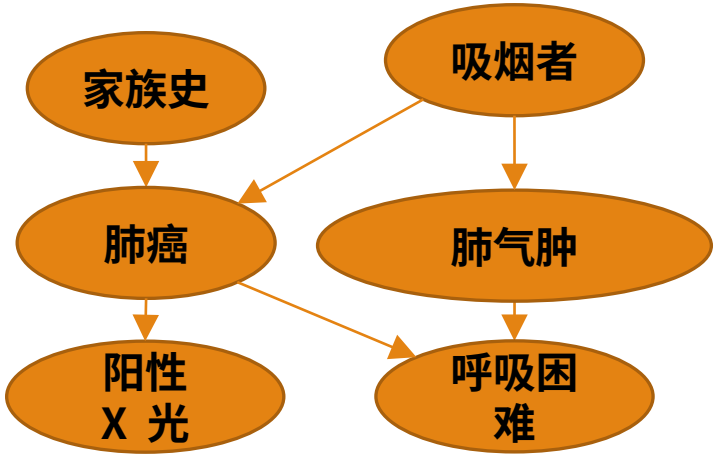
有向**无环**图模型



不是贝叶斯网络

有向**有环**图模型

$$p(A, B, C) = p(B) \cdot p(A|B) \cdot p(C|A, B)$$

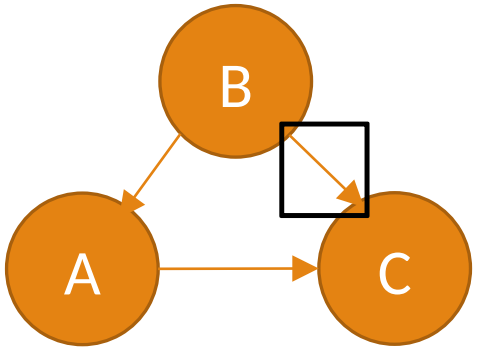


	FH, S	*FH, ~S*	*~FH, S*	*~FH, ~S*
LC	0.8	0.5	0.7	0.1
~LC	0.2	0.5	0.3	0.9

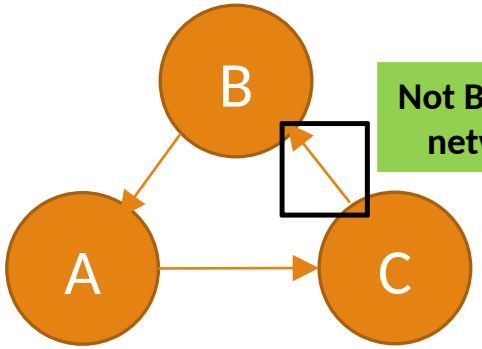
Bayesian Belief Networks

- **Bayesian belief network** (or **Bayesian network**, **probabilistic network**):
  - Allows *class conditional independencies* between *subsets* of variables
- Two components:
  - A *directed acyclic graph* (called a structure)
  - A set of *conditional probability tables* (CPTs)

Nodes: random variables Links: dependency



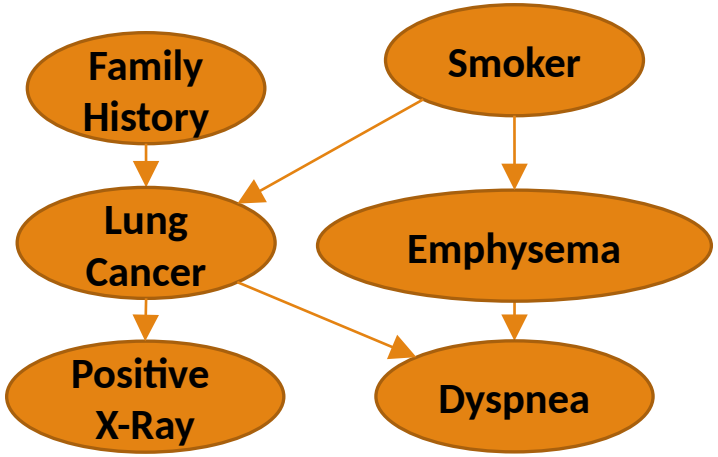
directed **acyclic** graphical model



Not Bayesian networks

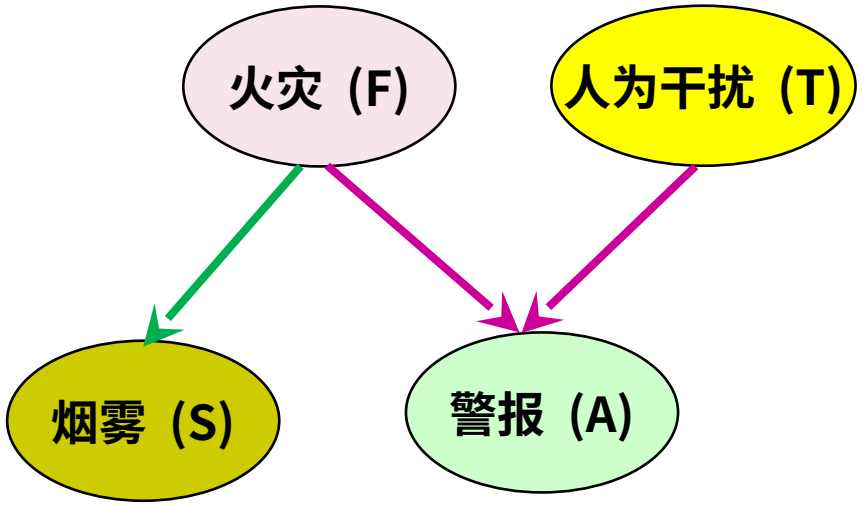
directed **cyclic** graphical model

$$p(A, B, C) = p(B) \cdot p(A|B) \cdot p(C|A, B)$$



	FH, S	*FH, ~S*	*~FH, S*	*~FH, ~S*
LC	0.8	0.5	0.7	0.1
~LC	0.2	0.5	0.3	0.9

贝叶斯网络及其 CPT



CPT 给出其父结点每种可能取值组合下的条件概率：

条件概率表（CPT)

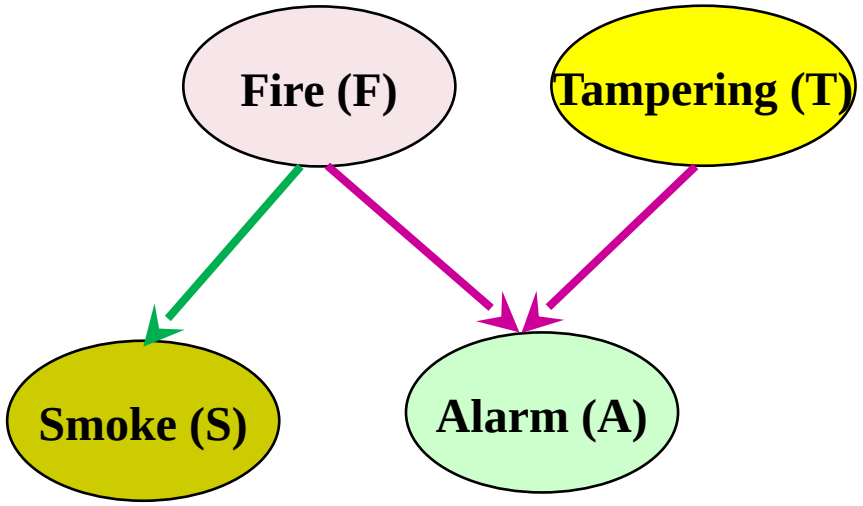
火灾	烟雾	$\Theta_{s\|f}$
真	真	.90
假	真	.01

火灾	人为干扰	警报	$\Theta_{a\|f,t}$
真	真	真	.5
真	假	真	.99
假	真	真	.85
假	假	真	.0001

$$p(X) = \prod_k p(x_k|\text{Parents}(x_k))$$

$$p(F, S, A, T) = p(F) \cdot p(T) \cdot p(S|F) \cdot p(A|F, T)$$

A Bayesian Network and Its CPTs



Conditional **P**robability **T**ables (CPT)

Fire	Smoke	$\Theta_{s\|f}$
True	True	.90
False	True	.01

Fire	Tampering	Alarm	$\Theta_{a\|f,t}$
True	True	True	.5
True	False	True	.99
False	True	True	.85
False	False	True	.0001

CPT shows the conditional probability for each possible combination of its parents:

$$p(X) = \prod_k p(x_k | \text{Parents}(x_k))$$

$$p(F, S, A, T) = p(F) \cdot p(T) \cdot p(S|F) \cdot p(A|F, T)$$

贝叶斯网络：从图到联合分布

有向无环图规定了需要哪些局部条件概率。

一般因子分解

$$P(X_1, \ldots, X_d) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

示例图	局部项	联合概率
F → S; F,T → A	P(F), P(T), P(S|F), P(A|F,T)	P(F,T,S,A) = P(F)P(T)P(S|F)P(A|F,T)

- 条件概率表（CPT）提供局部概率。
- 推断组合这些局部项，回答关于未观测变量的查询。

Bayesian Networks: From a Graph to a Joint Distribution

A directed acyclic graph specifies which local conditional probabilities are needed.

General factorization

$$P(X_1, \ldots, X_d) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

Example graph	Local terms	Joint probability
F → S; F,T → A	P(F), P(T), P(S|F), P(A|F,T)	P(F,T,S,A) = P(F)P(T)P(S|F)P(A|F,T)

- Conditional probability tables (CPTs) provide the local probabilities.
- Inference combines the local terms to answer queries about unobserved variables.

训练贝叶斯网络：几种情形

- 情形 1：已知网络结构且所有变量可观测：*只需计算 CPT 条目*
- 情形 2：网络结构已知，部分变量隐藏：*梯度下降*（贪心爬山）法，即沿准则函数最速下降方向搜索解
  - 权重初始化为随机概率值
  - 每次迭代都朝当前看来最好的解移动，不回溯
  - 权重每次迭代更新，并收敛到局部最优

Training Bayesian Networks: Several Scenarios

- Scenario 1: Given both the network structure and all variables observable: *compute only the CPT entries*

- Scenario 2: Network structure known, some variables hidden: *gradient descent* (greedy hill-climbing) method, i.e., search for a solution along the steepest descent of a criterion function
  - Weights are initialized to random probability values
  - At each iteration, it moves towards what appears to be the best solution at the moment, without backtracking
  - Weights are updated at each iteration & converge to local optimum

训练贝叶斯网络：几种情形

- 情形 3：网络结构未知，所有变量可观测：在模型空间中搜索以*重建网络拓扑*
- 情形 4：结构未知、变量全部隐藏：目前尚无好的算法

D. Heckerman. A Tutorial on Learning with Bayesian Networks. In Learning in Graphical Models, M. Jordan, ed. MIT Press, 1999

Training Bayesian Networks: Several Scenarios

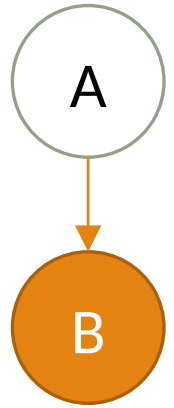
- Scenario 3: Network structure unknown, all variables observable: search through the model space to *reconstruct network topology*

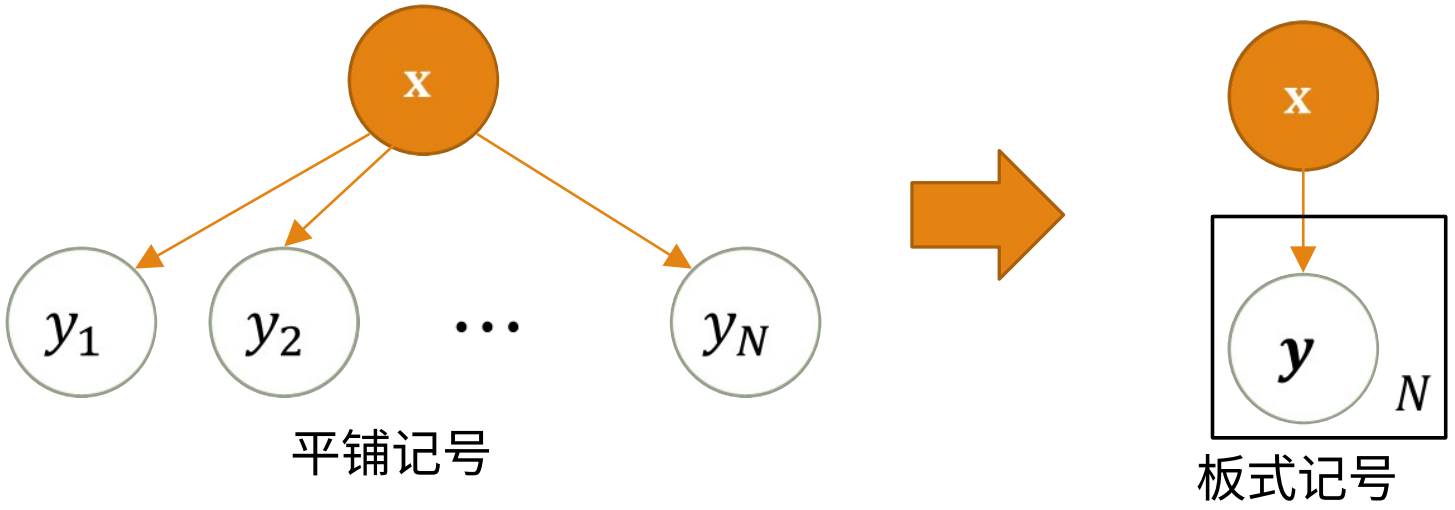
- Scenario 4: Unknown structure, all hidden variables: No good algorithms known for this purpose

D. Heckerman. A Tutorial on Learning with Bayesian Networks. In Learning in Graphical Models, M. Jordan, ed. MIT Press, 1999

概率图模型：板式记号

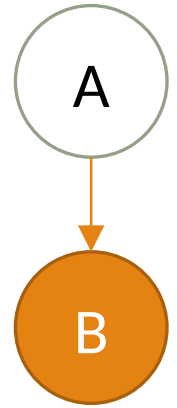
- 表示图模型中重复出现的变量
- 变量
  - 实心（或阴影）圆表示对应变量*可观测*；否则为*隐藏*
- 变量间的依赖：
  - 有向无环图（DAG）模型
- 用板式记号代替平铺记号

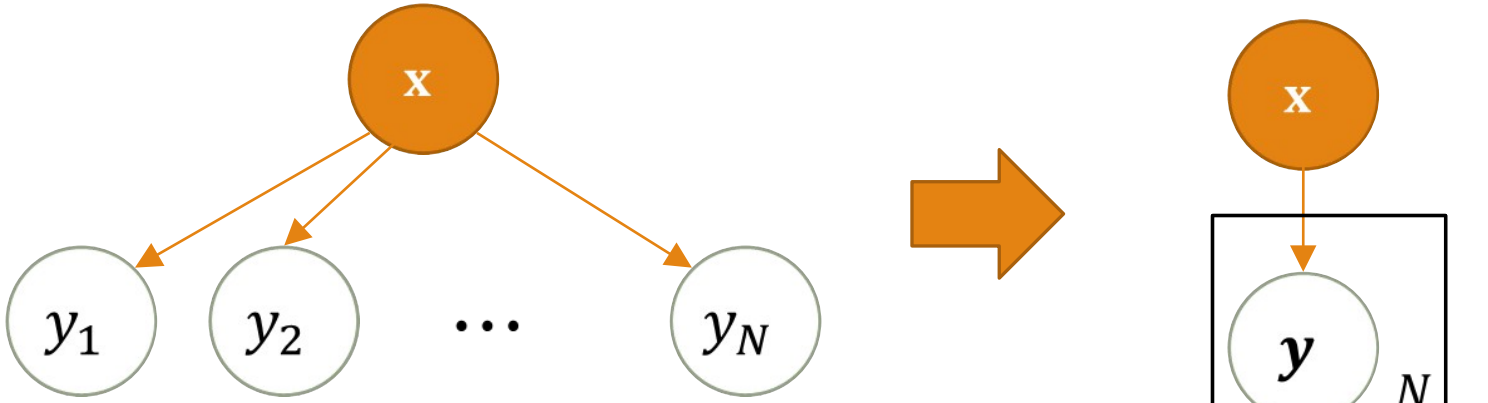




Probabilistic Graphic Model: Plate Notations

- Represent variables that repeat in a graphical model
- Variables
  - A solid (or shaded) circle means the corresponding variable is *observed*; otherwise it is *hidden*
- Dependency among variables:
  - A Directed Acyclic Graphical (DAG) model
- Using plate notation instead of flat notation

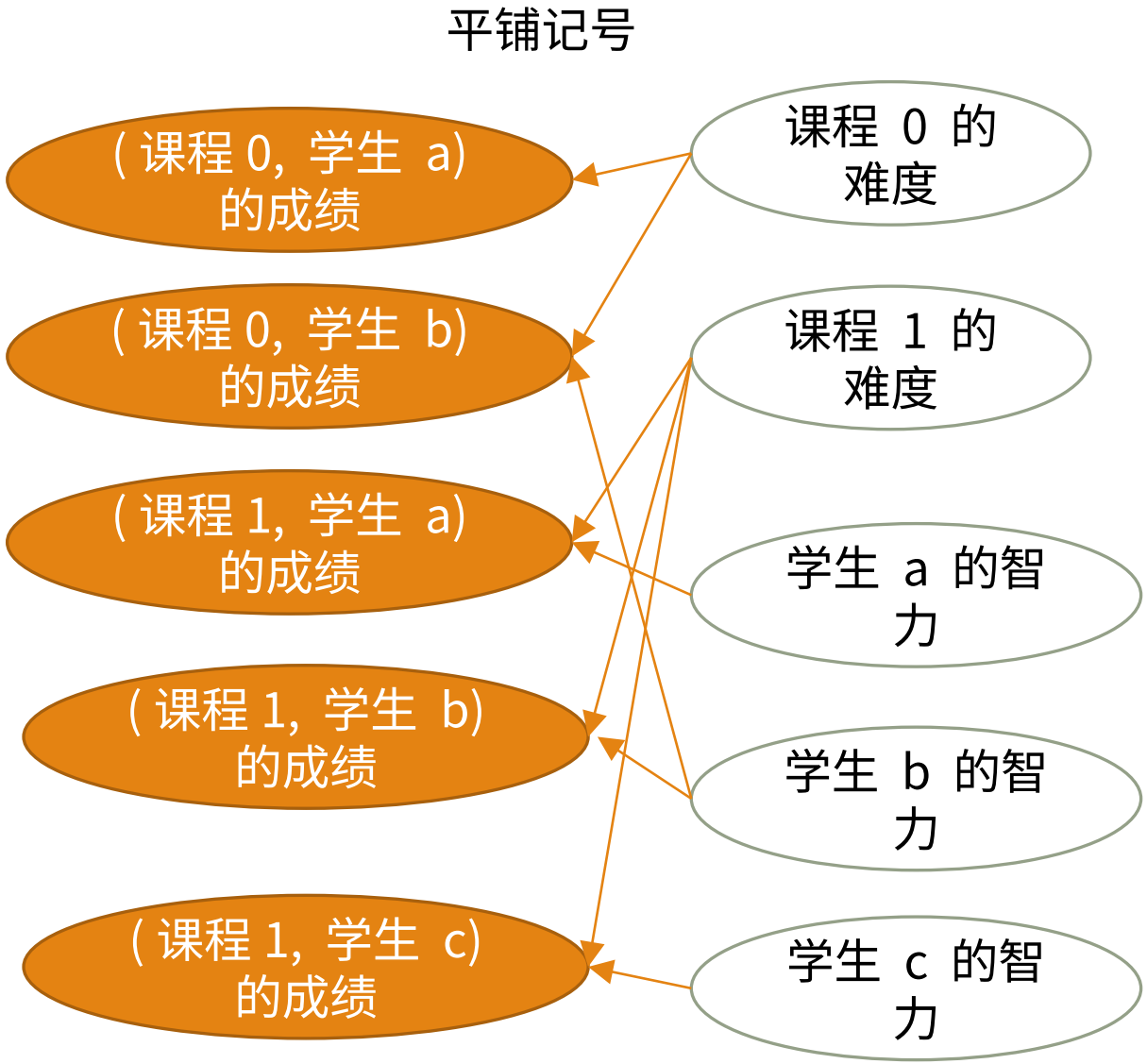


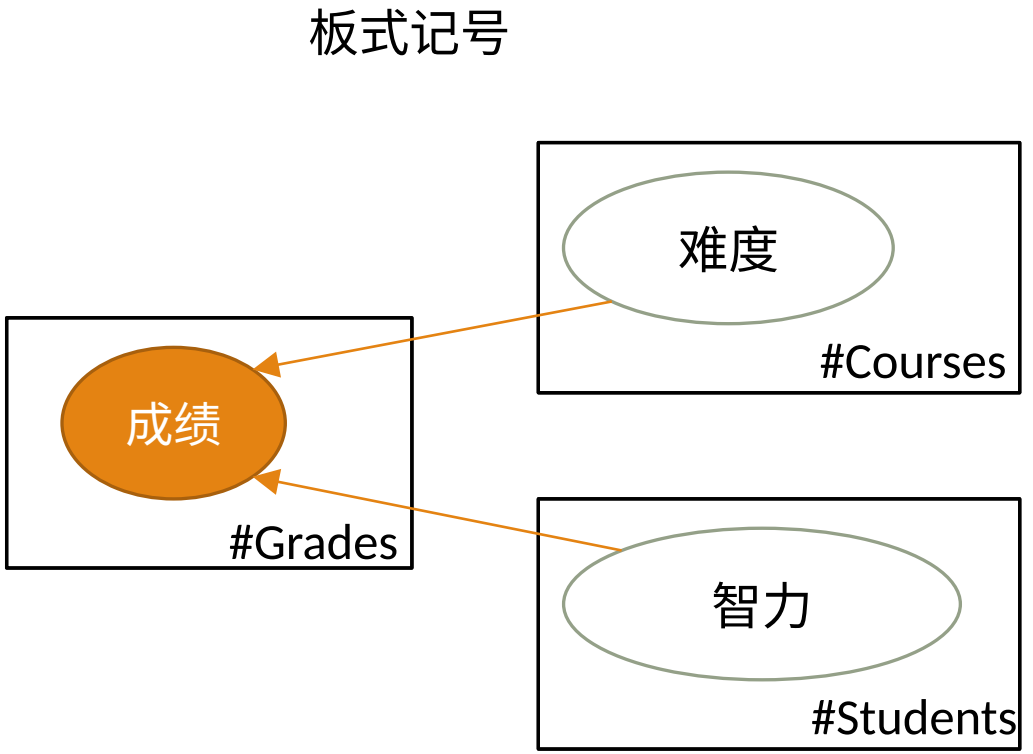


Flat notation

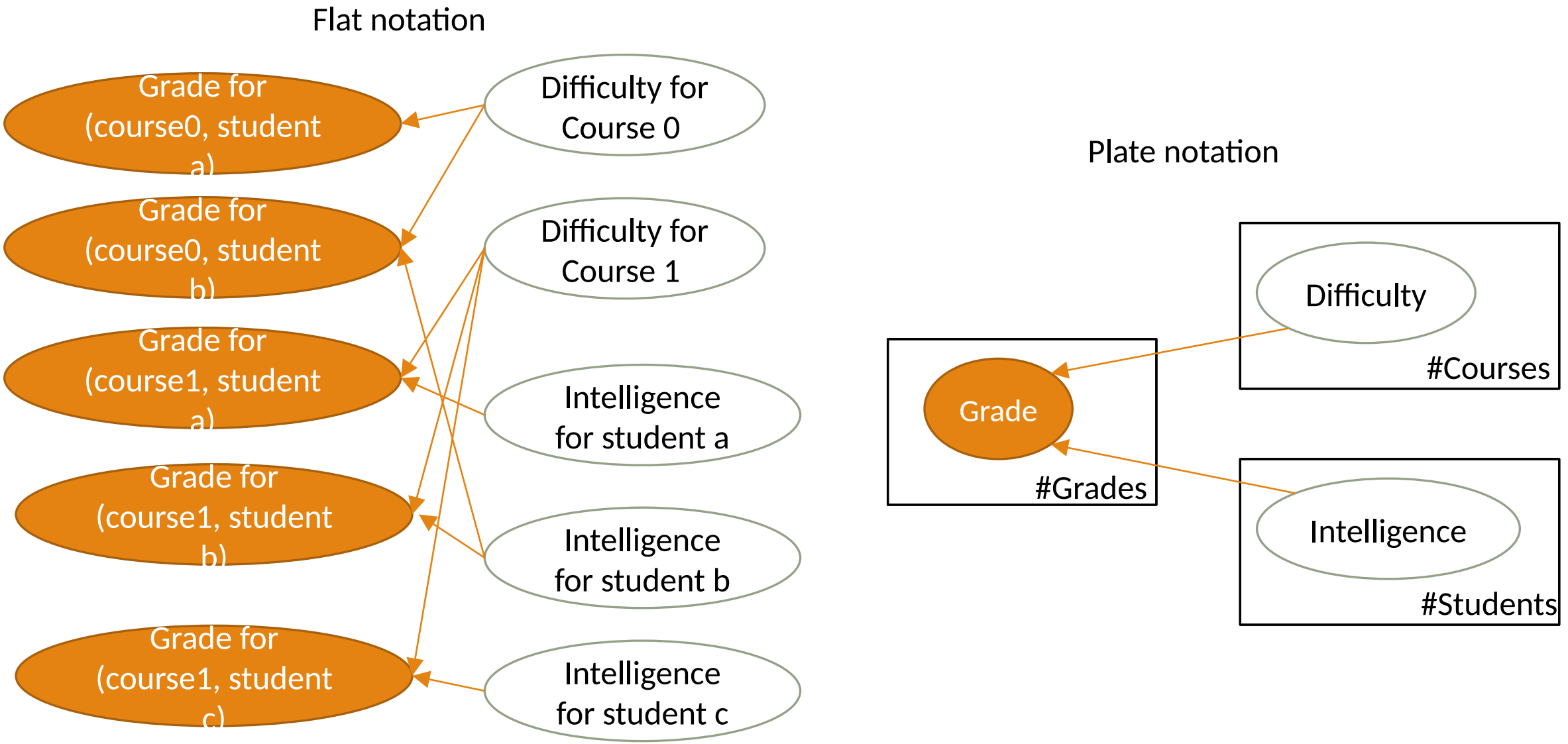
Plate notation

板式记号示例


平铺记号
(课程 0, 学生 a) 的成绩
(课程 0, 学生 b) 的成绩
(课程 1, 学生 a) 的成绩
(课程 1, 学生 b) 的成绩
(课程 1, 学生 c) 的成绩
课程 0 的难度
课程 1 的难度
学生 a 的智力
学生 b 的智力
学生 c 的智力


板式记号
成绩
#Grades
难度
#Courses
智力
#Students

An Example of Plate Notation



分类：一种数学映射

- **二分类问题：**
  - 示例：影评分类
    - $x_i = (x_1, x_2, x_3, ...)$， $y_i = +1$ 或 -1 （正面、负面）
    - x_1：词“awesome”的次数
    - x_2：词“disappointing”的次数

- 数学上， $x \in X = \Re^n, y \in Y = \{+1, -1\}$
  - 我们希望得到函数 $f: X \rightarrow Y$
  - 把输入样本映射到正确的标签

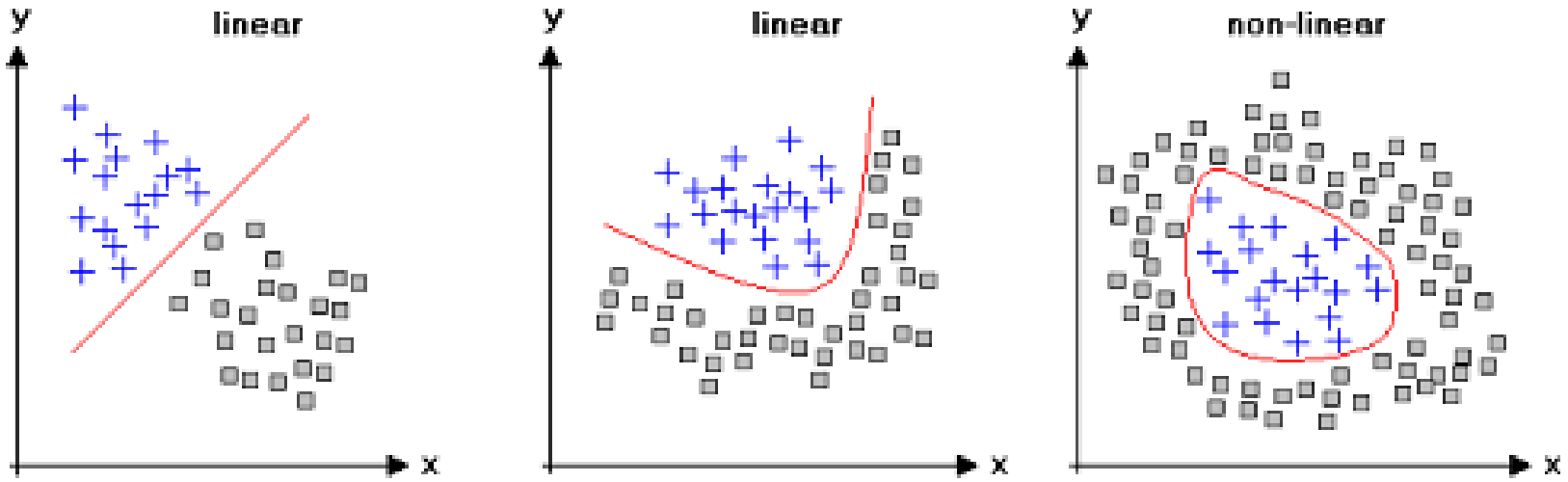
Classification: A Mathematical Mapping

- **The binary classification problem:**
  - Example: movie-review classification
    - $x_i = (x_1, x_2, x_3, \ldots)$, $y_i = +1$ or -1 (positive, negative)
    - x_1 : # of word "awesome"
    - x_2 : # of word "disappointing"

- Mathematically, $\boldsymbol{x} \in X = \Re^n, y \in Y = \{+1, -1\}$
  - We want to derive a function $f: X \rightarrow Y$
  - which maps input examples to their correct labels

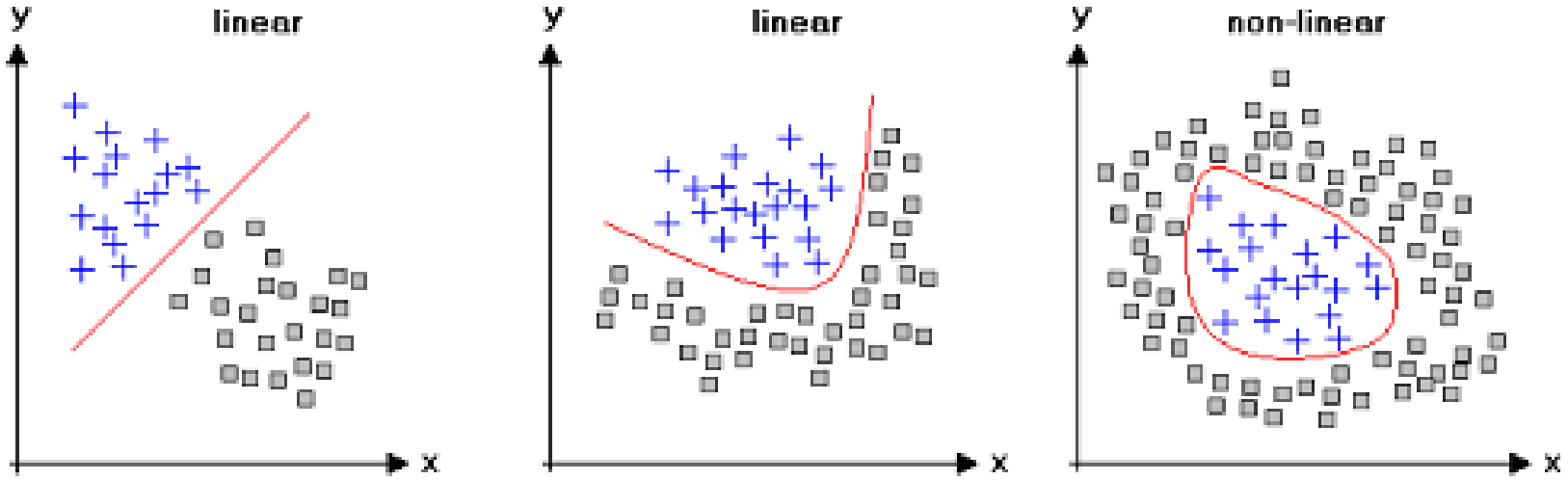
SVM — 支持向量机

- 对线性与非线性数据分类
  - Vapnik 等人（1992）——奠基于 Vapnik 与 Chervonenkis 在 1960 年代的统计学习理论



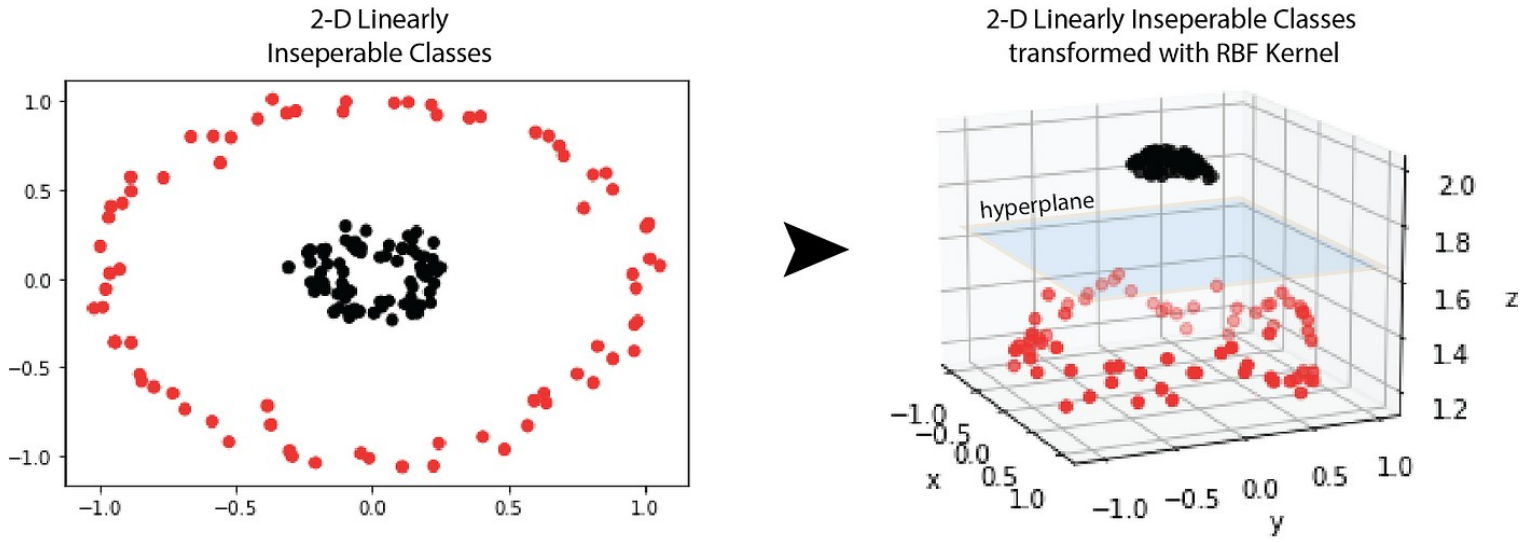
SVM — Support Vector Machines

- Classification of linear and nonlinear data
  - Vapnik and colleagues (1992)—groundwork from Vapnik & Chervonenkis' statistical learning theory in 1960s



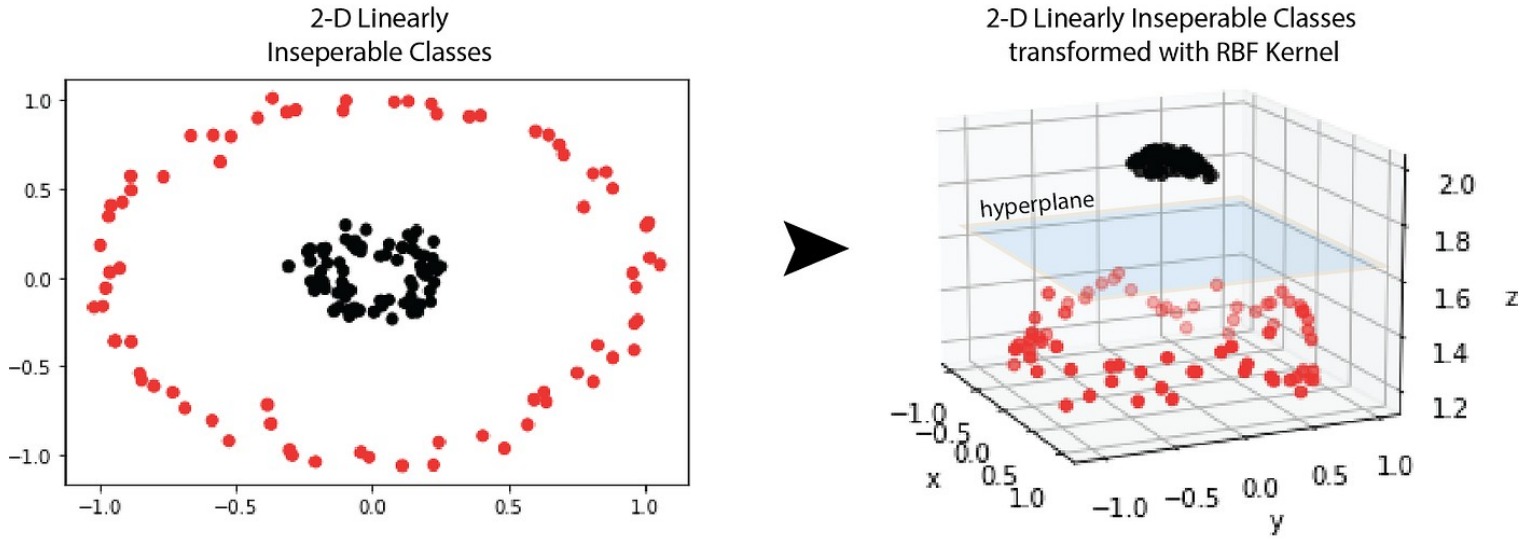
SVM — 支持向量机

- SVM 用非线性映射把原始数据映射到更高维的特征空间。在特征空间中，它寻找最优分离**超平面**（即“决策边界”）
- 借助合适的映射，各类别在特征空间中可能变得线性可分
- SVM 利用**支持向量**（“关键”训练元组）和**间隔**（由支持向量定义）找到该超平面

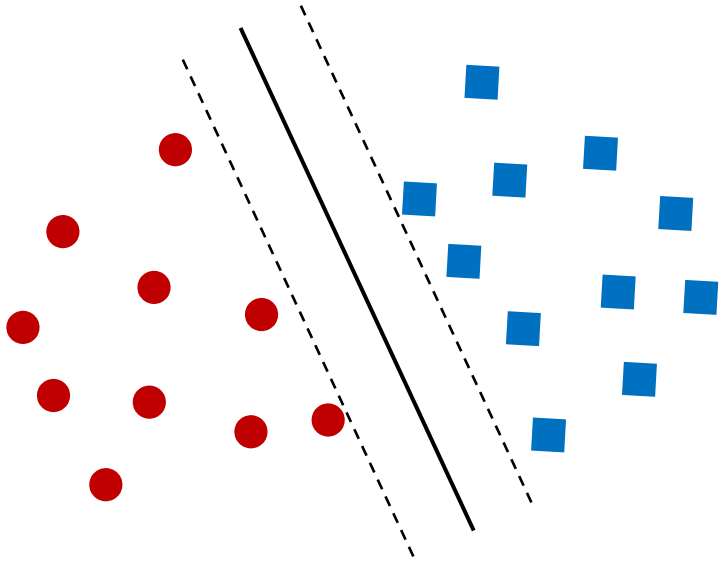


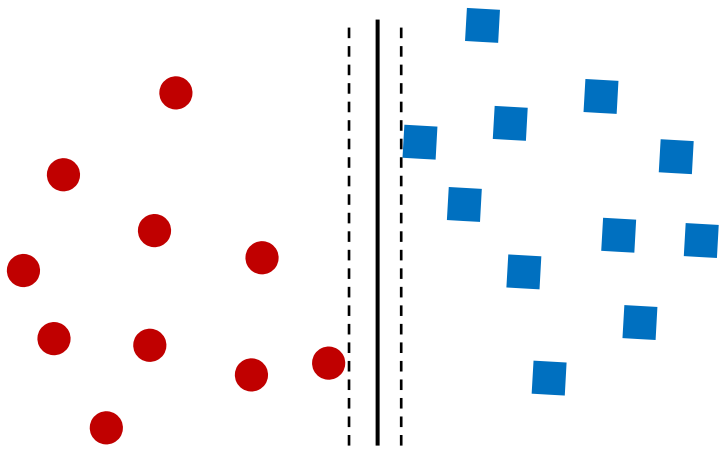
SVM — Support Vector Machines

- SVM uses a nonlinear mapping to map the original data into a higher-dimensional feature space. In the feature space, it searches for an optimal separating **hyperplane** (i.e., "decision boundary")
- With a suitable mapping, the classes may become linearly separable in the feature space
- SVM finds this hyperplane using **support vectors** ("essential" training tuples) and **margins** (defined by the support vectors)



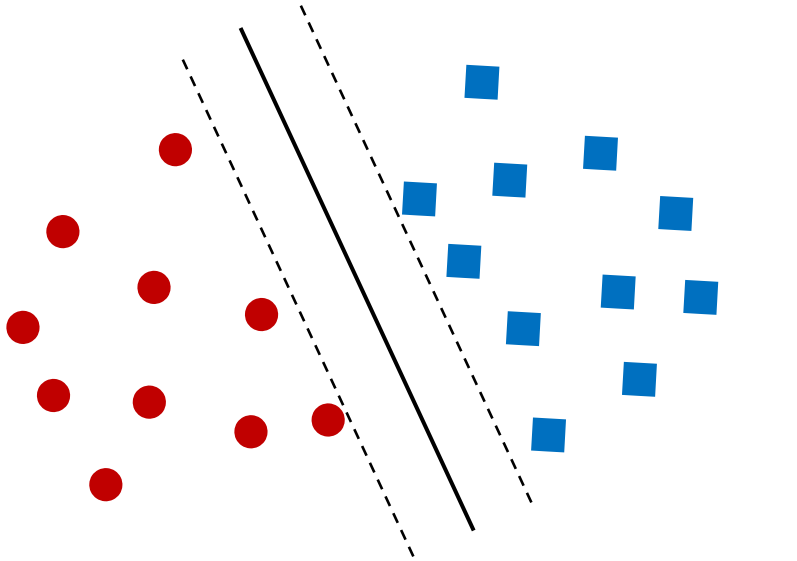
SVM — 基本思想

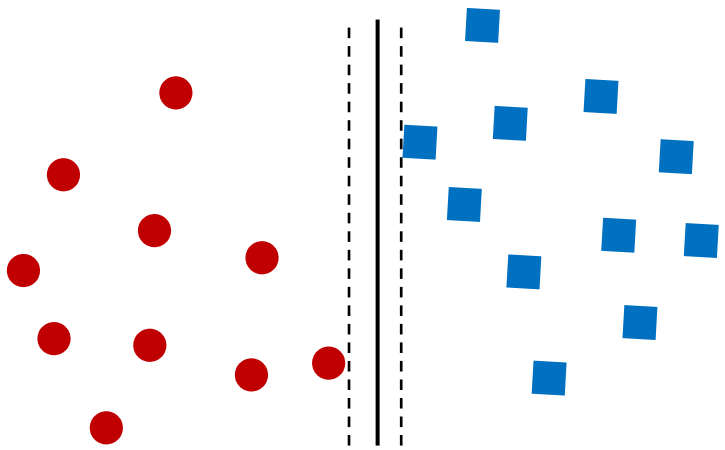




- **学习最大间隔分类器**
  - 在无穷多条分离两类的直线（超平面）中
  - 找出以**最大间隔**分离两类的那一条
    - 即**最大间隔超平面（MMH）**

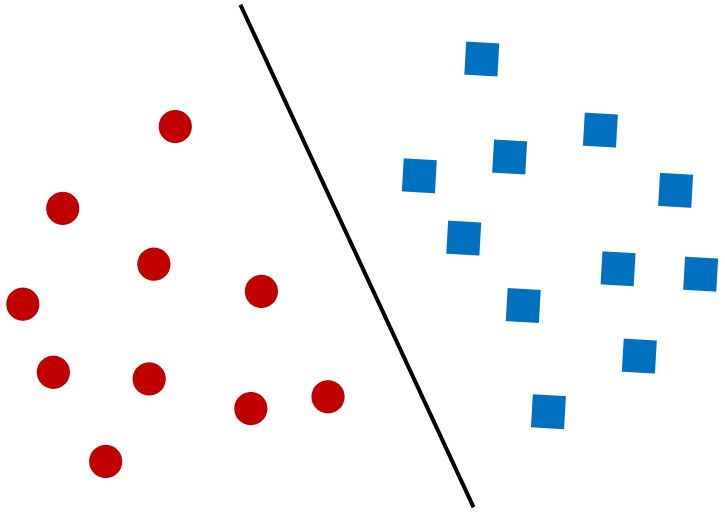
SVM — General Philosophy



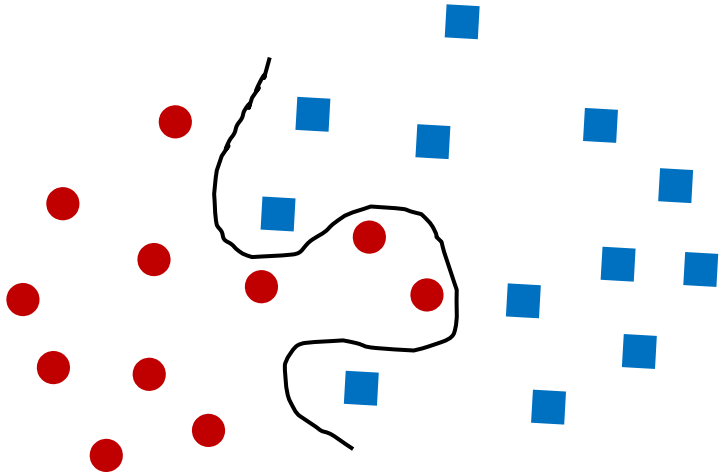


- **Learning a max-margin classifier**
  - From the infinite set of lines (hyperplanes) separating two classes
  - Find the one which separates two classes with the **largest margin**
    - i.e. a **maximum marginal hyperplane (MMH)**

SVM — 数据线性可分时



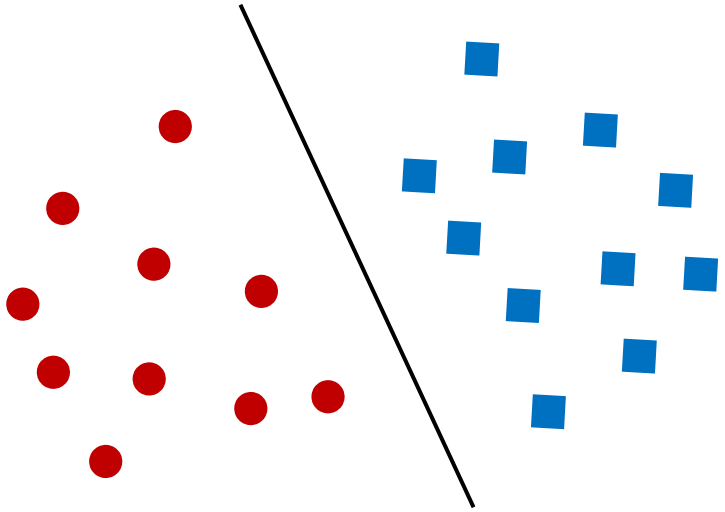
线性可分



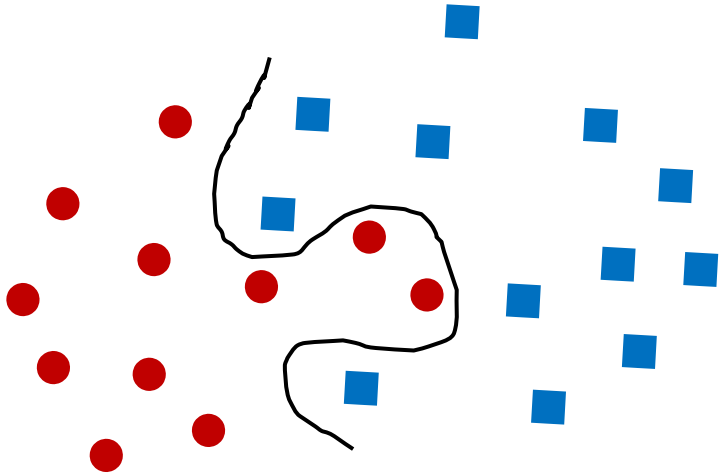
线性不可分

- 最简单的情形：数据**线性可分**
  - 类别能被线性决策面精确分开的数据集称为线性可分

SVM — When Data Is Linearly Separable



Linearly Separable



Linearly Inseparable

- The simplest case: When data is **linearly separable**
  - Data sets whose classes can be separated exactly by linear decision surfaces are said to be linearly separable

线性可分数据的线性 SVM

- 分离超平面可写为

  $w^{\wedge}Tx+b=0$

  其中 $w=(w_1,w_2,\ldots,w_n)^{\wedge}T$ 为权重向量， b 为标量（偏置）

  - 二维时可写为： $w1\ x1\ +\ w2\ x2+b=\ 0$

待学习的模型参数

- 定义间隔两侧的超平面：

  H_1: $w_0 + w_1 x_1 + w_2 x_2 \geq 1$ for $y_i = +1$, and

  H_2: $w_0 + w_1 x_1 + w_2 x_2 \leq -1$ for $y_i = -1$

- 落在超平面 H_1 或 H_2 （即定义间隔的两侧）上的训练元组称为**支持向量**

Linear SVM for Linearly Separable Data

- A separating hyperplane can be written as

$$\boldsymbol{w}^T \boldsymbol{x} + b = 0$$

where $w = (w_1, w_2, \ldots, w_n)^T$ is a weight vector and b a scalar (bias)

Model parameters to learn

  - For 2-D, it can be written as: $w_1 x_1 + w_2 x_2 + b = 0$

- The hyperplane defining the sides of the margin:

H_1: $w_0 + w_1 x_1 + w_2 x_2 \geq 1$ for $y_i = +1$, and

H_2: $w_0 + w_1 x_1 + w_2 x_2 \leq -1$ for $y_i = -1$

- Any training tuples that fall on hyperplanes H_1 or H_2 (i.e., the sides defining the margin) are **support vectors**

线性可分数据的线性 SVM

- 任一数据点 x 到分离超平面的距离为

$$\text{distance}(ax+by+c=0,(x_0,y_0))=\frac{|ax_0+by_0+c|}{\sqrt{a^2+b^2}}$$

$$r=(|f(x)|)/(\|w\|)=(y_i(w\text{^}Tx_i+b))/(\|w\|)$$

- 目标是最大化最近数据点到超平面的距离

$$\text{argmax_(w,b)}\{1/(\|w\|)\min[y_i(w\text{^}Tx_i+b)]\}$$

- 这很难直接求解，我们将其转化为更容易的问题

$$\arg\min_{w,b} \quad \|\boldsymbol{w}\|^2$$

$$\text{s.t.} \quad y_i(\boldsymbol{w}^T\boldsymbol{x}_i+b)\geq 1, \qquad i=1,2,\ldots,n$$

- 这是 SVM 的基本形式，可用*二次规划*求解

Linear SVM for Linearly Separable Data

- The distance from any data point $\boldsymbol{x}$ to the separating hyperplane is

$$\text{distance}(ax + by + c = 0, (x_0, y_0)) = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} \qquad r = \frac{|f(\boldsymbol{x})|}{\|\boldsymbol{w}\|} = \frac{y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b)}{\|\boldsymbol{w}\|}$$

- Our objective is to maximize the distance of the closest data point to the hyperplane

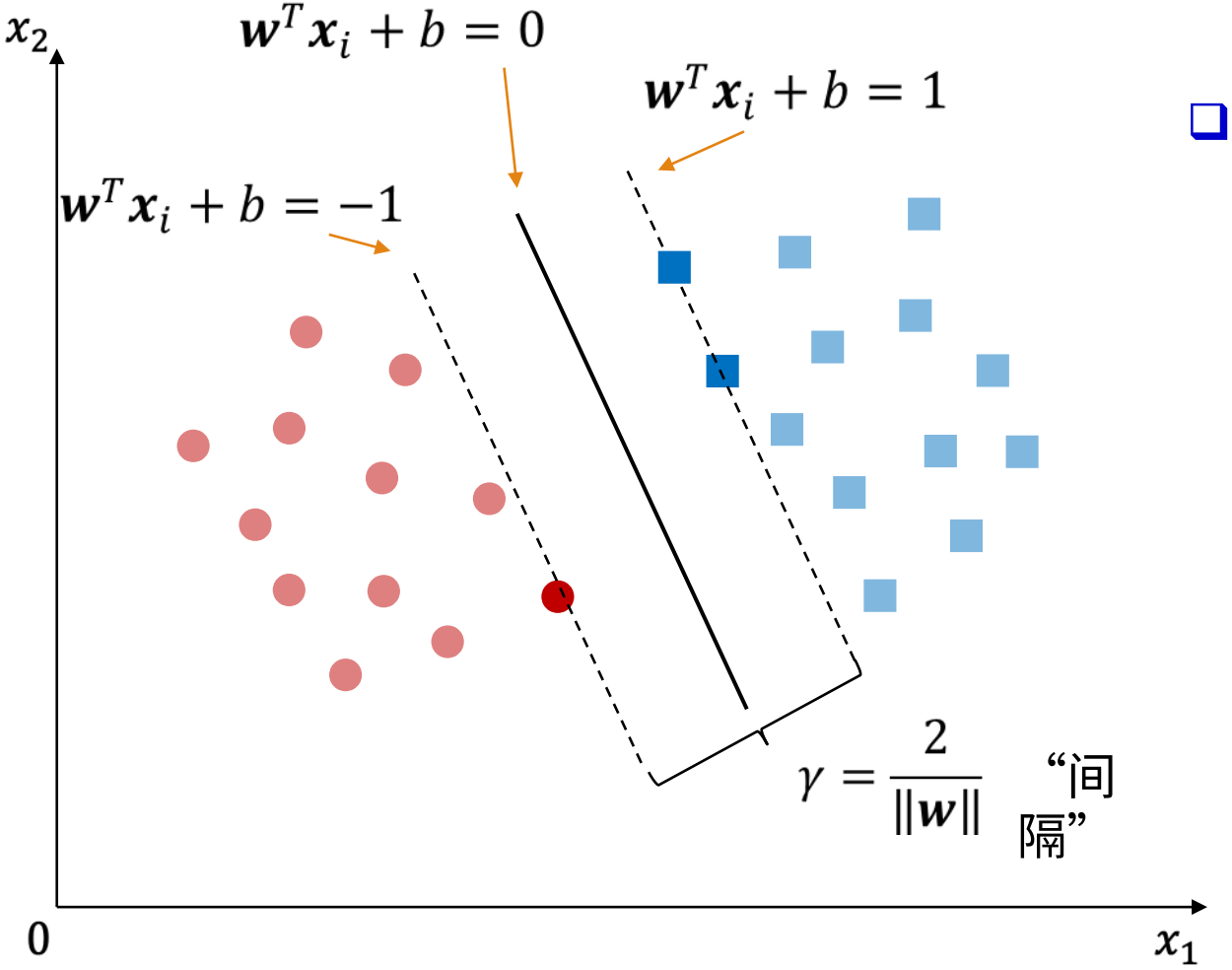
$$\arg\max_{w,b}\left\{\frac{1}{\|\boldsymbol{w}\|}\min[y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b)]\right\}$$

- This is hard to solve, we shall convert it to an easier problem

$$\begin{aligned}\arg\min_{w,b} \quad & \|\boldsymbol{w}\|^2 \\ \text{s.t.} \quad & y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) \geq 1, \qquad i = 1,2,\dots,n\end{aligned}$$

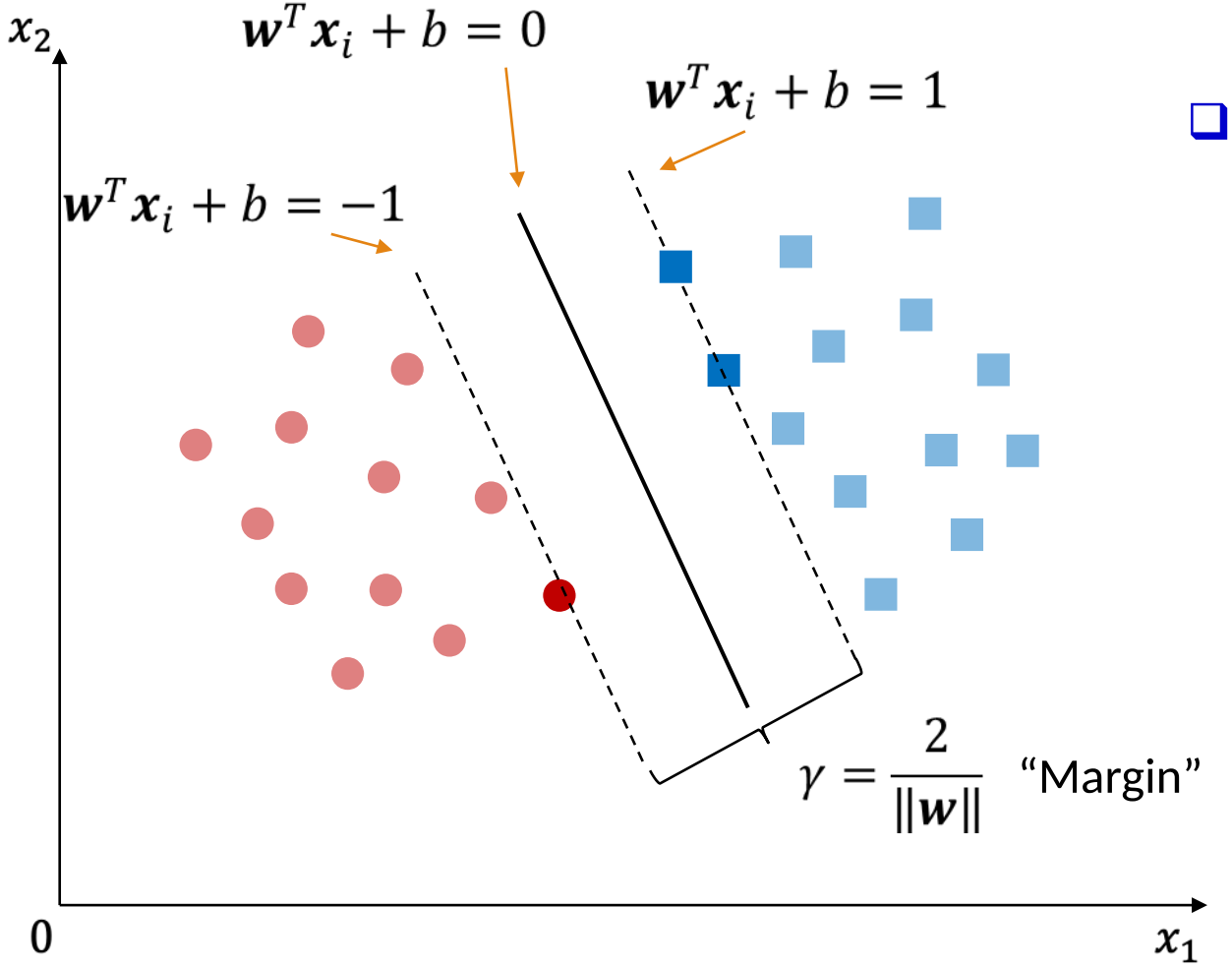
- This is the basic form of SVM, and it can be solved by using *quadratic programming*

线性可分数据的线性 SVM



- 距分离超平面最近的数据点称为**支持向量**

Linear SVM for Linearly Separable Data

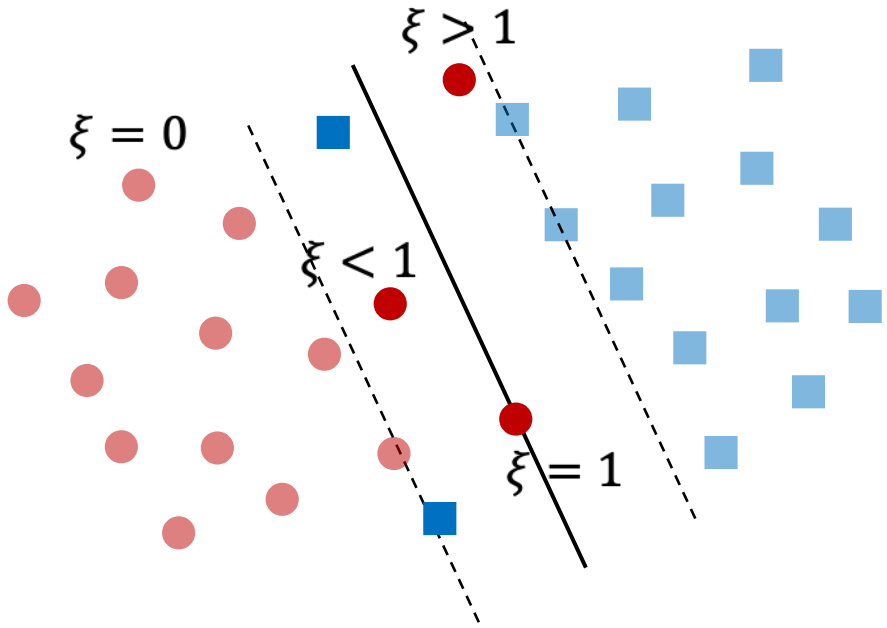


- The data points closest to the separating hyperplane are called **support vectors**

线性不可分数据的 SVM

- 允许数据点位于**间隔边界**的“错误一侧”

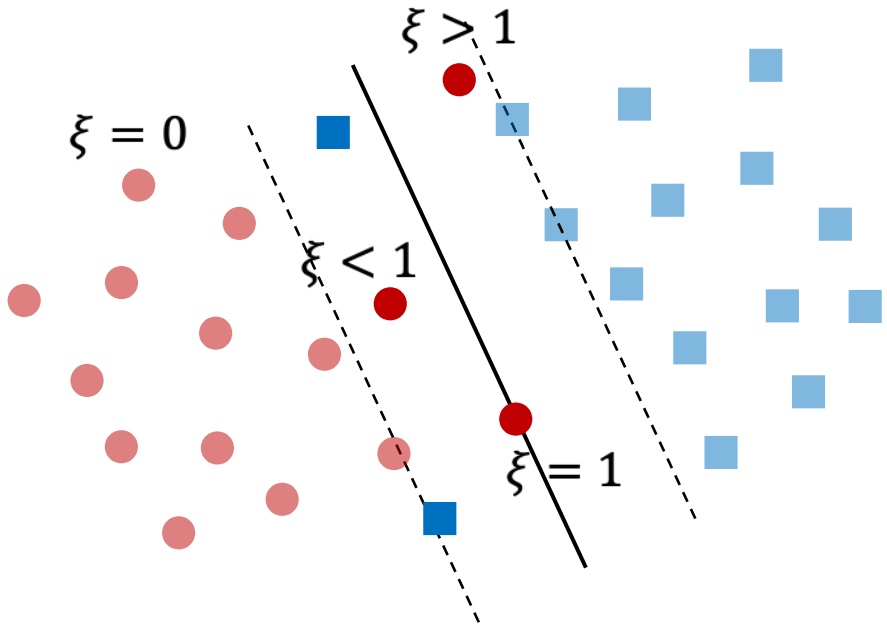
- 按到间隔边界的距离对错误一侧的点**施加惩罚**
  - ξ ： 松弛变量
  - **C (> 0)**： 控制惩罚与间隔之间的权衡
  - **C 越小：允许更多错误**
  - **C 越大：允许更少错误**
  - 这就是广泛使用的*软间隔 SVM*



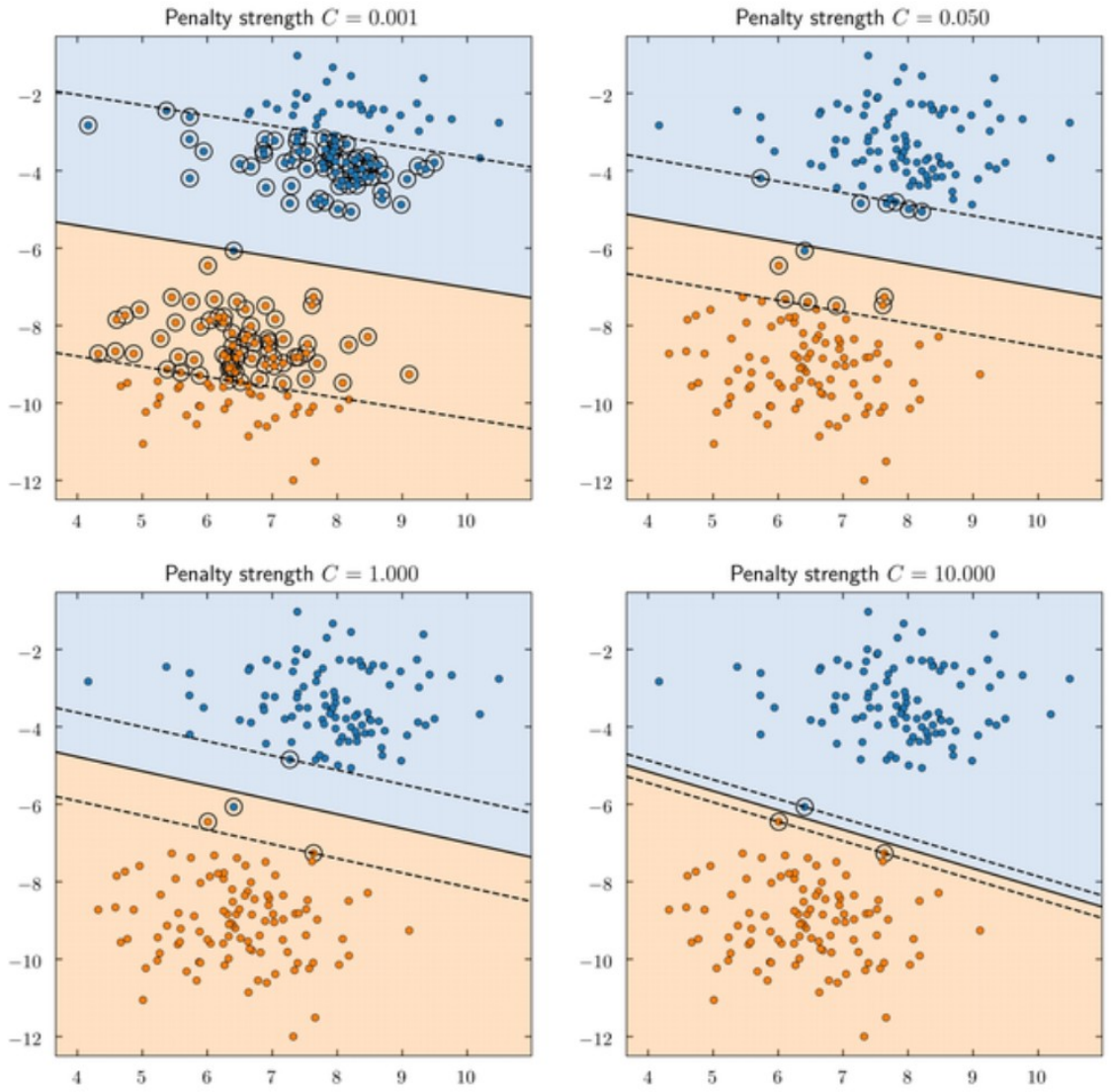
SVM for Linearly Inseparable Data

- We allow data points to be on the "wrong side" of the **margin boundary**

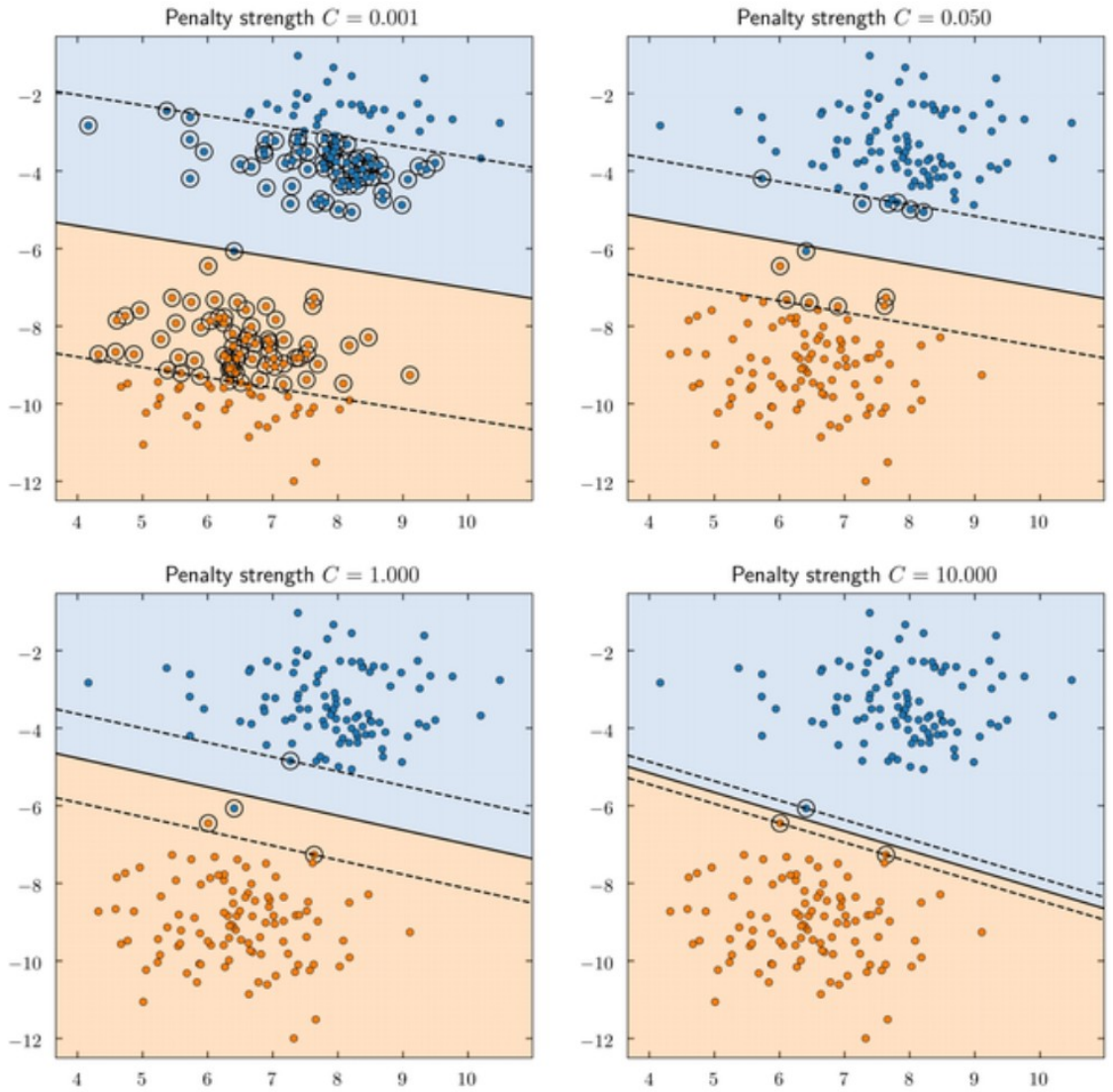
- **Penalize** points on the wrong side according to its distance to the margin boundary
  - ξ : slack variable
  - **C (> 0):** Controls the trade-off between the penalty and the margin
  - **Smaller C: allow more mistake**
  - **Larger C: allow less mistake**
  - This is the widely used *soft-margin SVM*



松弛变量的作用


Penalty strength $C = 0.001$
Penalty strength $C = 0.050$
Penalty strength $C = 1.000$
Penalty strength $C = 10.000$

Effect of slack variable



快速检查：软间隔 SVM 中 C 的作用

问题： 哪种设置通常允许更多训练样本违反间隔？

A	B	C	D
更大的 C	更小的 C	更多特征	更多支持向量

答案： B。更小的 C

较小的 C 对违反间隔的惩罚较轻，通常得到更宽的间隔。

Quick Check: Effect of C in a Soft-Margin SVM

Question: **Which setting usually permits more training examples to violate the margin?**

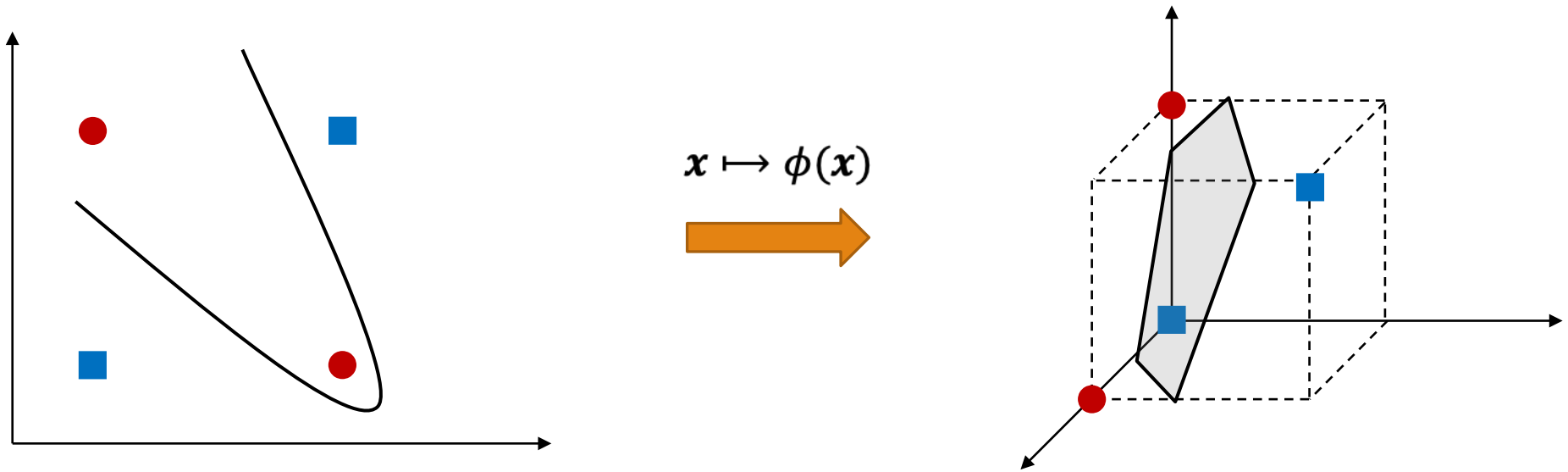
A	B	C	D
Larger C	Smaller C	More features	More support vectors

Answer: B. Smaller C

A smaller C penalizes margin violations less strongly and often yields a wider margin.

线性不可分数据的 SVM

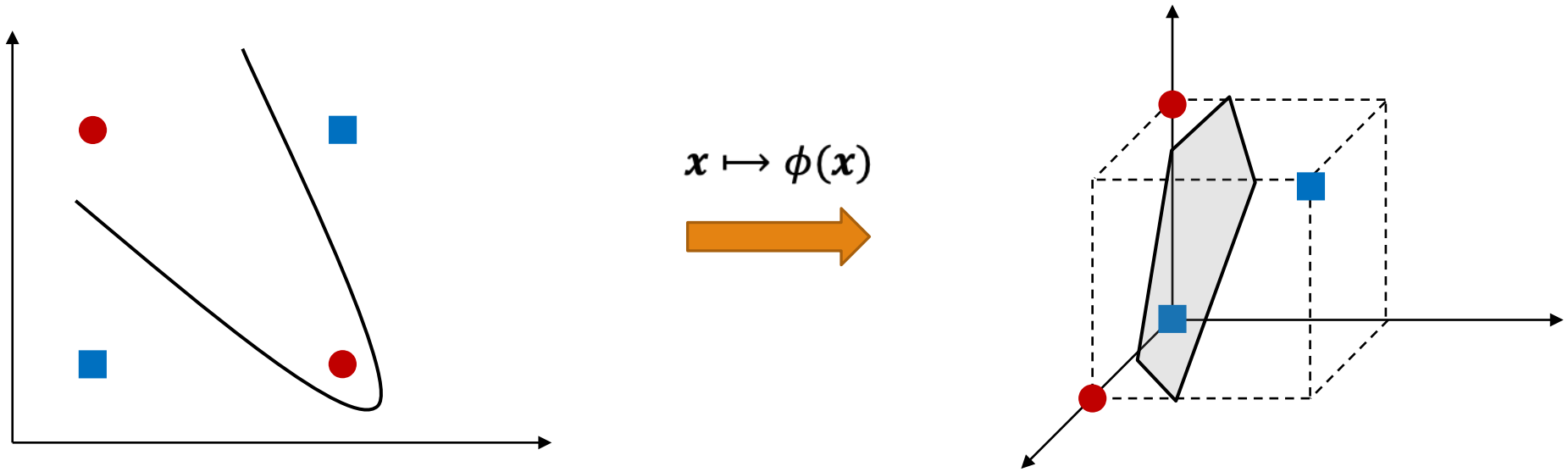
- 另一种做法：把线性不可分数据映射到更高维空间
- 在新空间中寻找线性分离超平面
  - 示例： XOR 问题



SVM for Linearly Inseparable Data

- Alternatively, for linearly inseparable data, we can map them to a higher dimensional space

- We search for a linear separating hyperplane in the new space
  - Example: The XOR problem



核函数用于非线性分类

- 不必在变换后的数据上计算点积，数学上等价于对原始数据应用核函数 $K(xi, xj)$ ，即

$$K(xi, xj) = \varphi(xi)\ \varphi(xj)$$

- 典型核函数

$$\text{Polynomial kernel of degree } h: \quad K(\boldsymbol{X_i}, \boldsymbol{X_j}) = (\boldsymbol{X_i} \cdot \boldsymbol{X_j} + 1)^h$$

$$\text{Gaussian radial basis function kernel}: \quad K(\boldsymbol{X_i}, \boldsymbol{X_j}) = e^{-\|\boldsymbol{X_i} - \boldsymbol{X_j}\|^2 / 2\sigma^2}$$

$$\text{Sigmoid kernel}: \quad K(\boldsymbol{X_i}, \boldsymbol{X_j}) = \tanh(\kappa \boldsymbol{X_i} \cdot \boldsymbol{X_j} - \delta)$$

- SVM 借助核函数可高效完成非线性分类，隐式地把输入映射到高维特征空间

https://www.youtube.com/watch?time_continue=42&v=3liCbRZPrZA

http://crsouza.com/2010/03/17/kernel-functions-for-machine-learning-applications/

Kernel Functions for Nonlinear Classification

- Instead of computing the dot product on the transformed data, it is mathematically equivalent to applying a kernel function $K(\boldsymbol{x}_i, \boldsymbol{x}_j)$ to the original data, i.e.,

$$K(\boldsymbol{x}_i, \boldsymbol{x}_j) = \phi(\boldsymbol{x}_i)\, \phi(\boldsymbol{x}_j)$$

- Typical Kernel Functions

$$\textbf{Polynomial kernel of degree } h: \quad K(\boldsymbol{X_i}, \boldsymbol{X_j}) = (\boldsymbol{X_i} \cdot \boldsymbol{X_j} + 1)^h$$

$$\textbf{Gaussian radial basis function kernel}: \quad K(\boldsymbol{X_i}, \boldsymbol{X_j}) = e^{-\|\boldsymbol{X_i} - \boldsymbol{X_j}\|^2 / 2\sigma^2}$$

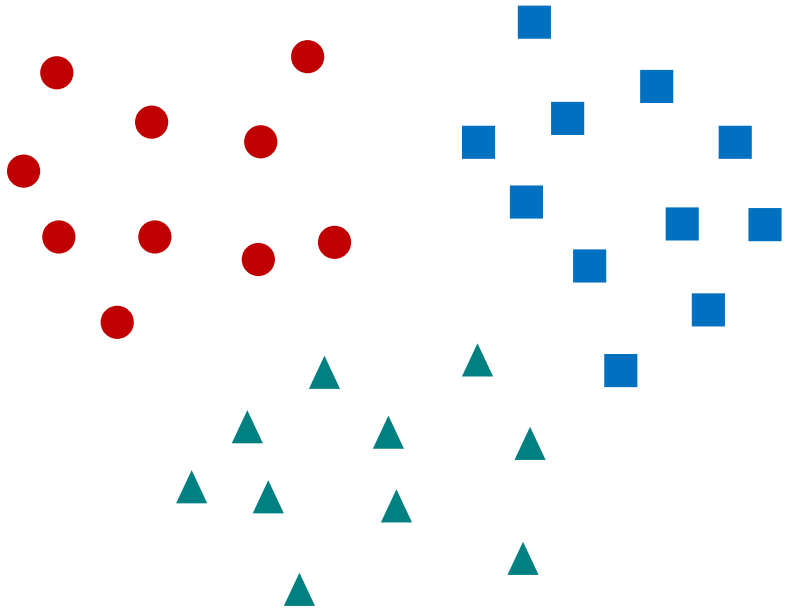
$$\textbf{Sigmoid kernel}: \quad K(\boldsymbol{X_i}, \boldsymbol{X_j}) = \tanh(\kappa \boldsymbol{X_i} \cdot \boldsymbol{X_j} - \delta)$$

- SVMs can efficiently perform a non-linear classification using kernel functions, implicitly mapping their inputs into high-dimensional feature spaces

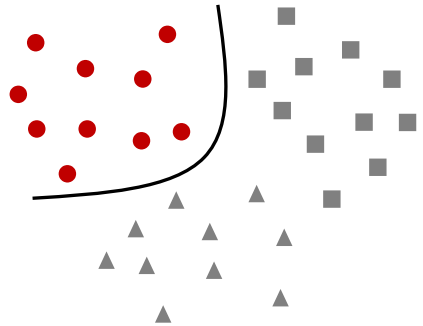
https://www.youtube.com/watch?time_continue=42&v=3liCbRZPrZA

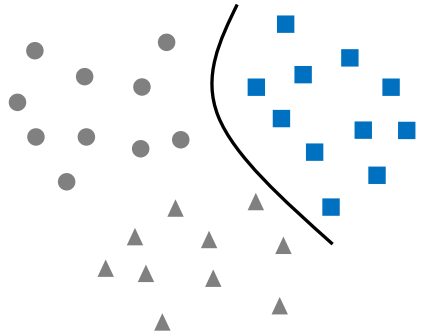
http://crsouza.com/2010/03/17/kernel-functions-for-machine-learning-applications/

用 SVM 做多类分类

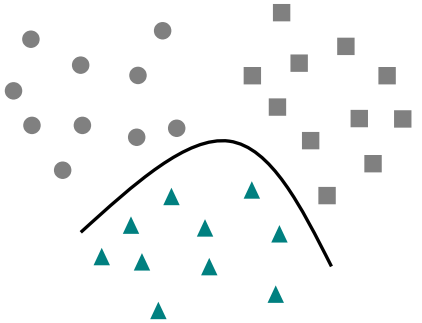


N 个类别

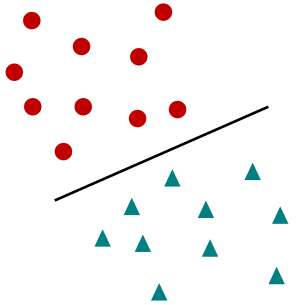


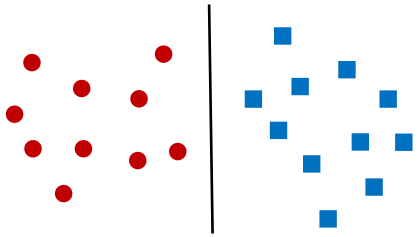


一对其余

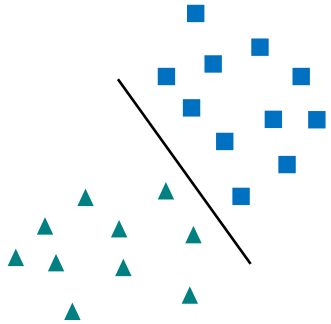


需要 N 个分类器



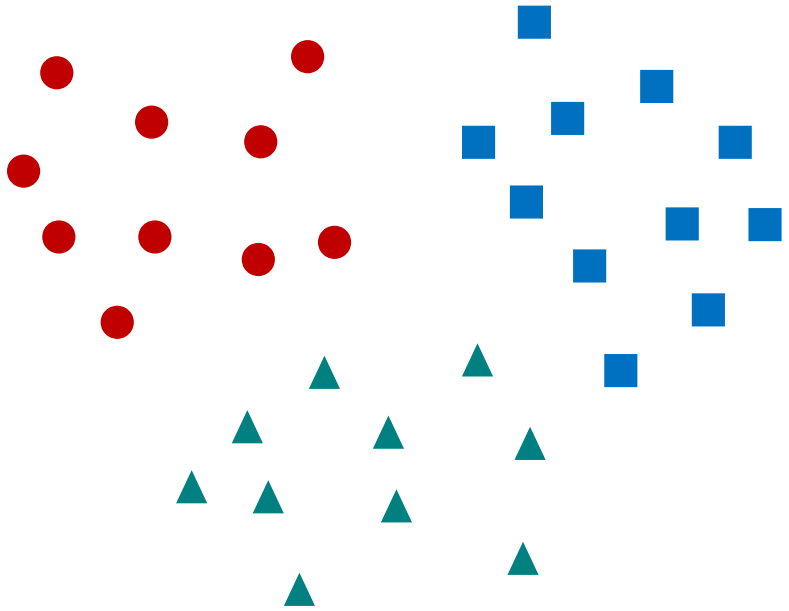


一对一

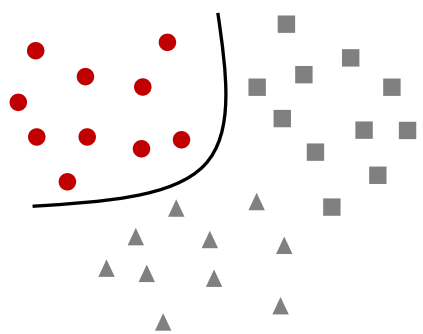


需要 N(N - 1)/2 个分类器

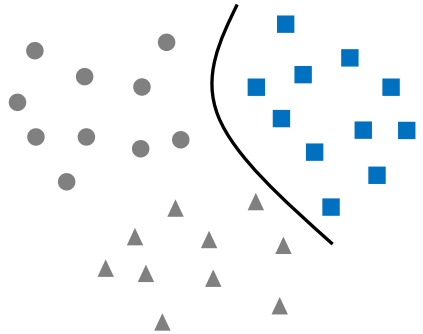
Multi-class Classification with SVM

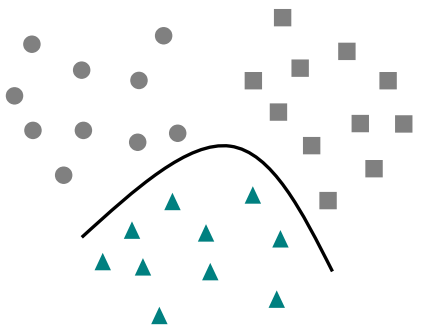


N classes

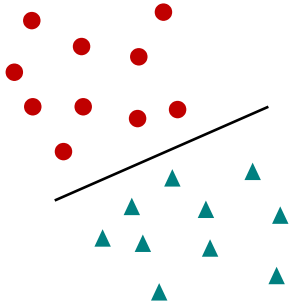


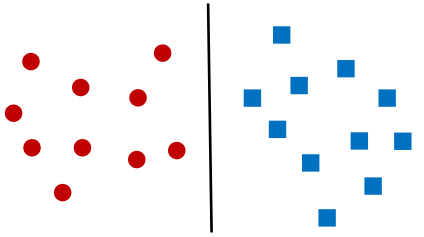
One-vs-Rest



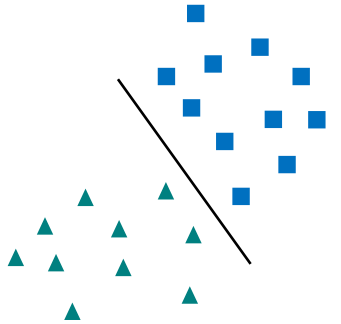


Requires N classifiers





One-vs-One



Requires N(N - 1)/2 classifiers

SVM 能扩展到海量数据吗?

- **SVM 在高维数据上很有效**
  - 训练所得分类器的**复杂度**由支持向量的个数而非数据维度决定
  - **支持向量**是关键的训练样本——它们离决策边界（MMH）最近
  - 因此，即使数据维度很高，支持向量少的 SVM 也能有良好的泛化

- 核 SVM 的训练时间和内存增长很快，可能难以扩展到超大数据集
  - 用层次微聚类方法扩展 SVM （H. Yu, J. Yang, and J. Han, "Classifying Large Data Sets Using SVM with Hierarchical Clusters", KDD' 03）

Is SVM Scalable on Massive Data?

- **SVM is effective on high-dimensional data**
  - The **complexity** of trained classifier is characterized by the # of support vectors rather than the dimensionality of the data
  - The **support vectors** are the essential or critical training examples—they lie closest to the decision boundary (MMH)
  - Thus, an SVM with a small number of support vectors can have good generalization, even when the dimensionality of the data is high

- Kernel SVMs may not scale to very large datasets because training time and memory grow quickly
  - Scaling SVM by a hierarchical micro-clustering approach (H. Yu, J. Yang, and J. Han, "Classifying Large Data Sets Using SVM with Hierarchical Clusters", KDD'03)

SVM：应用

- **特点**：训练可能较慢，但由于能刻画复杂的非线性决策边界（间隔最大化），准确率高

- 用途：分类与数值预测
  - SVM 也可用于多类（＞2）分类和回归分析（需额外参数）

- 应用：
  - 手写数字识别、物体识别、说话人识别、时间序列预测基准测试

SVM: Applications

- **Features**: training can be slow but accuracy is high owing to their ability to model complex nonlinear decision boundaries (margin maximization)

- Used for: classification and numeric prediction
  - SVM can also be used for classifying multiple (> 2) classes and for regression analysis (with additional parameters)

- Applications:
  - Handwritten digit recognition, object recognition, speaker identification, benchmarking time-series prediction tests

SVM 回顾

- **优点**
  - 凸目标：标准 SVM 训练存在全局最优
  - 在中小规模数据集上通常有效
  - 借助核函数获得灵活性
  - 与半监督训练相容

- **缺点**
  - 核 SVM 在超大数据集上代价可能很高

SVM Recap

- **Pros**
  - Convex objective: standard SVM training has a global optimum
  - Often effective on small and medium-sized datasets
  - Flexibility through kernel functions
  - Conformity with semi-supervised training

- **Cons**
  - Kernel SVMs can be costly on very large datasets

SVM 参数一览

选项	控制什么	实际效果
C	违反间隔的惩罚	小：间隔更宽 / 违反更多；大：拟合更紧
核函数	特征空间中相似度的定义	简单边界用线性核；非线性结构用 RBF 或多项式核
γ (RBF)	单个训练样本的影响范围	小：边界更平滑；大：更局部、更复杂
特征尺度	对距离和点积的相对贡献	训练前对数值特征标准化

用验证集选择 C 和核参数；测试集只用于最终报告。

SVM Parameters at a Glance

Choice	What it controls	Practical effect
C	Penalty for margin violations	Small: wider margin / more violations; large: tighter fit
Kernel	Definition of similarity in feature space	Linear for simple boundaries; RBF or polynomial for nonlinear structure
γ (RBF)	Reach of one training example	Small: smoother boundary; large: more local and complex
Feature scale	Relative contribution to distances and dot products	Standardize numeric features before training

Validation selects C and kernel parameters; the test set is used only for final reporting.

第 1 次课综合：三种不同的建模选择

方法	核心问题	核心机制	主要风险
特征选择	哪些变量携带有用信号?	打分、子集搜索或正则化	泄漏或选择不稳定
贝叶斯网络	变量之间如何相互依赖?	DAG 加局部 CPT	结构错误或计数稀疏
SVM	哪条边界使分离最大?	间隔优化与核函数	缩放不当或超参数选择不当

训练拟合表示与模型；验证在备选方案中选择；测试估计最终泛化能力。

Session 1 Synthesis: Three Different Modeling Choices

Method	Central question	Core mechanism	Main risk
Feature selection	Which variables carry useful signal?	Score, subset search, or regularization	Leakage or unstable selection
Bayesian network	How do variables depend on one another?	DAG plus local CPTs	Incorrect structure or sparse counts
SVM	Which boundary maximizes separation?	Margin optimization and kernels	Poor scaling or hyperparameter choice

Training fits the representation and model; validation chooses alternatives; testing estimates final generalization.

附录 1：更多 SVM 资源

- 代表性实现
  - **LIBSVM**：高效的 SVM 实现，支持多类分类、 nu-SVM 、单类 SVM ，并提供 java 、 python 等多种接口。
  - **SVM-light**：更简单但性能不优于 LIBSVM ，只支持二分类，且只有 C 版本
  - **SVM-torch**：另一个较新的实现，同样用 C 编写

Appendix 1: Further SVM Resources

- Representative implementations
  - **LIBSVM**: an efficient implementation of SVM, multi-class classifications, nu-SVM, one-class SVM, including also various interfaces with java, python, etc.
  - **SVM-light**: simpler but performance is not better than LIBSVM, support only binary classification and only in C
  - **SVM-torch**: another recent implementation also written in C

第 2 次课：规则、弱监督与复杂数据

模块	重点
1	基于规则与基于模式的分类
2	弱监督与少标签学习
3	数据流、序列与图
4	本次课综合

学习目标

选择与标签来源和输入数据结构相匹配的学习策略。

Session 2: Rules, Weak Supervision and Rich Data

Block	Focus
1	Rule-based and pattern-based classification
2	Weak supervision and limited-label learning
3	Streams, sequences and graphs
4	Session synthesis

Learning objective

Choose a learning strategy that matches the label source and the structure of the input data.

用 IF-THEN 规则分类

- 把知识表示为 **IF-THEN** 规则的形式

  R_1: IF *age* = youth AND *student* = yes THEN *buys_computer* = yes

- 规则的评估：*覆盖率*与*准确率*
  - coverage(R_1) = 被 R_1 **的条件**覆盖的元组比例（与 THEN 部分无关）
  - accuracy(R_1) = 被覆盖元组中被 R_1 正确分类的比例（IF 与 THEN 部分都要满足）

- 若触发多条规则，需要**冲突消解**
  - **规模排序**：给条件“最严格”（即*属性检验最多*）的被触发规则最高优先级
  - **基于类别排序**：按*各类的流行度或误分类代价*降序排列
  - **基于规则排序（决策列表）**：按某种规则质量度量或由专家把规则组织成一个长优先级列表

Using IF-THEN Rules for Classification

- Represent the knowledge in the form of **IF-THEN** rules

  R_1: IF *age* = youth AND *student* = yes THEN *buys_computer* = yes

- Assessment of a rule: *coverage* and *accuracy*
  - coverage(R_1) = ratio of tuples covered by **the condition of** R_1 (THEN-part is not important for this)
  - accuracy(R_1) = ratio of tuples correctly classified by R_1 in the covered ones (both IF-part and THEN-part counts)

- If more than one rule are triggered, need **conflict resolution**
  - **Size ordering**: assign the highest priority to the triggering rules that has the "toughest" requirement (i.e., with the *most attribute tests*)
  - **Class-based ordering**: decreasing order of *prevalence or misclassification cost per class*
  - **Rule-based ordering** (**decision list**): rules are organized into one long priority list, according to some measure of rule quality or by experts

规则覆盖率与准确率：算例

设在 100 条记录上评估一条规则：它在 20 条记录上触发，其中 16 条分类正确。

量	计算	值	含义
覆盖率	20 / 100	20%	规则适用的频率
规则准确率	16 / 20	80%	规则适用时正确的频率

解释

规则准确率高不代表用途广：*这条规则可靠但适用面窄。*

- **规则准确率只针对被覆盖的记录**；分类器总体准确率还取决于其余规则和默认规则。

Rule Coverage and Accuracy: Worked Example

Suppose a rule is evaluated on 100 records. It fires on 20 records and classifies 16 of them correctly.

Quantity	Calculation	Value	Meaning
Coverage	20 / 100	20%	How often the rule applies
Rule accuracy	16 / 20	80%	How often it is correct when it applies

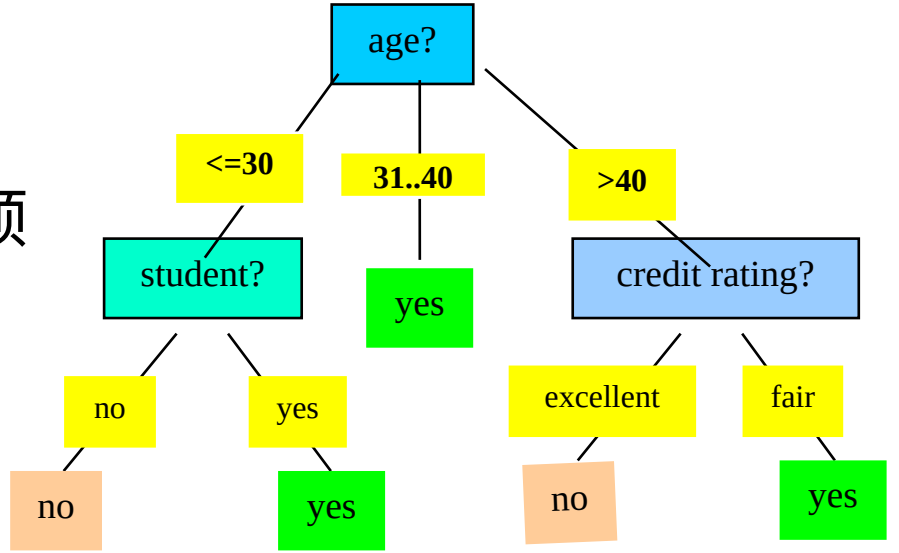
Interpretation

High rule accuracy does not imply broad usefulness: *this rule is reliable but specialized.*

- **Rule accuracy is local to covered records**; overall classifier accuracy also depends on the remaining rules and the default rule.

从决策树中提取规则

- 规则比大型树*更易理解*
- **从根到叶的*每条路径*生成一条规则**
- 路径上的各属性 - 值对构成**合取**；叶结点给出类别预测
- 规则互斥且穷尽



- 示例：从 *buys_computer* 决策树中提取规则

IF *age* = young AND *student* = *no* THEN *buys_computer* = *no*

IF *age* = young AND *student* = *yes* THEN *buys_computer* = *yes*

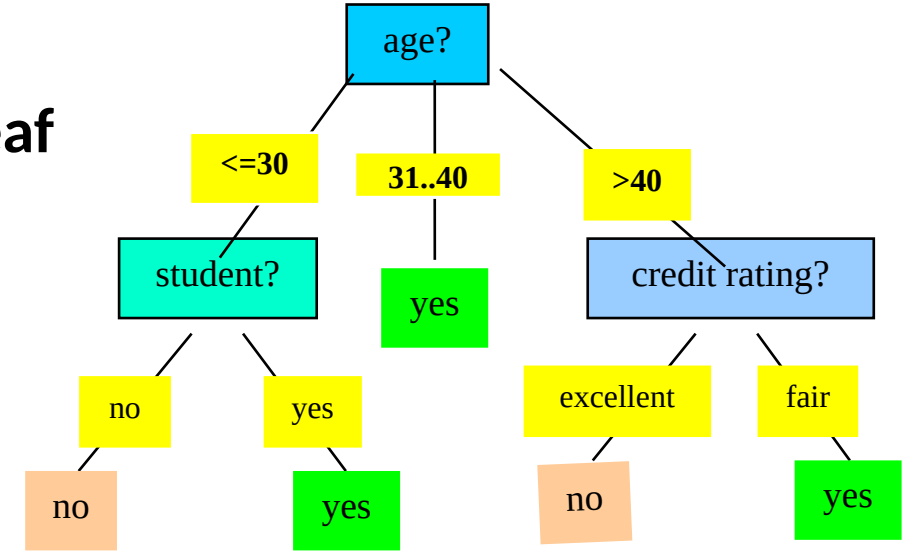
IF *age* = mid-age THEN *buys_computer* = *yes*

IF *age* = old AND *credit_rating* = *excellent* THEN *buys_computer* = *no*

IF *age* = old AND *credit_rating* = *fair* THEN *buys_computer* = *yes*

Rule Extraction from a Decision Tree

- Rules are *easier to understand* than large trees
- **One rule is created *for each path* from the root to a leaf**
- Each attribute-value pair along a path forms a **conjunction**: the leaf holds the class prediction
- Rules are mutually exclusive and exhaustive



- Example: Rule extraction from our *buys_computer* decision-tree

IF *age* = young AND *student* = *no* THEN *buys_computer* = *no*

IF *age* = young AND *student* = *yes* THEN *buys_computer* = *yes*

IF *age* = mid-age THEN *buys_computer* = *yes*

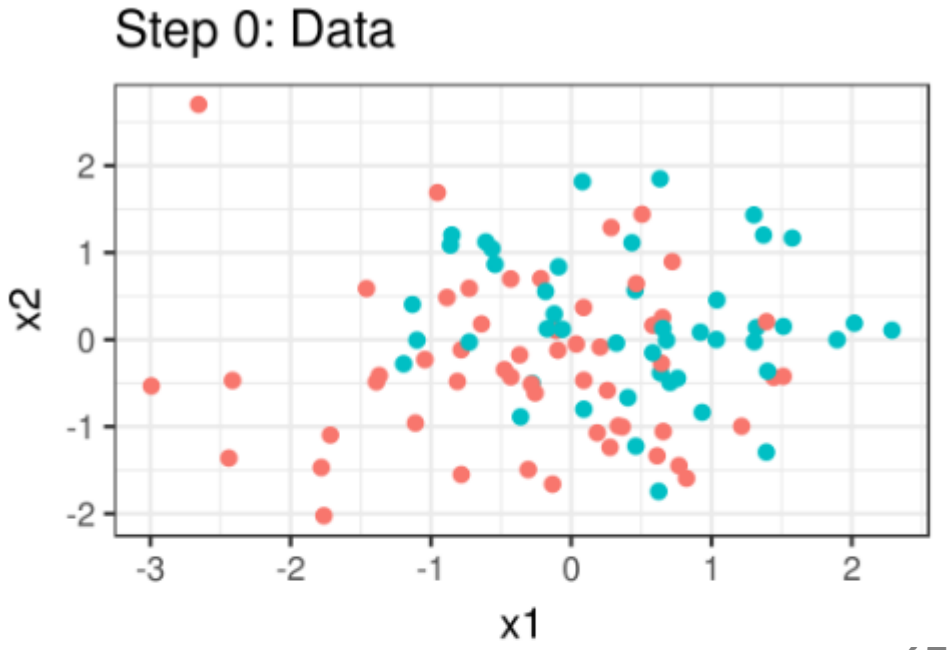
IF *age* = old AND *credit_rating* = *excellent* THEN *buys_computer* = *no*

IF *age* = old AND *credit_rating* = *fair* THEN *buys_computer* = *yes*

规则归纳：顺序覆盖法

- **顺序覆盖算法：直接从训练数据中提取规则**
- 规则被*顺序地*学习，针对类 C_i 的每条规则覆盖 C_i 的许多元组，而不覆盖（或很少覆盖）其他类的元组
- 对比决策树归纳：*同时*学习一组规则

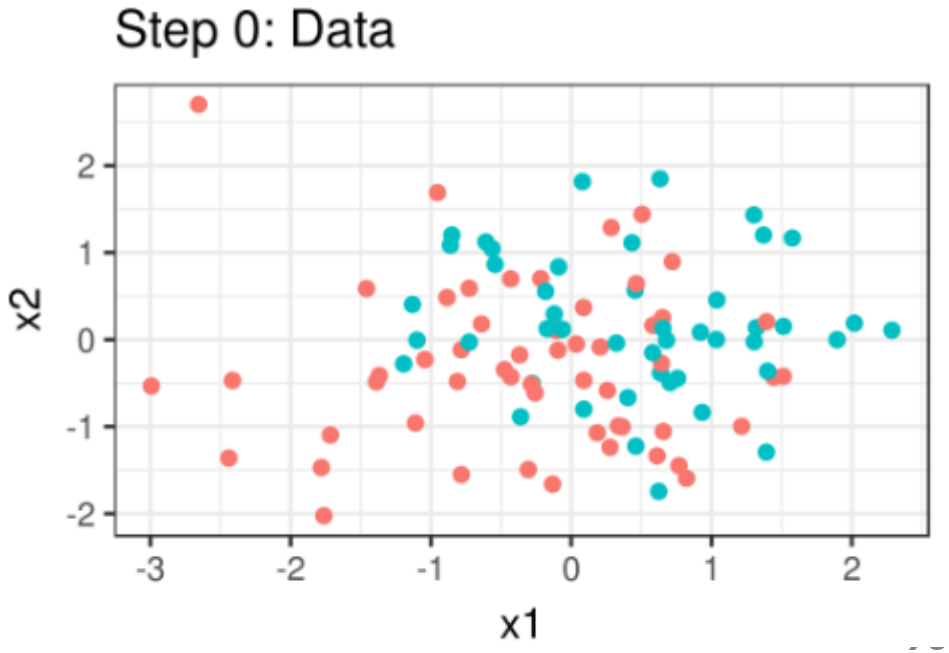
- *第 0 步：从空规则列表开始。*



Rule Induction: Sequential Covering Method

- **Sequential covering algorithm: Extracts rules directly from training data**
- Rules are learned *sequentially*, each for a given class C_i will cover many tuples of C_i but none (or few) of the tuples of other classes
- Comp. w. decision-tree induction: learning a set of rules *simultaneously*

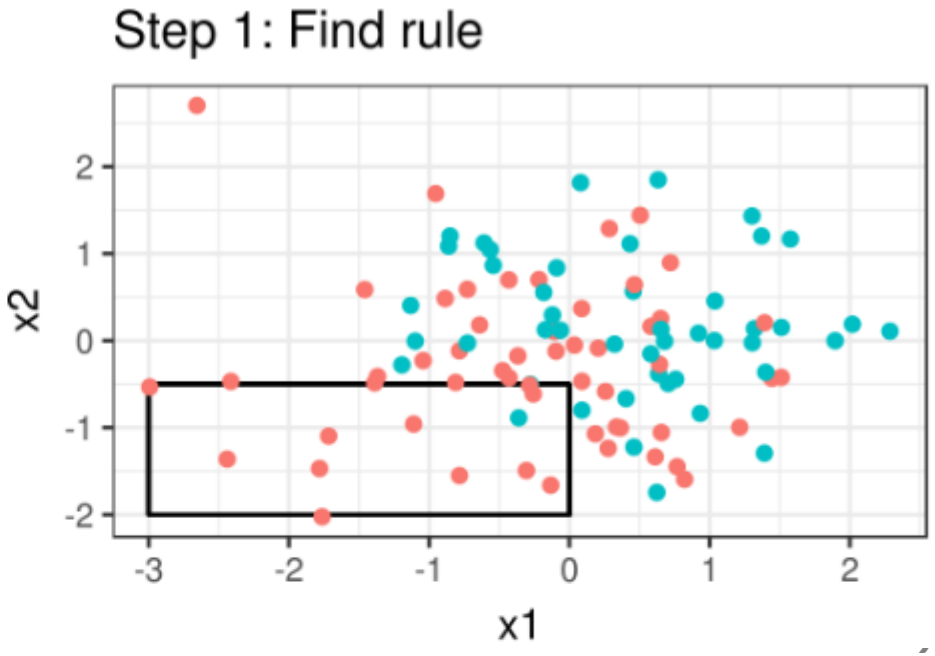
- *Step 0: Start with an empty list of rules.*



规则归纳：顺序覆盖法

- **顺序覆盖算法：直接从训练数据中提取规则**
- 规则被*顺序地*学习，针对类 C_i 的每条规则覆盖 C_i 的许多元组，而不覆盖（或很少覆盖）其他类的元组
- 对比决策树归纳：*同时*学习一组规则

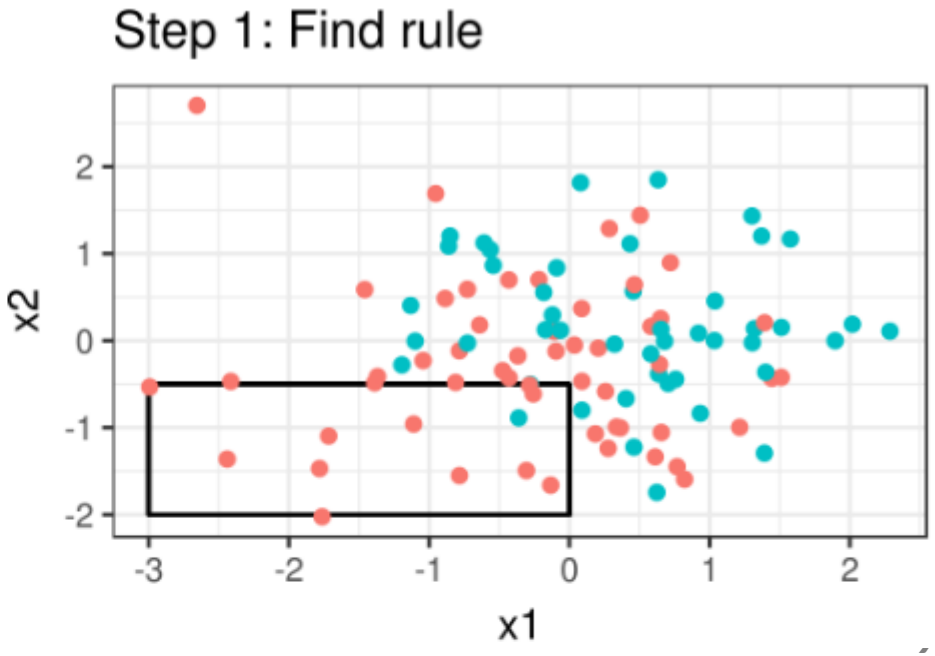
- *第 1 步：学习一条规则 r。*



Rule Induction: Sequential Covering Method

- **Sequential covering algorithm: Extracts rules directly from training data**
- Rules are learned *sequentially*, each for a given class C_i will cover many tuples of C_i but none (or few) of the tuples of other classes
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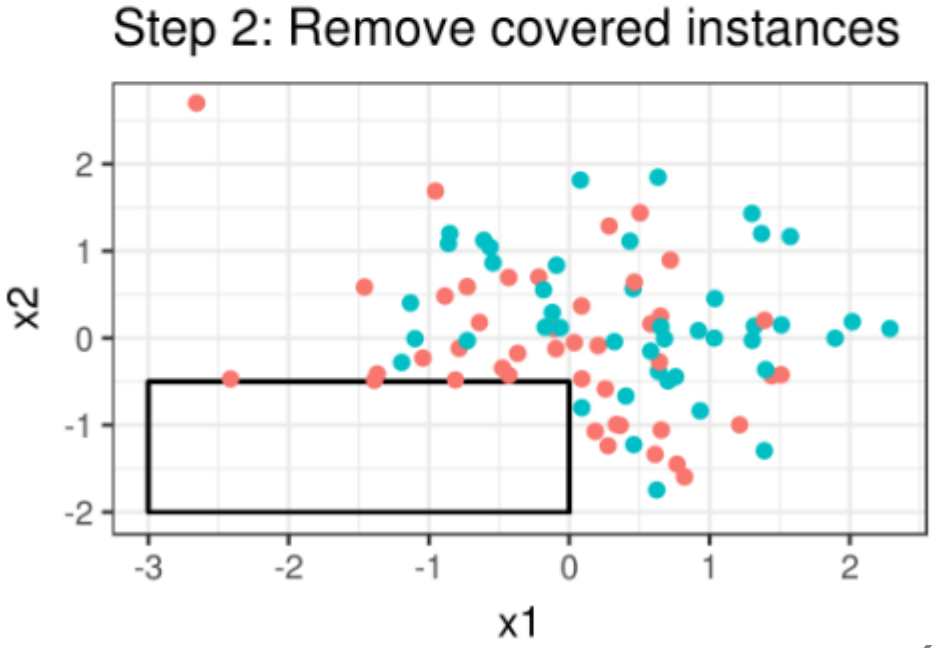
- *Step 1: Learn a rule r.*



规则归纳：顺序覆盖法

- **顺序覆盖算法：直接从训练数据中提取规则**
- 规则被*顺序地*学习，针对类 C_i 的每条规则覆盖 C_i 的许多元组，而不覆盖（或很少覆盖）其他类的元组
- 对比决策树归纳：*同时*学习一组规则

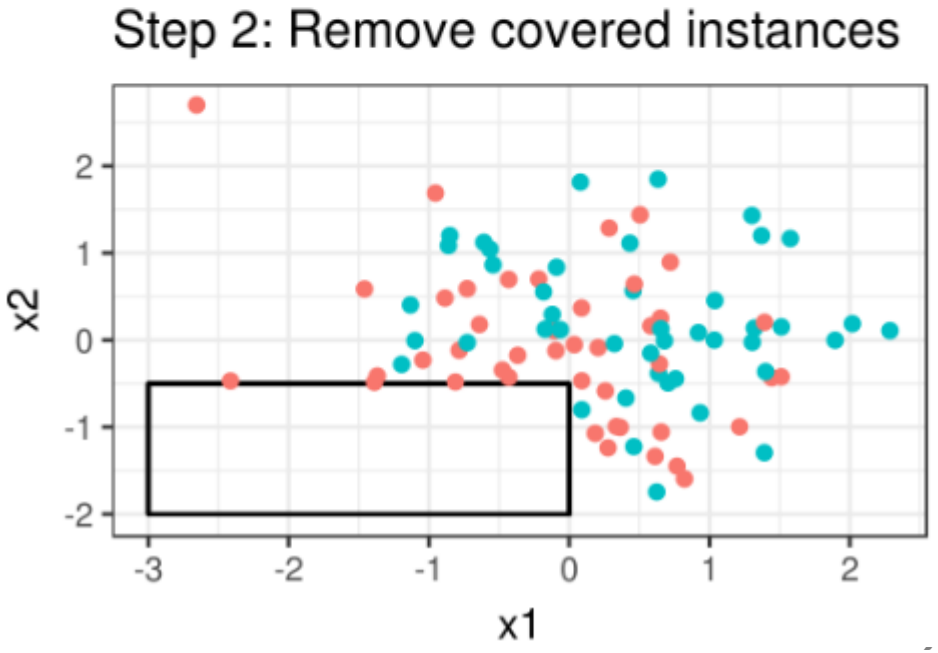
- *第 2 步：将规则覆盖的元组移除。*



Rule Induction: Sequential Covering Method

- **Sequential covering algorithm: Extracts rules directly from training data**
- Rules are learned *sequentially*, each for a given class C_i will cover many tuples of C_i but none (or few) of the tuples of other classes
- Comp. w. decision-tree induction: learning a set of rules *simultaneously*

- *Step 2: The tuples covered by the rules are removed.*



规则归纳：顺序覆盖法

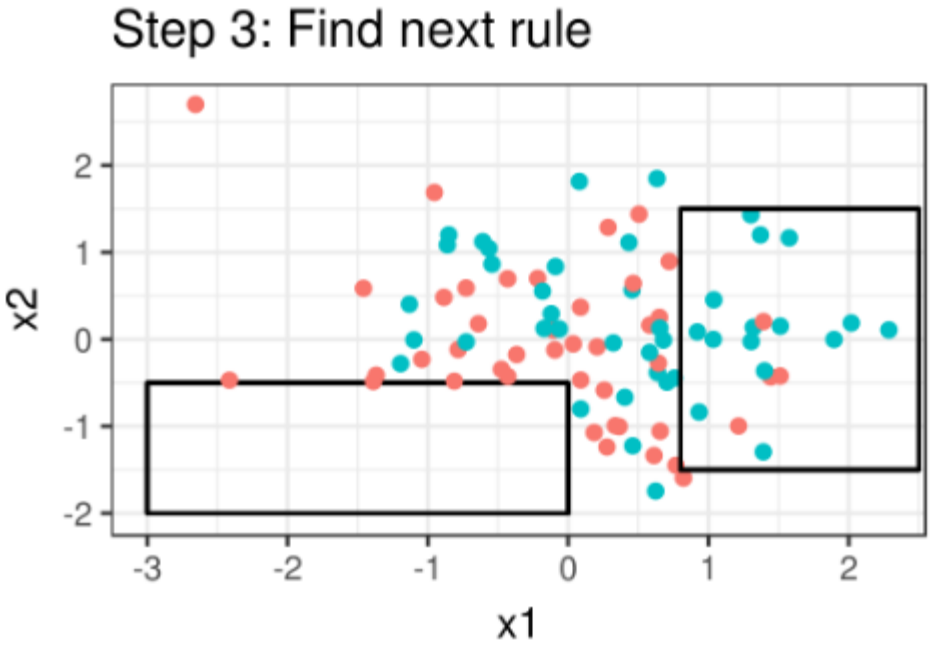
- **顺序覆盖算法：直接从训练数据中提取规则**
- 规则被*顺序地*学习，针对类 C_i 的每条规则覆盖 C_i 的许多元组，而不覆盖（或很少覆盖）其他类的元组
- 对比决策树归纳：*同时*学习一组规则

- *第 3 步：*对剩余

元组重复该过程，直到满足*终止条件*，例如

没有更多训练样本，或返回规则的质量

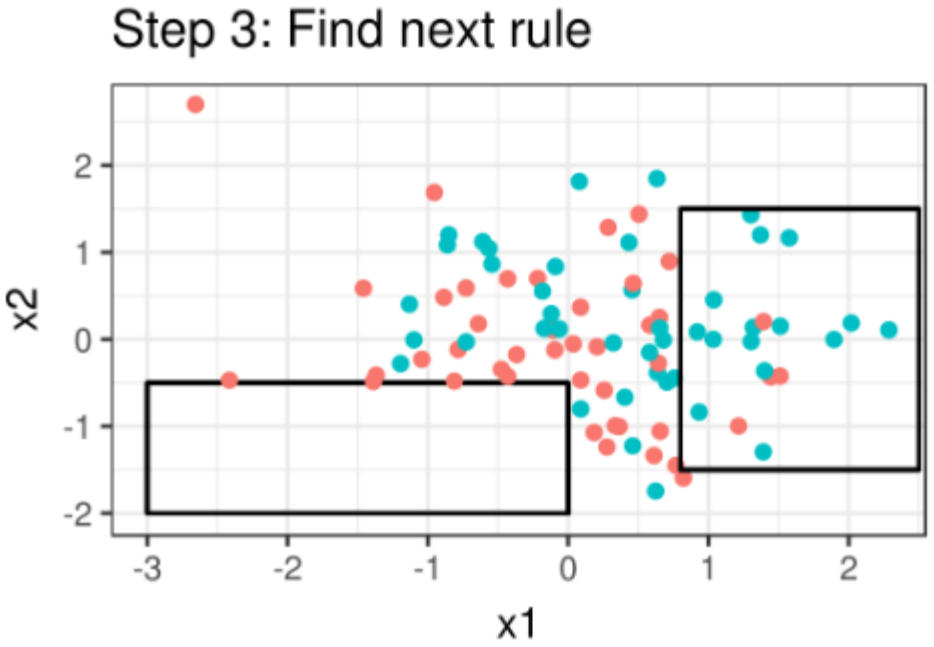
低于阈值。



Rule Induction: Sequential Covering Method

- **Sequential covering algorithm: Extracts rules directly from training data**
- Rules are learned *sequentially*, each for a given class C_i will cover many tuples of C_i but none (or few) of the tuples of other classes
- Comp. w. decision-tree induction: learning a set of rules *simultaneously*

- *Step 3: Repeat the process* on the remaining tuples until *termination condition*, e.g., when no more training examples or when the quality of a rule returned is below a threshold.



顺序覆盖：一次一条规则

步骤	操作	用途
1	为目标类学习一条高质量规则	刻画一个连贯区域
2	移除已覆盖的正例	下一次搜索聚焦于剩余正例
3	重复直到满足停止条件	构建有序规则列表
4	添加默认规则	对之前未覆盖的记录分类

规则是局部学习的，但其顺序决定最终分类器。

Sequential Covering: One Rule at a Time

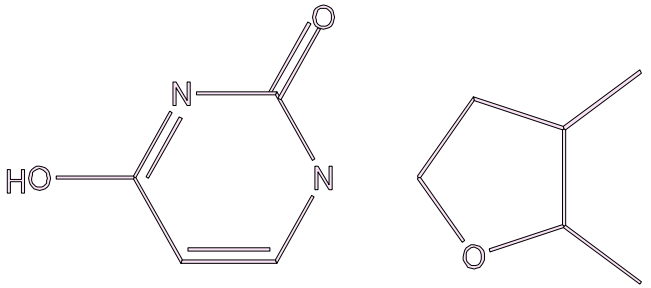
Step	Action	Purpose
1	Learn a high-quality rule for the target class	Capture one coherent region
2	Remove covered positive examples	Focus the next search on remaining positives
3	Repeat until a stopping condition is met	Build an ordered rule list
4	Add a default rule	Classify records not covered earlier

The rules are learned locally, but their order determines the final classifier.

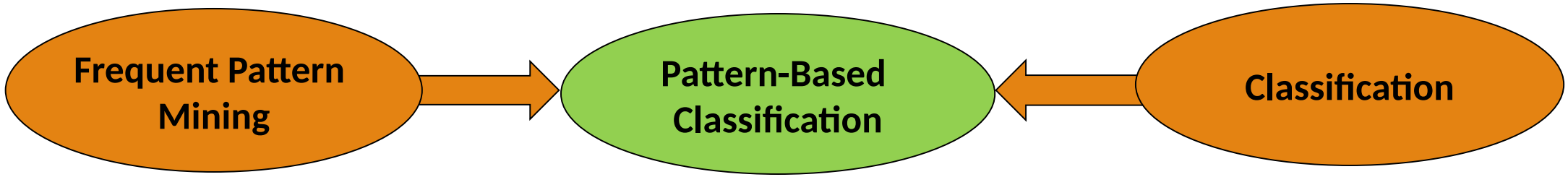
为什么要基于模式分类?



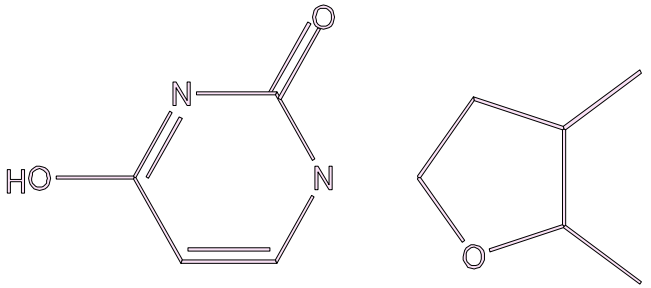
- **基于模式的分类：**两大主题的融合
- **为什么要基于模式分类?**
  - **特征构造**
    - 高阶；紧凑；有判别力
    - 例如，单词 → 短语（ Apple pie 、 Apple i-pad ）
  - **复杂数据建模**
    - 图（没有预定义的特征向量）
    - 序列
    - 半结构化 / 非结构化数据



Pattern-Based Classification, Why?



- **Pattern-based classification:** An integration of both themes
- **Why pattern-based classification?**
  - **Feature construction**
    - Higher order; compact; discriminative
    - E.g., single word → phrase (Apple pie, Apple i-pad)
  - **Complex data modeling**
    - Graphs (no predefined feature vectors)
    - Sequences
    - Semi-structured/unstructured Data



CBA： 基于关联的分类

- CBA： 基于关联的分类（Liu, Hsu and Ma, KDD’ 98）
- **方法**
  - 挖掘高置信度、高支持度的类关联规则
  - LHS： 属性 - 值对的合取； RHS： 类别标签

    $p_1 \wedge p_2 \ldots \wedge p_l \rightarrow$ “$A_{\text{class-label}} = C$”（置信度，支持度）
  - 按置信度和支持度降序排列规则

- **分类**：应用第一条匹配测试样本的规则；否则应用默认规则
- **效果**：常比某些传统分类方法（如 C4.5）更准确
- **为什么?** ——挖掘多个属性间的高置信度关联，可以克服某些一次只考虑一个属性的分类器带来的限制

CBA: Classification Based on Associations

- CBA: Classification Based on Associations (Liu, Hsu and Ma, KDD'98)
- **Method**
  - Mine high-confidence, high-support class association rules
  - LHS: conjunctions of attribute-value pairs); RHS: class labels

    $p_1 \wedge p_2 \ldots \wedge p_l \rightarrow$ "$A_{\text{class-label}} = C$" (confidence, support)
  - Rank rules in descending order of confidence and support

- **Classification**: Apply the first rule that matches a test case; otherwise apply the default rule
- **Effectiveness**: Often found more accurate than some traditional classification methods, such as C4.5
- **Why?** — Exploring high confident associations among multiple attributes may overcome some constraints introduced by some classifiers that consider only one attribute at a time

快速检查：强但窄的规则

问题：一条规则的规则准确率为 95%，但只覆盖 2% 的记录。最恰当的解读是什么?

A	B	C	D
用途广泛	适用时准确，但很少适用	必然过拟合	默认规则

答案： B

覆盖率与规则准确率回答不同的问题。

Quick Check: A Strong but Narrow Rule

Question: **A rule has 95% rule accuracy but covers only 2% of the records. What is the best interpretation?**

A	B	C	D
Broadly useful	Accurate when it applies, but rarely applies	Necessarily overfit	A default rule

Answer: B

Coverage and rule accuracy answer different questions.

弱监督学习

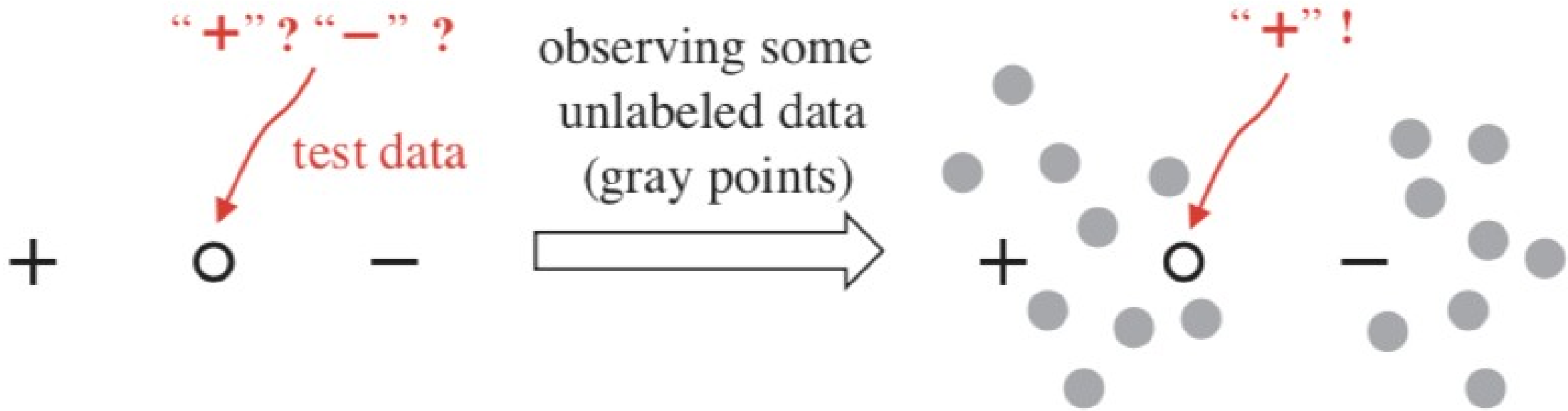
- 半监督学习
- 主动学习
- 迁移学习
- 远程监督
- 零样本学习

Weakly Supervised Learning

- Semi-supervised learning
- Active learning
- Transfer learning
- Distant supervision
- Zero-shot learning

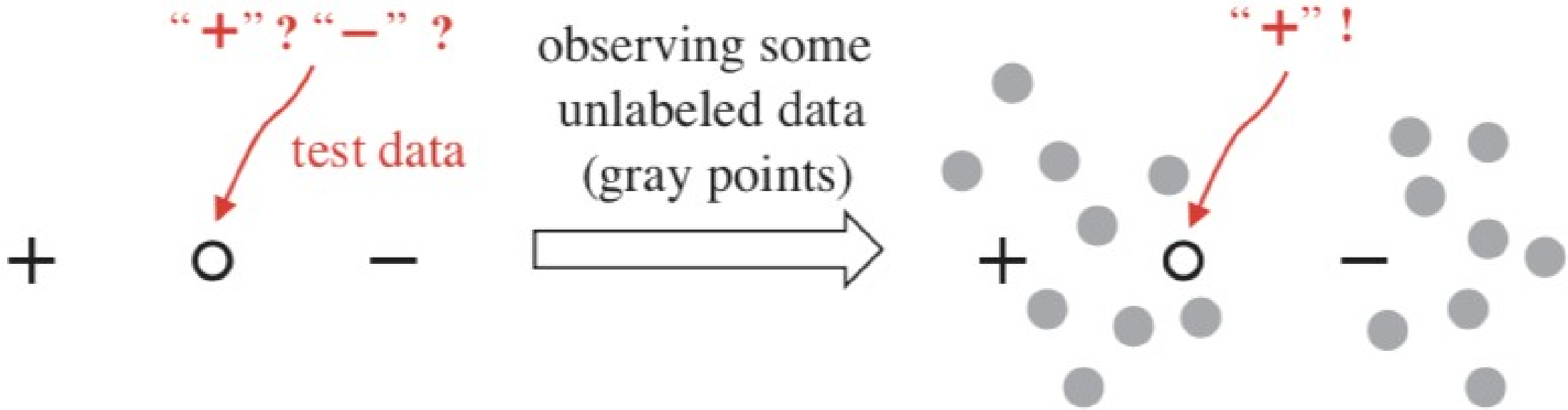
半监督学习

- **目标**：利用有标签与无标签数据构建分类器

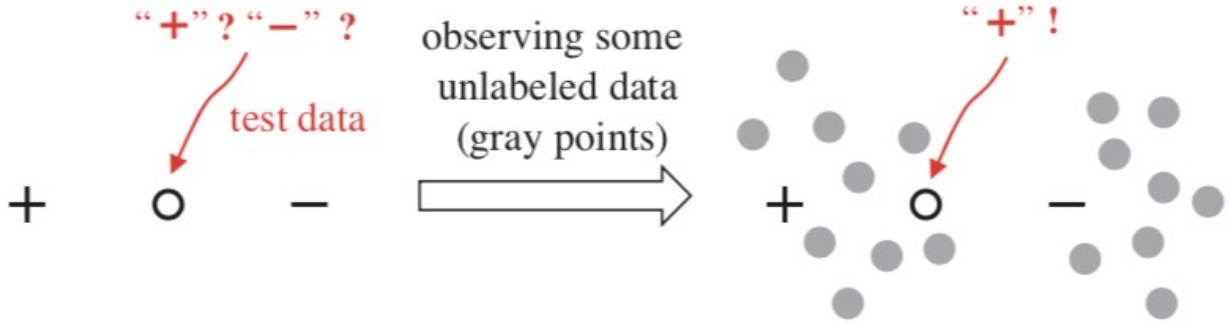


Semi-supervised Learning

- **Goal**: uses labeled data and unlabeled data to build a classifier

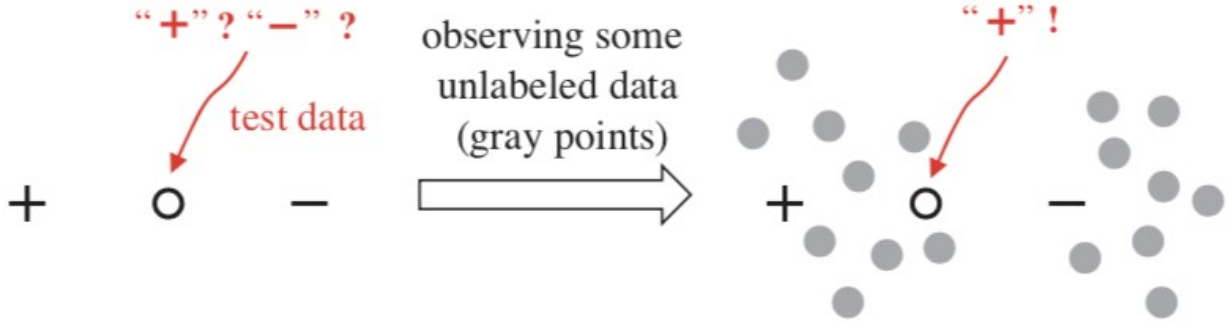


SSL： 自训练



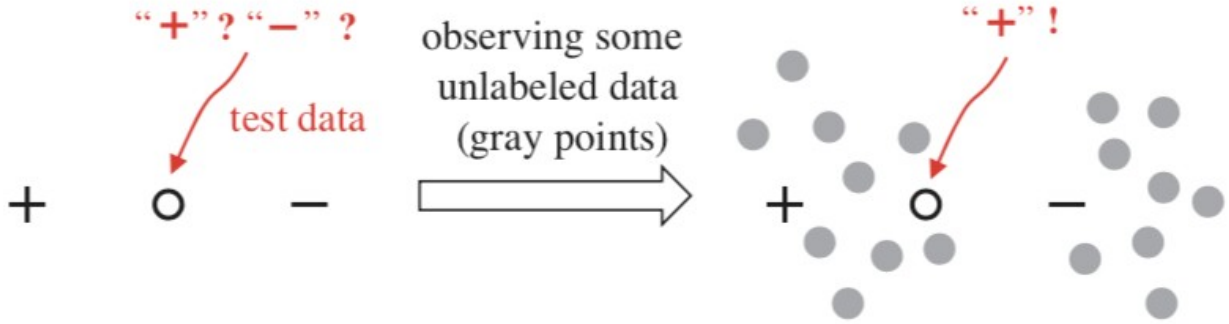
- 一般流程
  - 选择一种学习方法，如贝叶斯分类。
  - 用有标签数据构建分类器。
  - 用分类器给无标签数据打标签。
  - 选出置信度最高的无标签元组。
  - 把它及其预测标签加入有标签集合。
  - 重复

SSL: Self-training



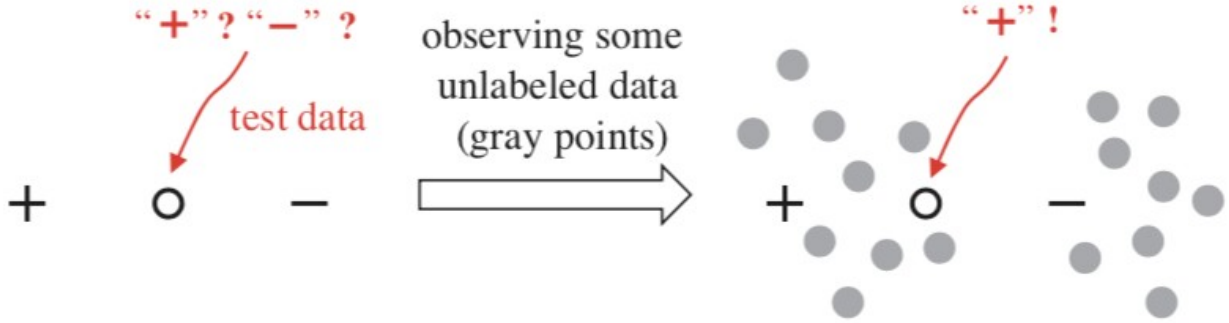
- **General procedure**
  - Select a learning method such as Bayesian classification.
  - Build the classifier using the labeled data.
  - Use the classifier to label the unlabeled data.
  - Select the unlabeled tuple having the highest confidence.
  - Add it and its predicted label to label set.
  - Repeat

SSL： 协同训练



- 一般流程
  - 为有标签数据定义两个互不重叠的特征集
  - 在有标签数据上训练两个分类器 f_1 和 f_2，各用一个特征集。
  - 分别用 f_1 和 f_2 对无标签元组分类。
  - 选出 f_1 置信度最高的无标签元组，加入 f_2 的有标签集合
  - 选出 f_2 置信度最高的无标签元组，加入 f_1 的有标签集合
  - 重复

SSL: Co-training



- **General procedure**
  - Define two separate non-overlapping feature sets for the labeled data
  - Train two classifiers, f_1 and f_2, on the labeled data, each trained on one feature set.
  - Classify un-labeled tuples by f_1 and f_2 separately.
  - Select the unlabeled tuple having the highest confidence by f_1 and add it to the labeled set for f_2
  - Select the unlabeled tuple having the highest confidence by f_2 and add it to the labeled set for f_1
  - Repeat

无标签数据何时有帮助?

半监督学习只有在输入的几何结构与标签有实质关联时才有效。

假设	直观理解	失效情形
平滑性	相近的样本往往有相同标签	相近的点可能标签不同
簇	同一簇中的样本标签相同	一个簇混合了多个类别
流形	标签在低维结构上平滑变化	流形与任务不一致

当这些假设不成立时，无标签数据可能强化错误的边界。

When Can Unlabeled Data Help?

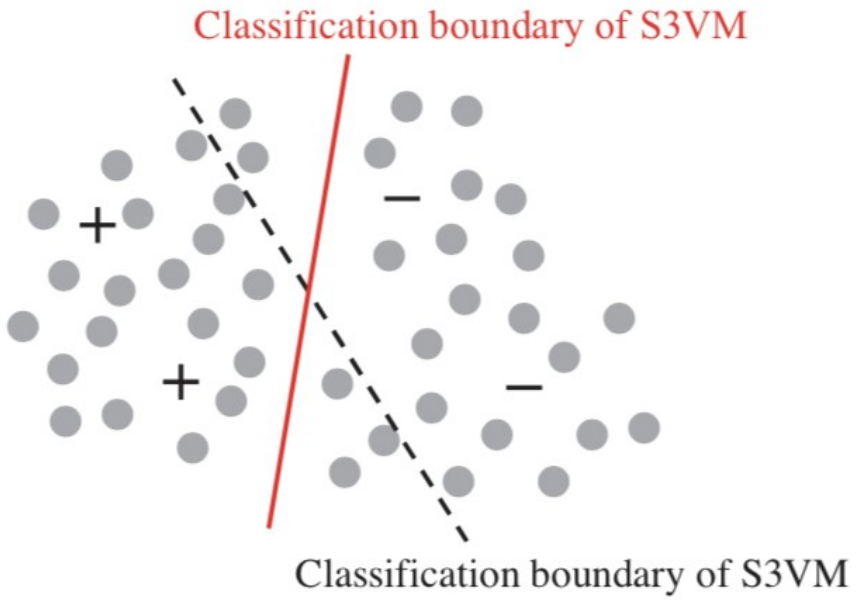
Semi-supervised learning works only when the geometry of the inputs is meaningfully related to the labels.

Assumption	Working intuition	Failure mode
Smoothness	Nearby examples tend to share a label	Nearby points may have different labels
Cluster	Examples in the same cluster share a label	A cluster mixes several classes
Manifold	Labels vary smoothly on a low-dimensional structure	The manifold does not align with the task

Unlabeled data can reinforce the wrong boundary when these assumptions are violated.

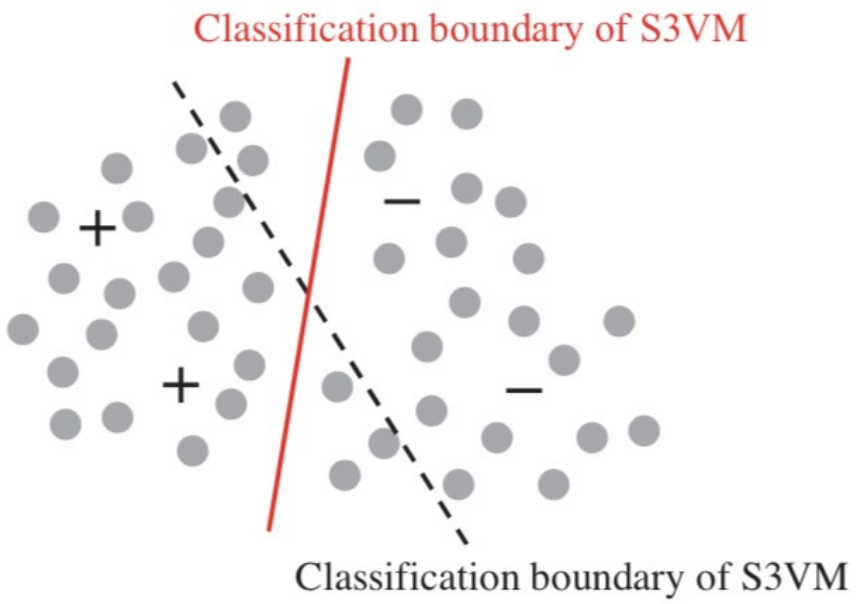
SSL 何时有效？—— 聚类假设

- **同一簇中的数据元组很可能标签相同**
- 示例： S3VM
  - 寻找最大间隔超平面（与 SVM 相同）
  - 同时避免破坏无标签元组的聚类结构
    - 例如穿过无标签数据的低密度区域



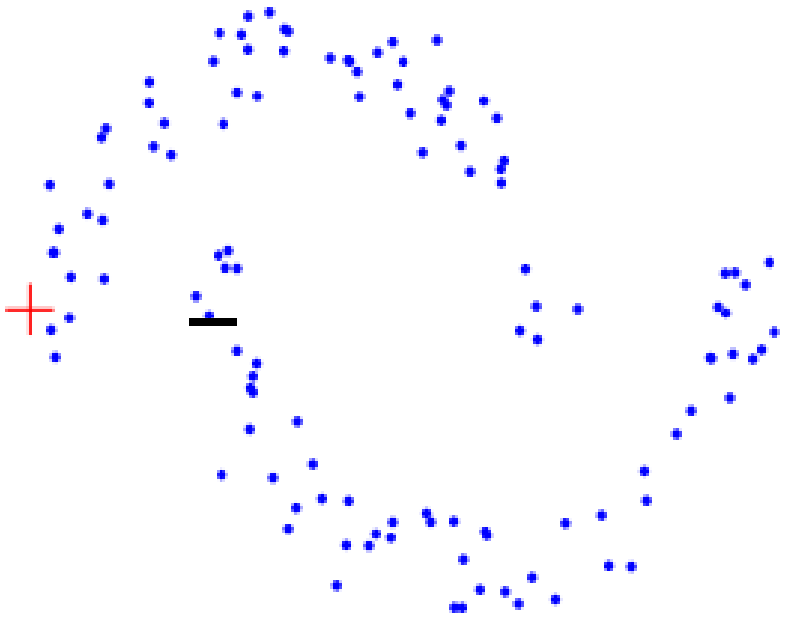
When does SSL Work? -- Clustering Assumption

- **Data tuples from same cluster are likely to share same label**
- An example: S3VM
  - Seeking a max-margin hyperplane (same as SVM)
  - avoiding to disrupt the clustering structure of unlabeled tuples
    - E.g. go through low-density region of unlabeled data



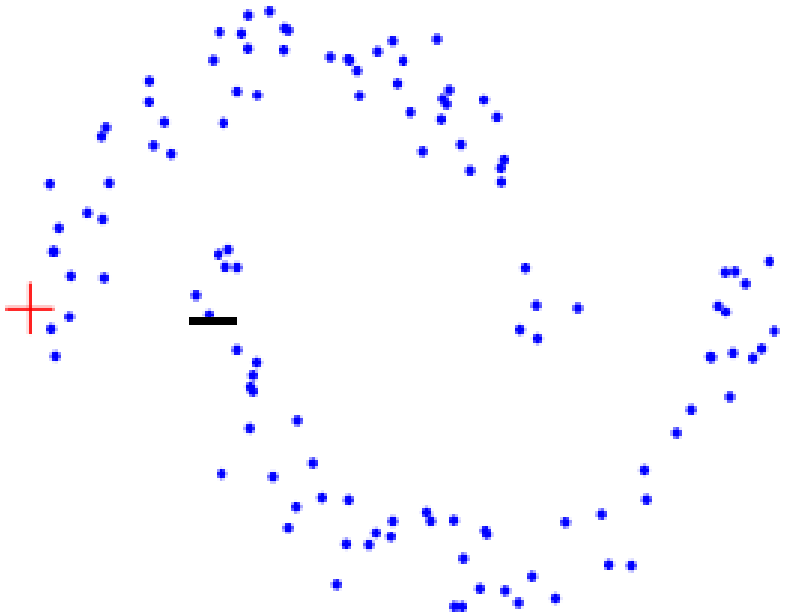
SSL 何时有效？—— 流形假设

- **相近的一对元组很可能有相同的类别标签**
- 示例：基于图的 SSL



When does SSL Work? -- Manifold Assumption

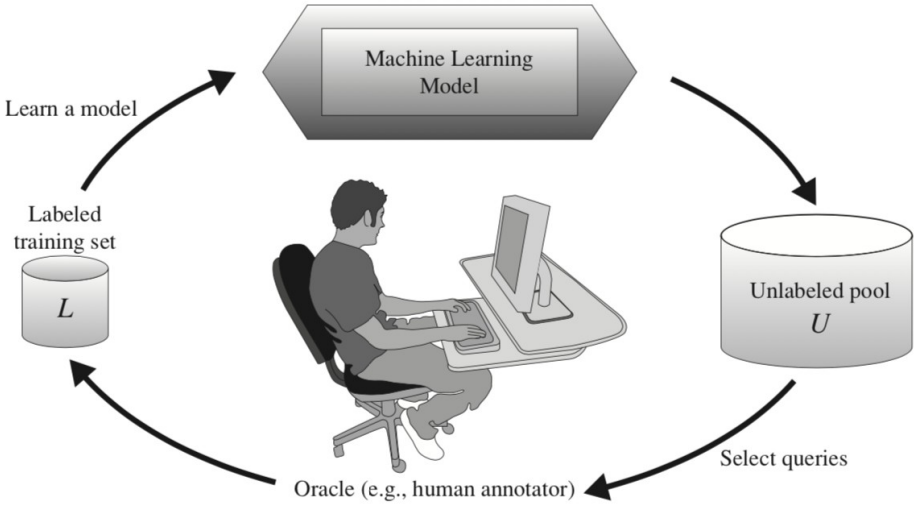
- **A pair of close tuples are likely to share the same class label**
- An example: graph-based SSL



主动学习

- **目标**：找出 *“最好的”无标签数据*样本，向**标注者（oracle）**询问其标签，以最大程度提升分类性能。

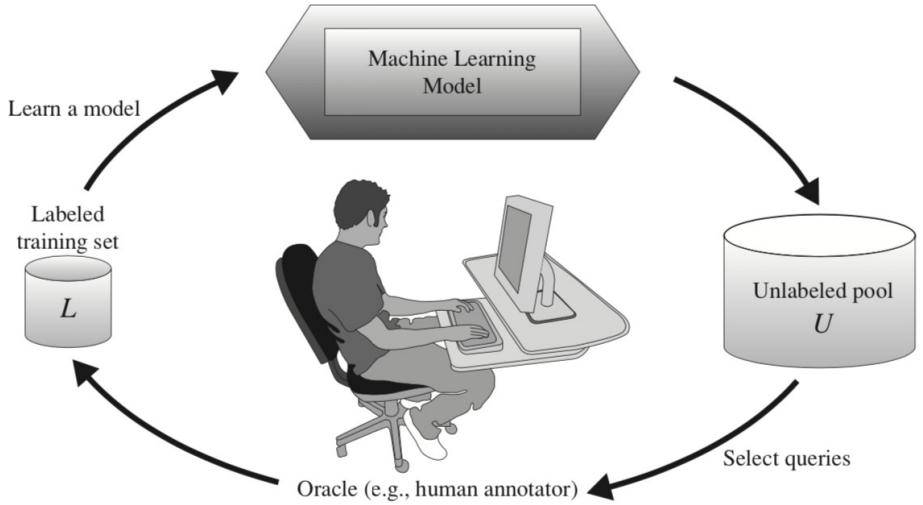
- **基于池的主动学习**
- **关键**：如何选择要查询的数据元组
  - 不确定性采样
  - 委员会查询
  - 版本空间
  - 决策论方法



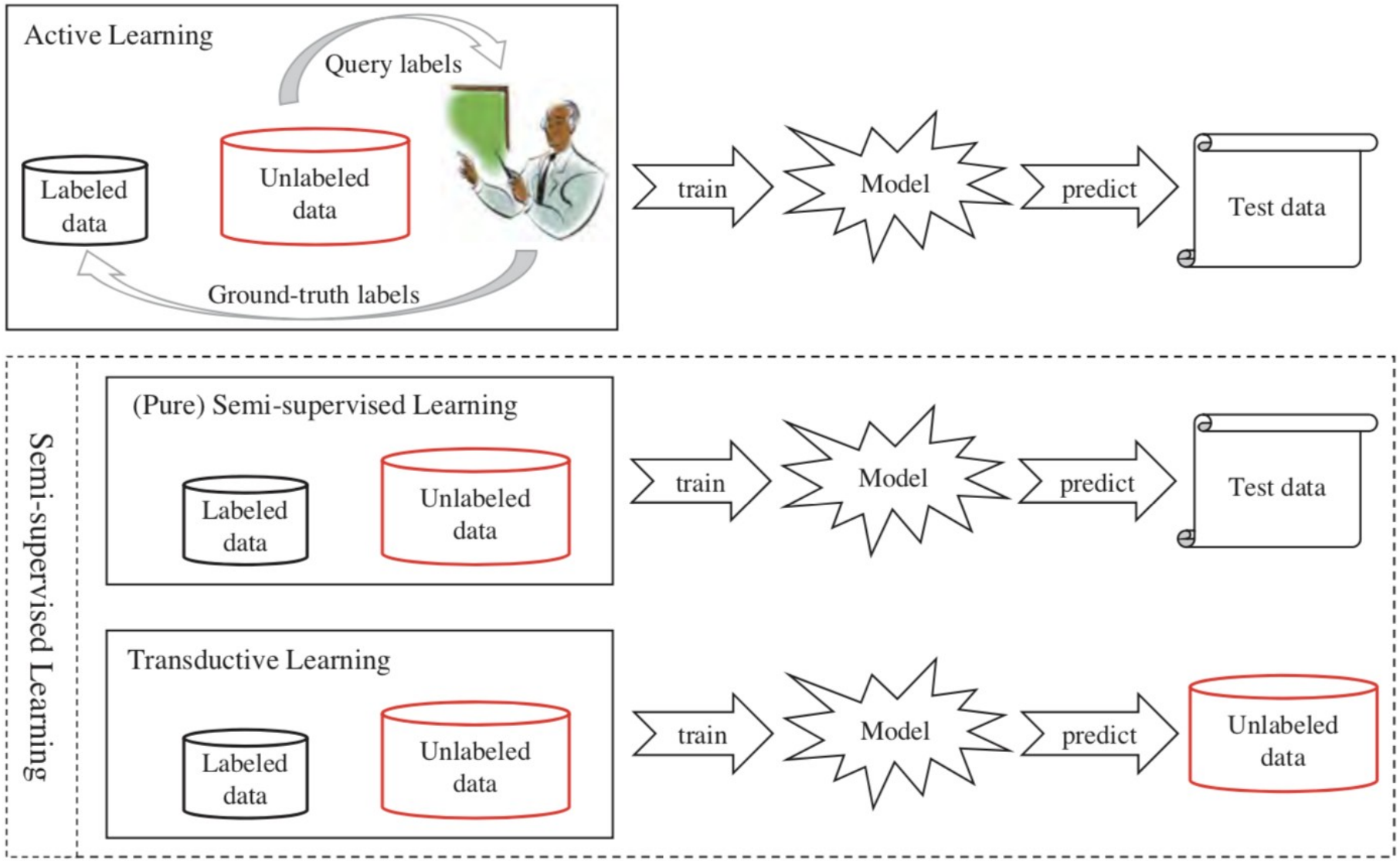
Active Learning

- **Goal**: find *'the best' unlabeled data* samples to ask the **oracle** for their labels, so as to maximally improve the classification performance.

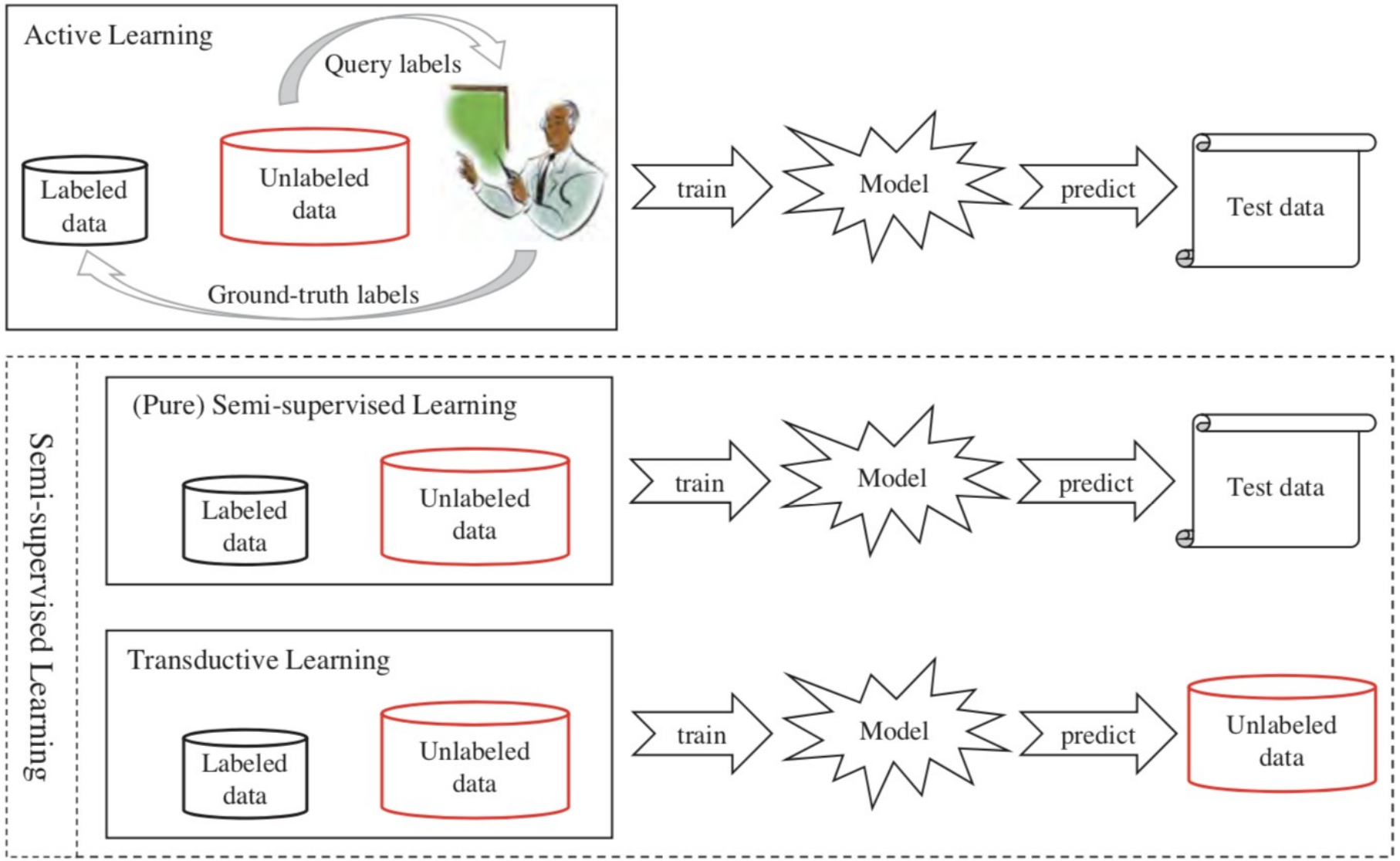
- **Pool-based active learning**
- **Key:** how to choose the data tuples to be queried
  - Uncertainty sampling
  - Query-by-committee
  - Version space
  - Decision-theoretic approach



主动学习 vs. SSL （ vs. 直推学习）

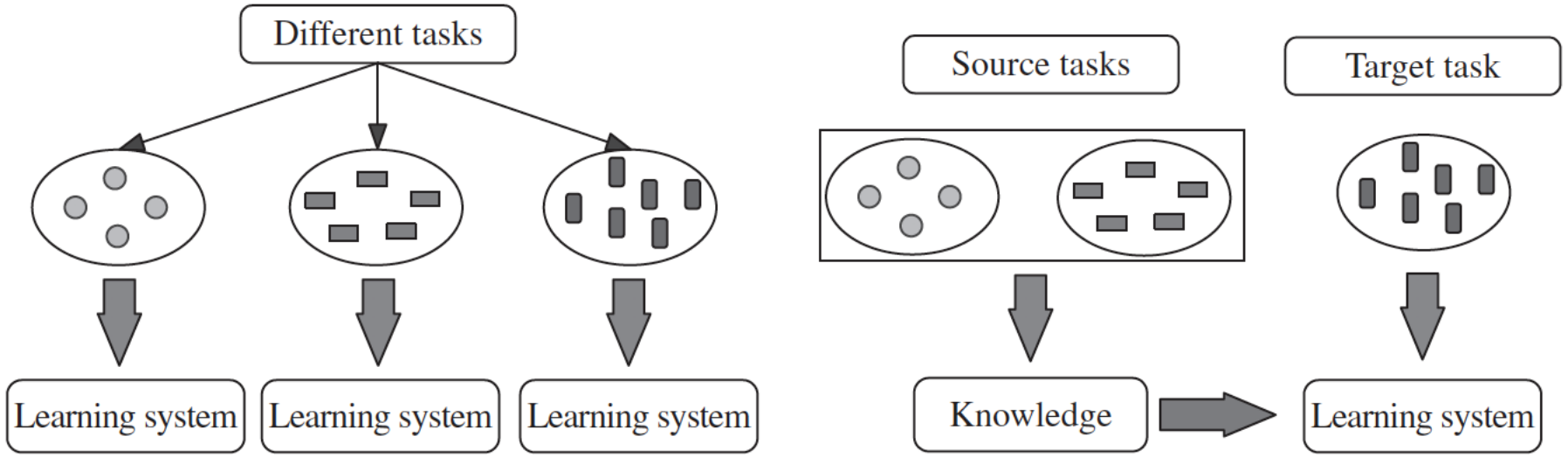


Active Learning vs. SSL (vs. transductive learning)



迁移学习

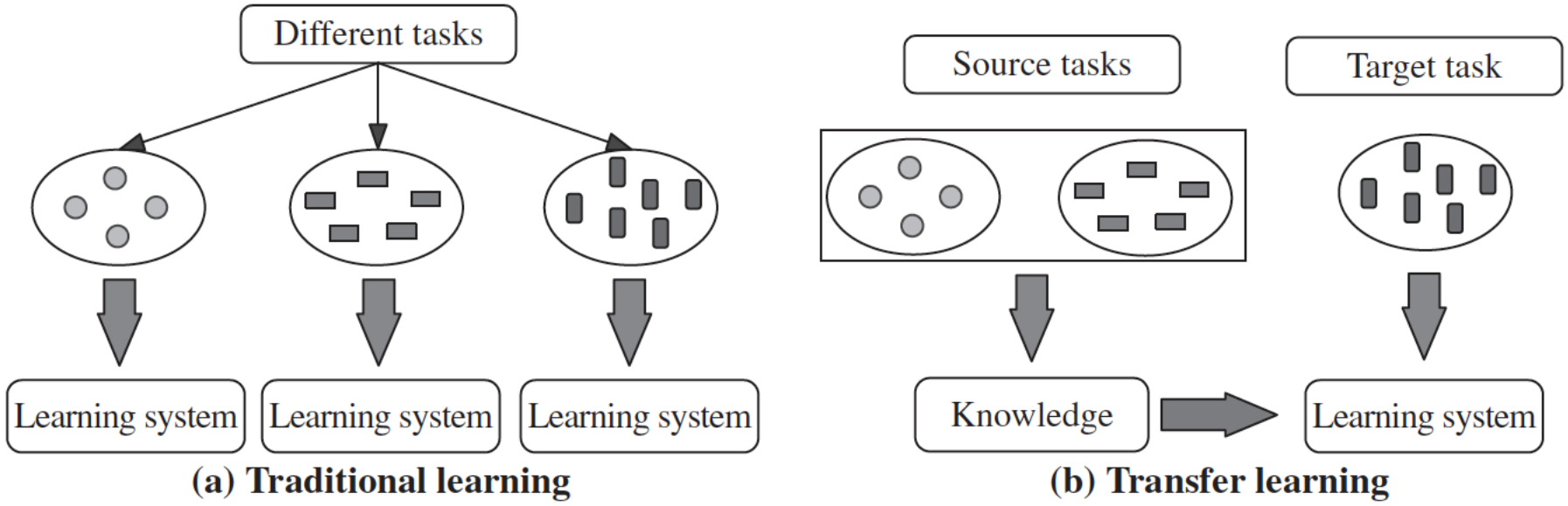
- **目标**：从一个或多个源任务中提取知识，并应用于目标任务



- **示**
  - 源任务：电子产品评论的情感分类
  - 目标任务：电影评论的情感分类

Transfer Learning

- **Goal**: aims to extract the knowledge from one or more source tasks and apply the knowledge to a target task



- **An example**
  - Source task: review sentiment classification on electronics
  - Target task: review sentiment classification on movies

TrAdaBoost

- **一般策略：基于实例的迁移学习**：对源任务的部分数据重新加权，用于学习目标任务。
  - 直觉：把最相关 / 最相似的数据从源“迁移”到目标
- **细节**
  - **Boosting**：每个基学习器更关注困难的训练样本（如被上一个基学习器误分类的样本）
  - **TrAdaBoost**：若源元组被误分类，则降低其权重（如与目标任务不太相关）
- **迁移学习的关键挑战**：负迁移
  - 量化源任务与目标任务的差异
  - 迁移间隔、散度度量
- **相关问题**
  - 多任务学习；预训练 + 微调

TrAdaBoost

- **General strategy: Instance-based transfer learning**: reweight some of the data from the source task and uses it to learn the target task.
  - Intuition: `transfer' most relevant/similar data from source to target
- **Details**
  - **Boosting:** each base learner pays more attention to difficult training examples (e.g., misclassified by the previous base learner)
  - **TrAdaBoost:** if a source tuple is misclassified, reduce its weight (e.g., not very relevant to the target task)
- **Key challenge in transfer learning**: negative transfer
  - Quantify the difference between the source and target tasks
  - Transfer margin, divergence metric
- **Related problems**
  - Multi-task learning; pre-training + fine-tuning

远程监督

- **目标**：自动生成大量有标签元组。
- **所生成标签的特点**
  - 有噪声，但数量大
- **示例 1：** Twitter 情感分类（正面 vs. 负面）
  - if a tweet contains :-), treat it as a positive tuple
  - 若推文含有 :-(，视为负例元组
- **示例 2：** Twitter 类别分类（如新闻、健康、科学、游戏等）
  - 若推文含有 URL，则用该 URL 的 ODP （开放目录项目）类别作为推文标签
  - 若推文含有 Youtube 视频链接，则把该视频的标签作为推文标签

Distant Supervision

- **Goal**: automatically generate a large number of labelled tuples.
- **Characteristic of the generated labels**
  - Noisy, but large in volume
- **Example #1:** Twitter sentiment classification (positive vs. negative)
  - if a tweet contains :-), treat it as a positive tuple
  - If a tweet contains :-(, treat it as a negative tuple
- **Example #2:** Twitter category classification (e.g., news, health, science, games, etc.)
  - If a tweet contains a URL, use the ODP (open directory project) category of the URL as the label of the tweet
  - If a tweet contains a Youtube video link, treat the label of Youtube video as the label of the tweet

远程监督

- **应用**
  - 社交媒体帖子分类
  - NLP 中的关系抽取

- **活跃研究方向：**如何获得*标注函数*

- **主要挑战：噪声标签**
  - 推文中的 :) 可能是中性甚至负面情感

Distant Supervision

- **Applications**
  - Social media post classification
  - Relation extraction for NLP

- **Active research area:** how to obtain *labeling function*

- **Main Challenge: noisy labels**
  - :) in a tweet post could have neutral or even negative sentiment

零样本学习

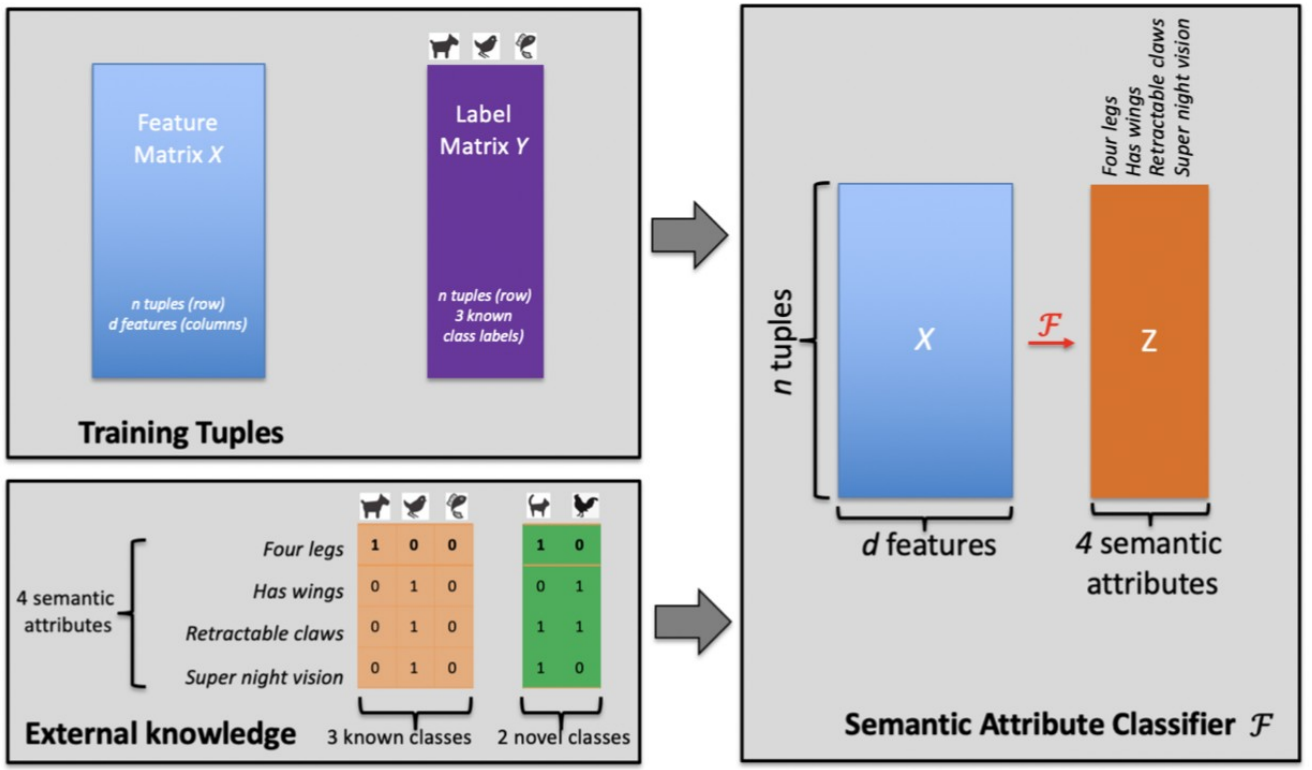
- **目标：**预测类别标签在训练阶段从未出现过的测试元组
- **一个启发性示例：**
  - 训练好的分类器把动物图像分为**猫头鹰** vs. **狗** vs. **鱼**。
  - 但测试图像实际上是**猫**。
- **一般策略：**利用外部知识或辅助信息
- **广义零样本学习：**测试图像可能来自已知类或新类
- **应用：**
  - 图像识别
  - 神经活动识别
  - 图异常检测

Zero-shot Learning

- **Goal:** predict a test tuple whose class label was never observed during the training stage
- **A motivating example**:
  - A trained classifier to classify an animal image into **owl** vs. **dog** vs. **fish**.
  - But, the test image is actually about **cat.**
- **General strategies**: leverage external knowledge or side information
- **Generalized zero-shot learning**: the test image might come from known or novel classes
- **Applications:**
  - Image recognition
  - Neural activity recognition
  - Graph anomaly detection

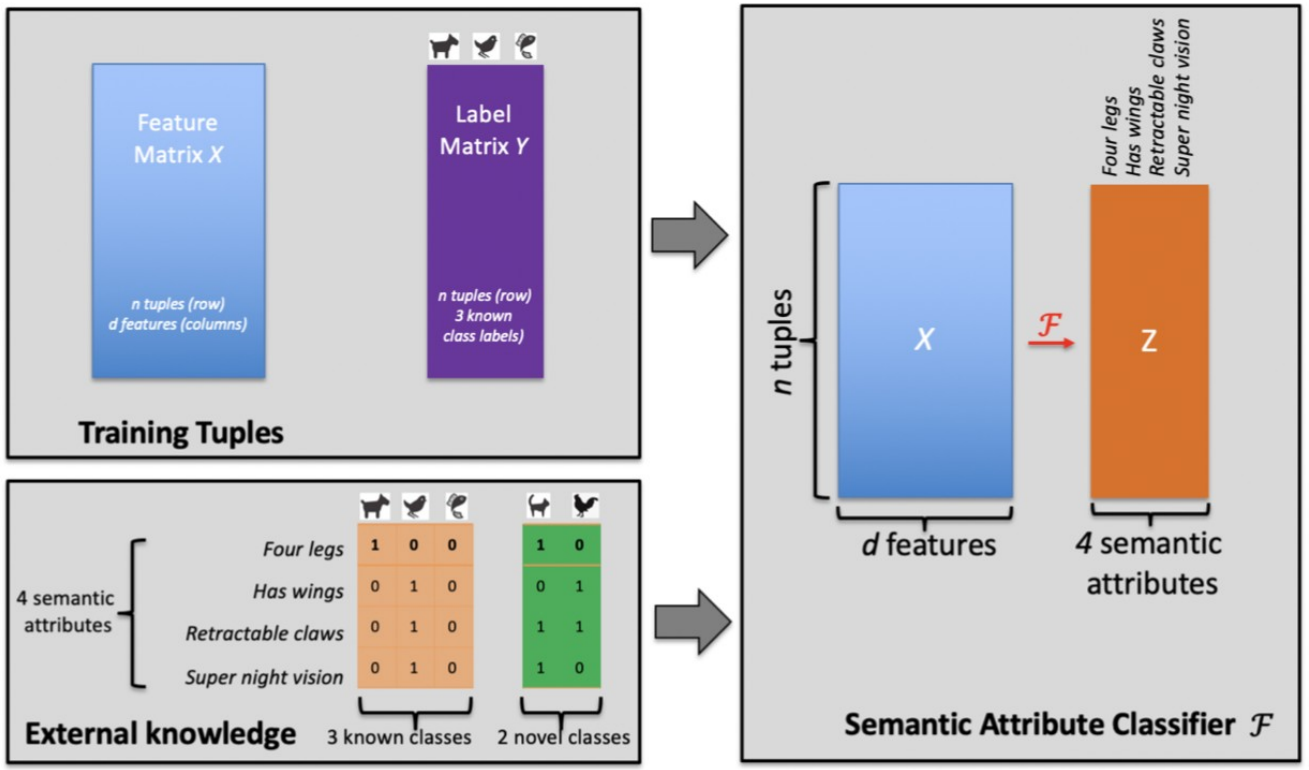
语义属性分类器

- **细节**
  - 用训练元组构建语义属性分类器
  - 用语义属性分类器推断语义属性
  - 用语义属性预测新类别
- **核心思想**：利用语义属性迁移语义分类器的输出



Semantic attribute classifier

- **Details**
  - Use the training tuples to build semantic attribute classifier
  - Use the semantic attribute classifier to infer semantic attributes
  - Use the semantic attributes to predict the novel class
- **Key Idea**: leverage the semantic attributes to transfer the output of the semantic classifier



快速检查：哪种学习设置？

问题：你只能请人从大量无标签样本中标注 20 个。哪种方法会挑选信息量最大的样本？

A	B	C	D
自训练	主动学习	迁移学习	远程监督

答案： B 。

主动学习：学习器选择查询，以最大化有限人工标注的价值。

Quick Check: Which Learning Setting?

Question: You can ask a human to label only 20 examples from a large unlabeled pool. Which method chooses the most informative examples?

A	B	C	D
Self-training	Active learning	Transfer learning	Distant supervision

Answer: B.

Active learning The learner selects queries to maximize the value of limited human labels.

复杂数据类型的分类

- 数据流分类
- 序列分类
- 图数据分类

Classification with Rich Data Type

- Stream Data Classification
- Sequence Classification
- Graph Data Classification

数据流分类

- **一个启发性示例：**
  - 检测欺诈交易的数据挖掘工具
  - 交易以流的形式在不同时间依次到达
  - 训练集 ✉ 构建分类模型 ✉ 接收新数据 ✉ 重新训练 …

- **挑战：**
  - 到达速度快
  - 长度无限
  - 单遍扫描约束
  - 概念漂移

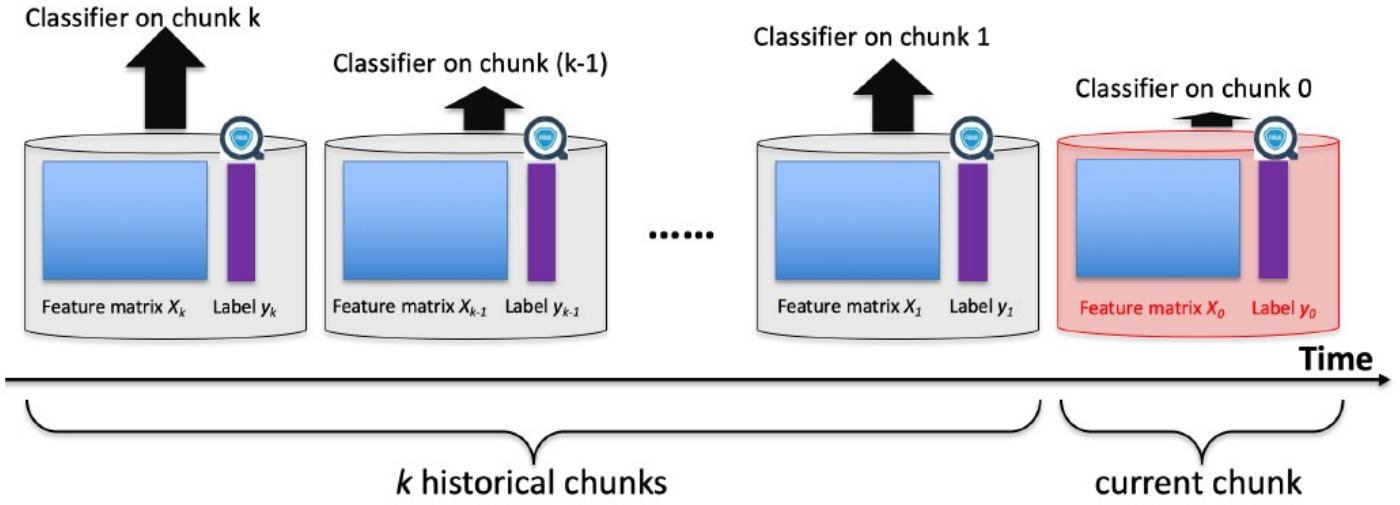
Stream Data Classification

- **A motivating example**:
  - Data mining tool to detect fraudulent transaction
  - the transactions arrive sequentially at different times in a stream fashion
  - Training set → build classification model → receiving new data → retraining …

- **Challenges:**
  - High arrival speed
  - Infinite length
  - One-pass constraint
  - Concept drifting

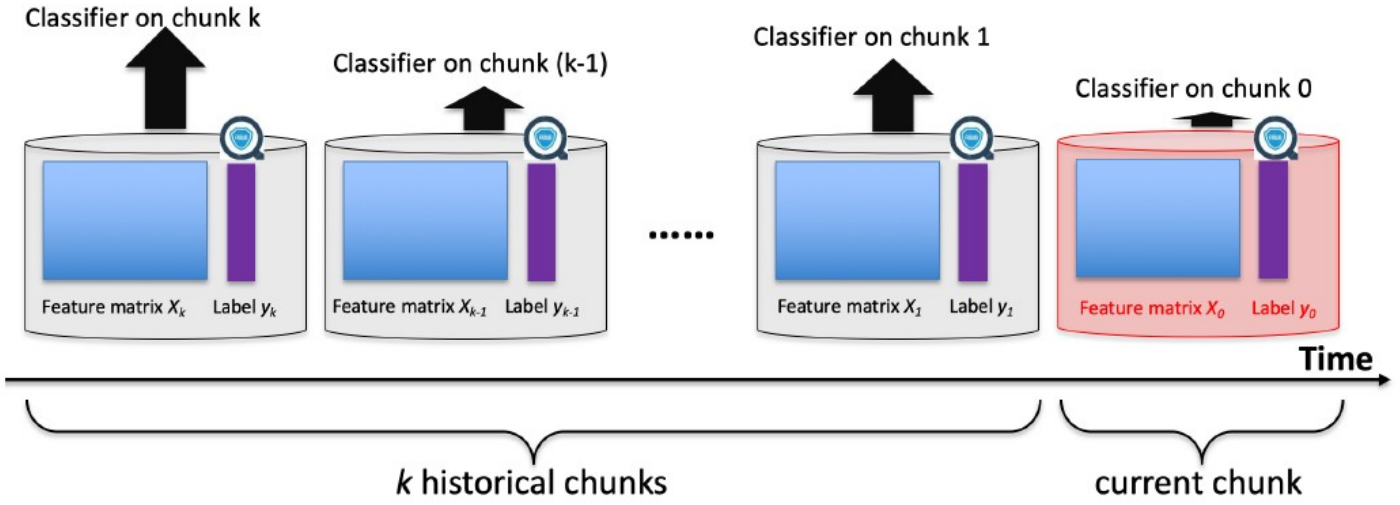
基于集成的数据流分类

- **核心思想：**
  - 只用最新的数据块训练新分类器 ✉ 应对数据流的高到达速率。
  - 每个到达的数据元组只访问一次，用于训练当前分类器并更新各分类器的权重 ✉ 满足单遍约束。
  - 调整各分类器的权重，更关注最相关的数据块 ✉ 捕捉随时间漂移的类别概念。



Stream Data Classification via Ensemble

- **Key Ideas:**
  - Only use the most recent chunk to train the new classifier → high arrival rate of the stream data.
  - Each incoming data tuple is accessed once to train the current classifier as well as to update the weights of individual classifiers → one pass constraint.
  - Adjust the weights of the individual classifiers to pay more attention to the most relevant chunks → captures the drifted class concept over time.



数据流分类

- **VFDT：极速决策树**
  - Hoeffding 树：用训练元组的抽样子集构建决策树，
  - 滑动窗口机制使分类器聚焦于最新的流数据。

- **其他应用**
  - 市场营销
  - 网络监控
  - 传感器网络

Stream Data Classification

- **VFDT: very fast decision tree**
  - Hoeffding trees: build the decision tree by using a sampled subset of training tuples,
  - Sliding window mechanism to obtain classifier to focus on the most recent stream data.

- **Other applications**
  - Marketing
  - Network monitoring
  - Sensor networks

序列分类

- **序列：有序的值列表** $(\boldsymbol{x}^1, \boldsymbol{x}^2, ..., \boldsymbol{x}^T)$
  - 一个句子
  - 一段 DNA
  - 顾客随时间的交易列表

特征向量与序列有什么区别?

- **序列分类：**构建分类器预测给定序列的标签
  - NLP 中句子的正面 vs. 负面情感
  - 基因组分析中的编码区 vs. 非编码区，
  - 市场分析中的高价值顾客 vs. 普通顾客。

- **序列分类的其他形式：**为每个时间戳预测标签

Sequence Classification

- **Sequence: an ordered list of values** $(x^1, x^2, ..., x^T)$
  - A sentence
  - A DNA segment
  - List of transactions of a customer over time

What is the difference between a feature vector and a sequence?

- **Sequence Classification:** build a classifier to predict the label of a given sequence
  - the positive vs. negative sentiment of the sentence in NLP
  - the gene coding area vs. non-coding area in genomic analysis,
  - high-value vs. ordinary customer in market analysis.

- **Other forms of sequence classification:** predict label for each time stamp

基于特征工程的序列分类

- **基本思路**
  - 把输入序列转换为特征向量
  - 训练常规分类器
- **对符号序列：** n-gram
  - 输入 DNA 片段：`ACCCCCGT’
  - 输出：

Unigrams　　Bigrams

Index	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Candidate features	A	C	G	T	AC	AG	AT	CG	CT	CA	GA	GC	GT	TA	TC	TG	AA	CC	GG	TT
Feature vector (binary)	1	1	1	1	1	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0
Feature vector (frequency)	1	5	1	1	1	0	0	1	0	0	0	0	1	0	0	0	0	4	0	0

- **对数值序列：通过离散化转换为符号序列**
- **较新的方法：** RNN 及相关技术

Sequence Classification via Feature Engineering

- **General Ideas**
  - Convert the input sequence to a vector of features
  - Train a conventional classifier
- **For symbolic sequence: n-gram**
  - Input DNA segment: `ACCCCCGT'
  - Output:

Unigrams | Bigrams

Index	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Candidate features	A	C	G	T	AC	AG	AT	CG	CT	CA	GA	GC	GT	TA	TC	TG	AA	CC	GG	TT
Feature vector (binary)	1	1	1	1	1	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0
Feature vector (frequency)	1	5	1	1	1	0	0	1	0	0	0	0	1	0	0	0	0	4	0	0

- **For numerical sequence: convert to symbolic sequence by discretization**
- **More recent approaches:** RNN and related techniques

基于距离 / 核函数的序列分类

- **观察：**
  - K-NN 分类器：依赖距离或相似度度量
  - 非线性 SVM ：依赖核函数
- **一般策略：在序列上定义合适的距离度量或核函数**
- **序列上的距离度量**
  - 欧氏距离
  - 动态时间规整（DTW)：处理时间上的扭曲
- **序列上的核函数**
  - 字符串核

Sequence Classification via Distance/Kernel Function

- **Observations:**
  - K-NN classifier: rely on distance or similarity measure
  - Non-linear SVM: rely on a kernel function
- **General strategy: define appropriate distance measure or kernel function on sequence**
- **Distance measure on sequence**
  - Euclidean distance
  - Dynamic time warping (DTW): to account for distortion in time
- **Kernel function on sequence**
  - String kernel

图数据分类

- **图数据（又称网络数据）：由边连接的结点集合**
  - 社交网络
  - 电网
  - 交易网络
  - 生物网络

- **结点级 vs. 图级分类**
  - 结点级分类：网页分类
  - 图级分类：毒性分类

Graph Data Classification

- **Graph Data (aka network data): collection of nodes linked by edges**
  - Social networks
  - Power grid
  - Transaction networks
  - Biological networks

- **Node-level vs. graph-level classification**
  - Node-level classification: webpage classification
  - Graph-level classification: toxicity classification

图数据分类方法

- **基于特征工程的图数据分类**
  - 提取描述每个结点或每个图的一组特征
  - 输入常规分类器，构建结点级或图级分类器
    - 结点级特征：度、三角形个数、中心性、 pagerank 等
    - 图级特征：规模、直径、三角形个数等
    - 替代方法：用深度学习 (GNN) “自动”提取结点 / 图级特征

- **基于邻近度度量的图数据分类**
  - 度量结点或图之间的邻近度
  - 构建 KNN 类分类器

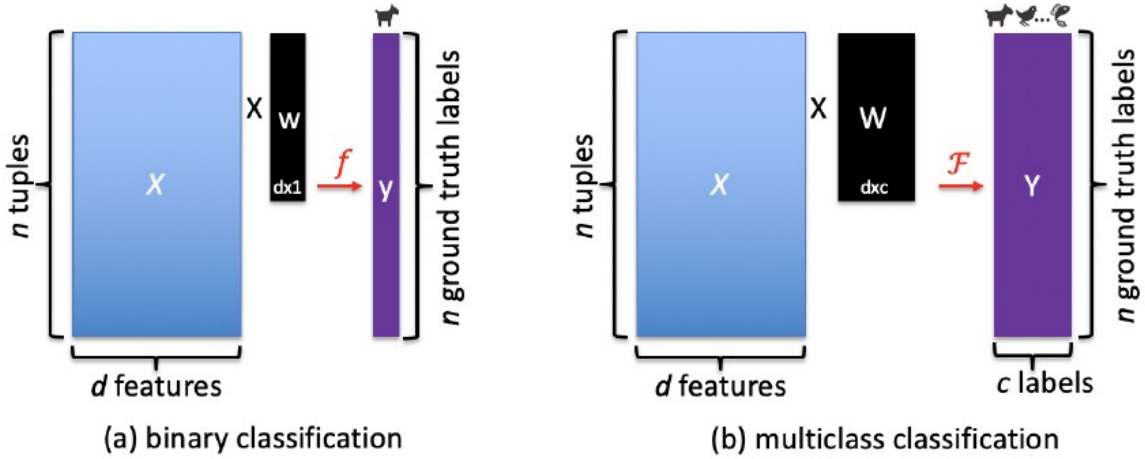
Graph Data Classification Methods

- **Graph Data Classification via Feature engineering**
  - Extract a set of features describing each node or each graph
  - Fed into a conventional classier to build a node-level or graph-level classifier
    - Node-level features: degree, # of triangles, centrality, pagerank, etc.
    - Graph-level features: size, diameter, # of triangles, etc.
    - Alternative methods: using deep learning (GNN) to "automatically" extract node/graph level features

- **Graph Data Classification via Proximity measure**
  - Measure proximity between nodes or graphs
  - Build KNN type of classifier

多类分类

- 二分类 vs. 多类分类

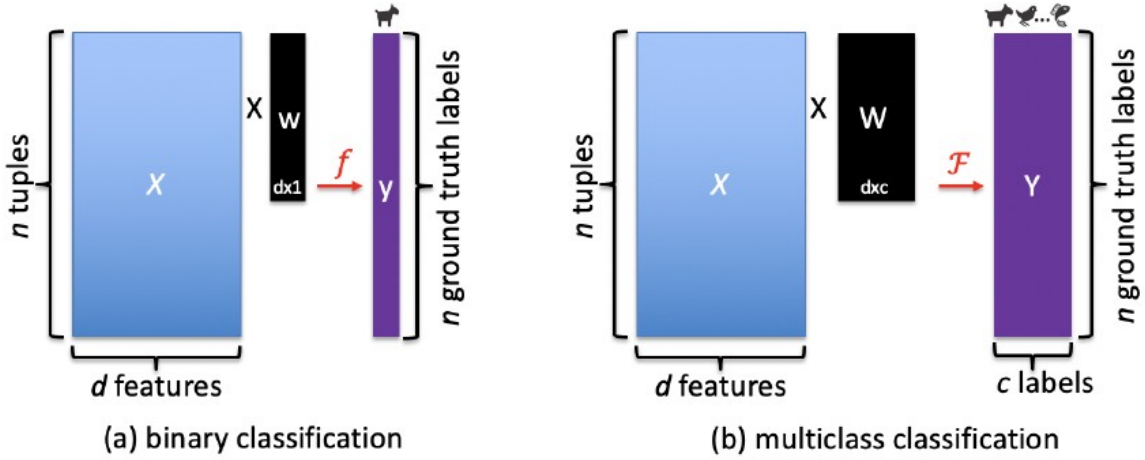


- 多类分类方法
  - 一对全部（OVA）：训练 m 个二分类器，每类一个（类 j vs. 其余）
  - 全部对全部（AVA）：为任意两类训练一个二分类器
  - 纠错编码：
    - 为每个类分配一个纠错码
    - 提高多类分类的准确率

Class	*Error-correcting codeword*
C_1	1 1 1 1 1 1 1
C_2	0 0 0 0 1 1 1
C_3	0 0 1 1 0 0 1
C_4	0 1 0 1 0 1 0

- 相关问题：多标签分类（每个数据元组可属于一个或多个类别）

Multiclass Classificati



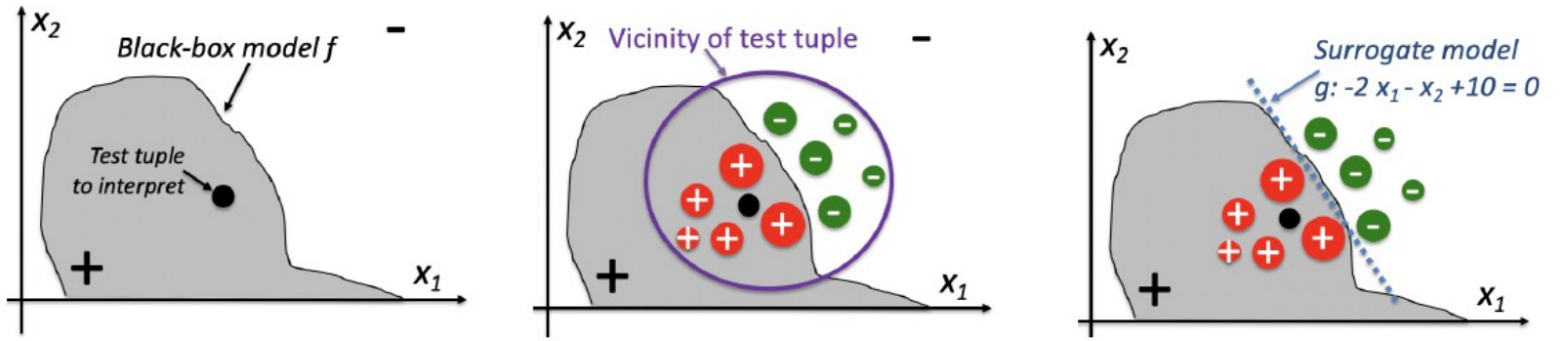
- **Binary vs. multiclass classification**
- **Methods for multiclass classification**
  - **One-versus-all (OVA):** train *m* binary classifiers, one for each class (class *j* vs. the remaining)
  - **All-versus-all (AVA):** train a binary classifier for any two classes
  - **Error-correcting coding:**
    - Assign each class an error-correcting code
    - Improve the accuracy of multiclass classification

Class	*Error-correcting codeword*
C_1	1 1 1 1 1 1 1
C_2	0 0 0 0 1 1 1
C_3	0 0 1 1 0 0 1
C_4	0 1 0 1 0 1 0

- **A related problem:** multi-label classification (each data tuple could belong to one or more classes)

分类的可解释性

- **设定：**以人能理解的方式解释模型的预测或决策过程的能力
- **易于解释的分类器：**决策树归纳、线性分类器
- **LIME：局部可解释的模型无关解释**
  - **核心思想：**训练局部代理模型（如稀疏线性模型），平衡局部保真度与可解释性

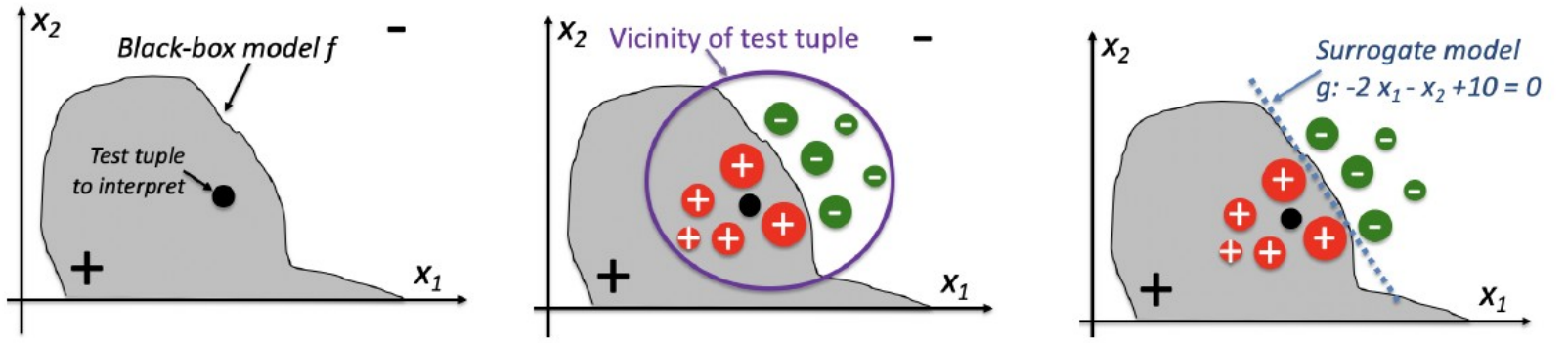


(a) black-box model f (b) local weighted sampling (c) training surrogate model

- **其他方法：**决策类解释、影响函数

Interpretability of Classification

- **Setting:** the ability to explain a model's predictions or decision process in a way people can understand
- **Interpretability-friendly classifiers:** decision tree induction, linear classifiers
- **LIME: Local Interpretable Model-agnostic Explanations**
  - **Key idea:** train a local surrogate model (e.g., a sparse linear model) to balance the local fidelity and interpretability



(a) black-box model f (b) local weighted sampling (c) training surrogate model

- **Alternative approaches:** counterfactual explanation, influence function

第 2 次课综合：让方法匹配学习设置

什么在变化?	代表性方法	首先要问的问题
期望输出	IF–THEN 规则；顺序覆盖； CBA	预测是否必须可直接检查?
标签可得性	SSL ；主动；迁移；远程；零样本	可靠的监督从哪里来?
输入结构	数据流；序列；图分类器	模型必须保留哪些依赖关系?
预测范围	多类策略与可解释性	如何评估错误与解释?

依据监督来源、数据结构和评估目标选择分类器——而不只看算法名称。

Session 2 Synthesis: Match the Method to the Learning Setting

What changes?	Representative methods	Question to ask first
Desired output	IF–THEN rules; sequential covering; CBA	Must predictions be directly inspectable?
Label availability	SSL; active; transfer; distant; zero-shot	Where can reliable supervision come from?
Input structure	Stream; sequence; graph classifiers	Which dependencies must the model preserve?
Prediction scope	Multiclass strategies and interpretability	How will errors and explanations be evaluated?

Choose the classifier by supervision source, data structure, and evaluation goal - not by algorithm name alone.

第 2 次课附录

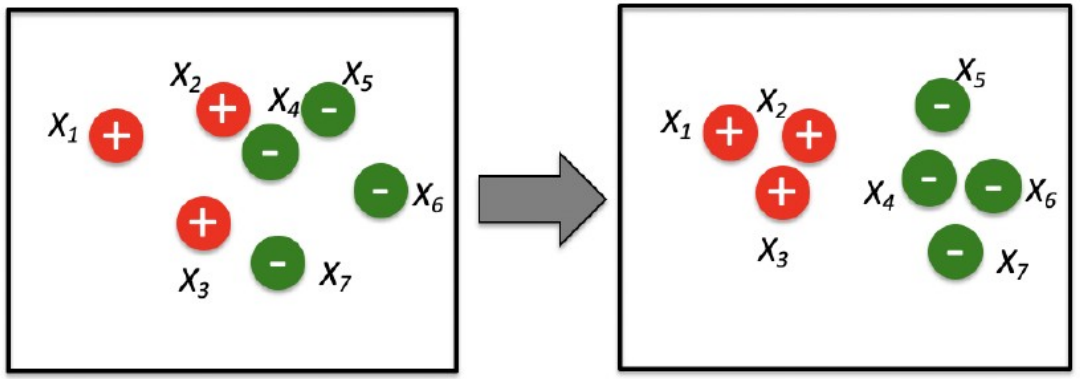
Appendix for Session 2

附录 2：度量学习

- **设定：**为给定分类任务（如 k 近邻分类器）自动学习最佳距离度量。
- **学习马氏距离**
  - 示例：
  - 详细形式

$$\max_{M \in R^{p \times p}} \sum_{(X_i, X_j) \in \mathcal{D}} d_M^2(X_i, X_j)$$

$$\text{s.t.} \sum_{(X_i, X_j) \in \mathcal{S}} d_M^2(X_i, X_j) \leq 1, \quad M \succeq 0$$



(a) Euclidean distance　　(b) Learnt Mahalanobis distance

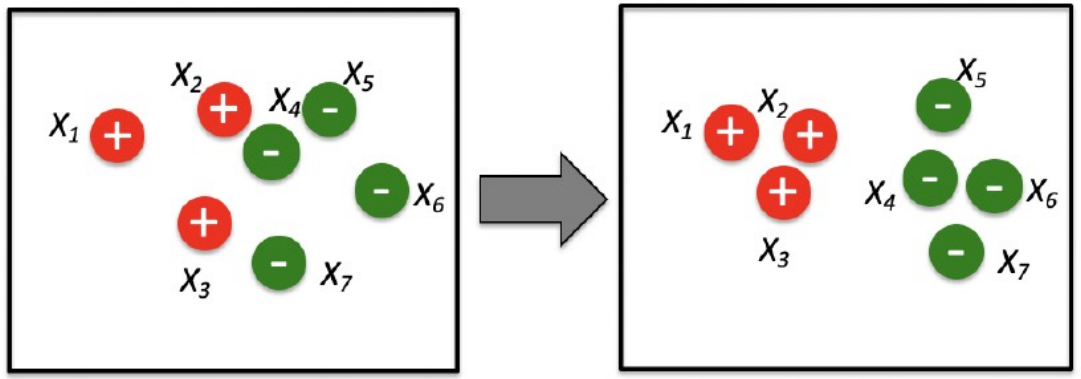
  - 直观：(1) S 中任意一对相似元组的距离小；(2) D 中任意一对不相似元组的距离大
  - M 为半正定矩阵；凸形式

Appendix 2: Metric Learning

- **Setting:** automatically learn the best distance metrics for a given classification task (e.g., k-nearest-neighbor classifiers).
- **Learning Mahalanobis distance**
  - An example:
  - Detailed formulation

$$\max_{M \in R^{p \times p}} \sum_{(X_i, X_j) \in \mathcal{D}} d_M^2(X_i, X_j)$$

$$\text{s.t.} \sum_{(X_i, X_j) \in \mathcal{S}} d_M^2(X_i, X_j) \leq 1, \quad M \succeq 0$$



(a) Euclidean distance (b) Learnt Mahalanobis distance

  - *Intuition*: (1) the distance between any pair of similar tuples in *S* is small and (2) the distance between any pair of dissimilar tuples in *D* is large
  - *M* is positive semi-definite; convex formulation

遗传算法

- **核心思想：**借鉴自然进化的思想
- **一般流程：**
  - 生成由随机规则组成的初始种群
    - 每条规则用一个比特串表示
    - 示例：规则“IF A1 AND NOT A2 THEN C2 "；比特串：“100"
  - 依据适者生存形成新种群
    - 规则的适应度：以分类准确率评估
    - 通过遗传算子（如交叉、变异）产生后代
  - 持续进化，直到种群中每条规则都达到适应度阈值
- **在数据挖掘中的应用：**评估其他算法的适应度

Genetic Algorithms

- **Key idea:** incorporate ideas of natural evolution
- **General procedure:**
  - Create an initial population consisting of random generated rules
    - each rule represented by a string of bits
    - Example: The rule: “IF A1 AND NOT A2 THEN C2 "; bit string: “100"
  - Form a new population based on survival of the fittest
    - Fitness of a rule: assessed by classification accuracy
    - Create offspring by genetic operators (e.g., crossover, mutation)
  - Evolve until a fitness threshold is satisfied for each rule in the population
- **Applications to data mining:** evaluate the fitness of other algorithms

强化学习

- **与分类的关系**
  - **分类：**接收指导性反馈（即真实类别标签）来构建最佳分类器
  - **强化学习：**接收评价性反馈（如某动作的即时奖励）来寻找最佳动作
- **多臂老虎机问题：**
  - *设定*：一台有多个摇臂的老虎机，每个摇臂对应一个价值未知的动作（如某产品的一则广告）。要确定每次采取哪个动作。
  - *算法*： ε- 贪心算法、上置信界方法
- **应用：**在线广告、机器人、国际象棋

Reinforcement Learning

- **Relationship with classification**
  - **Classification:** receives the instructive feedback (i.e., the true class label) to construct the best classifier
  - **Reinforcement learning:** receives evaluative feedback (e.g., the immediate reward of an action) to find the best action
- **Multi-armed bandit problem:**
  - *Setting*: a slot machine with multiple arms, and each arm corresponds to an action (an advertisement on a product) with an unknown value. Find out each action to take.
  - *Algorithms*: ε-greedy algorithm, upper-confidence-bound method
- **Applications:** online advertisement, robotics, chess